Lecture 15 / 16:

Cameras & Lenses

Computer Graphics and Imaging
UC Berkeley CS184

Rendering with Realistic Camera Model



Credit: Bertrand Benoit. “Sweet Feast,” 2009. [Blender /VRay]

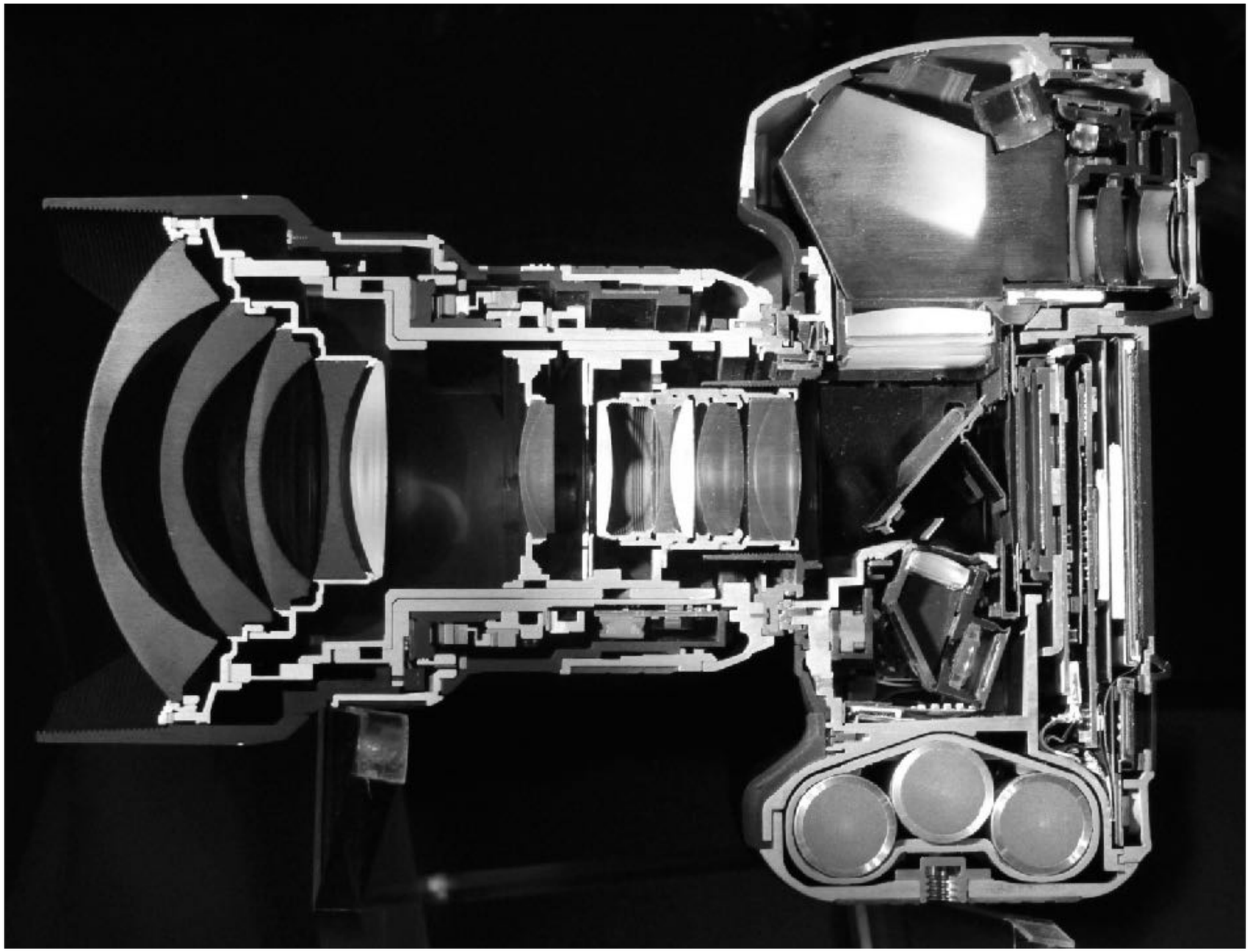
Rendering with Realistic Camera Model



Credit: Giuseppe Albergo. “Colibri” [Blender]

Image Capture Overview

What's Happening Inside the Camera?



Cross-section of Nikon D3, 14-24mm F2.8 lens

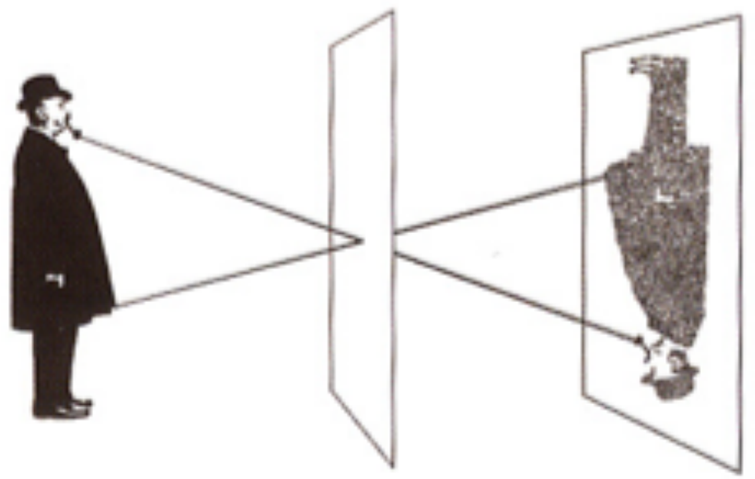
Ren Ng

Presented by David McAllister

Pinholes & Lenses Form Image on Sensor

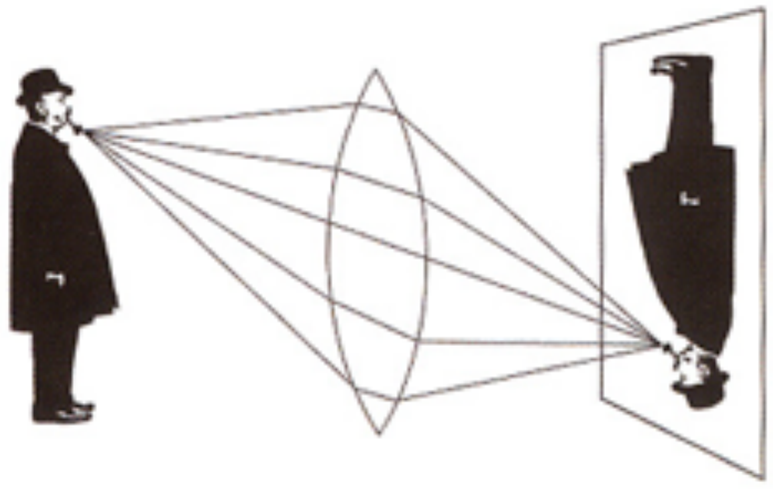
Photograph made with small pinhole





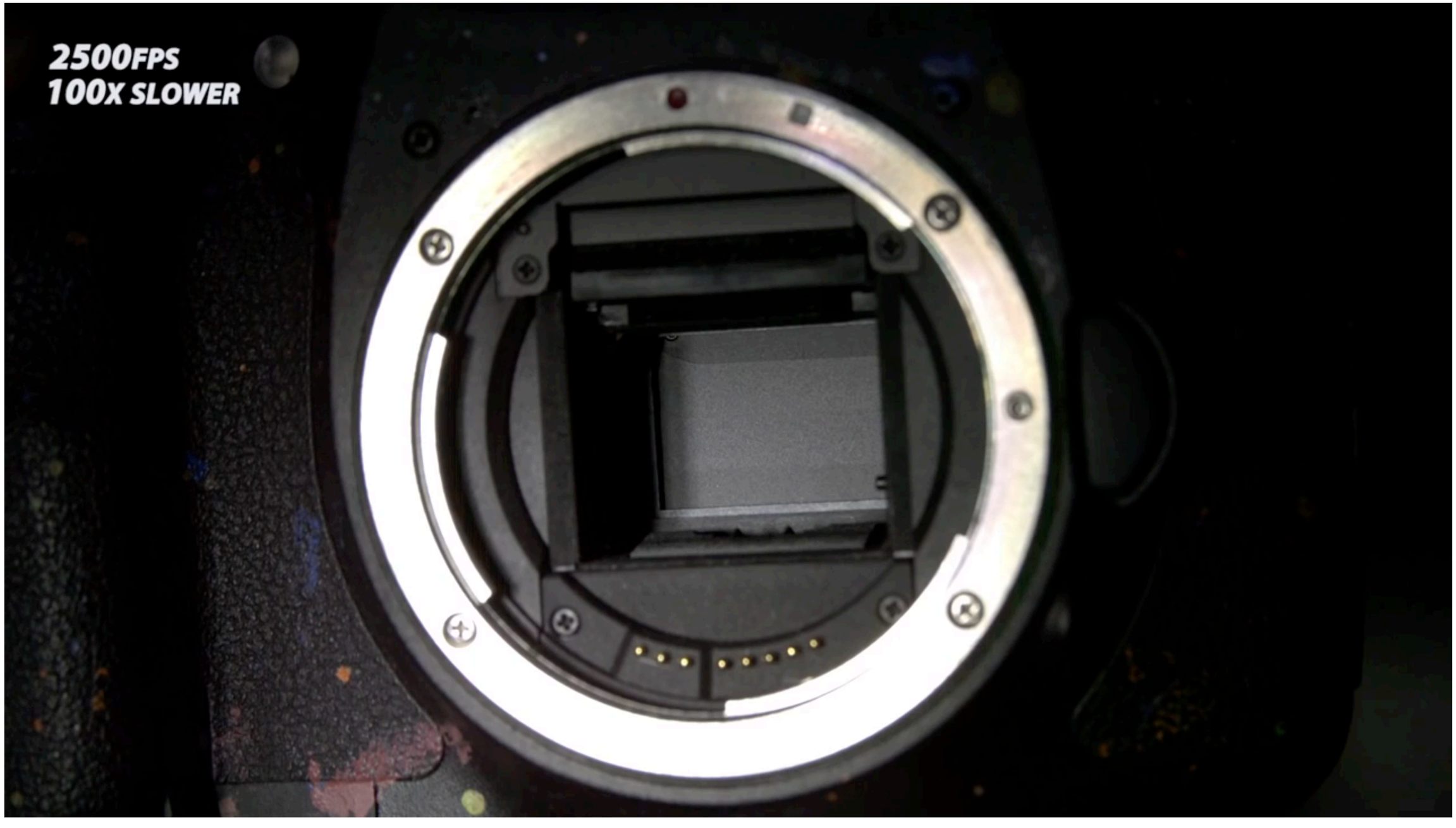
Photograph made with lens





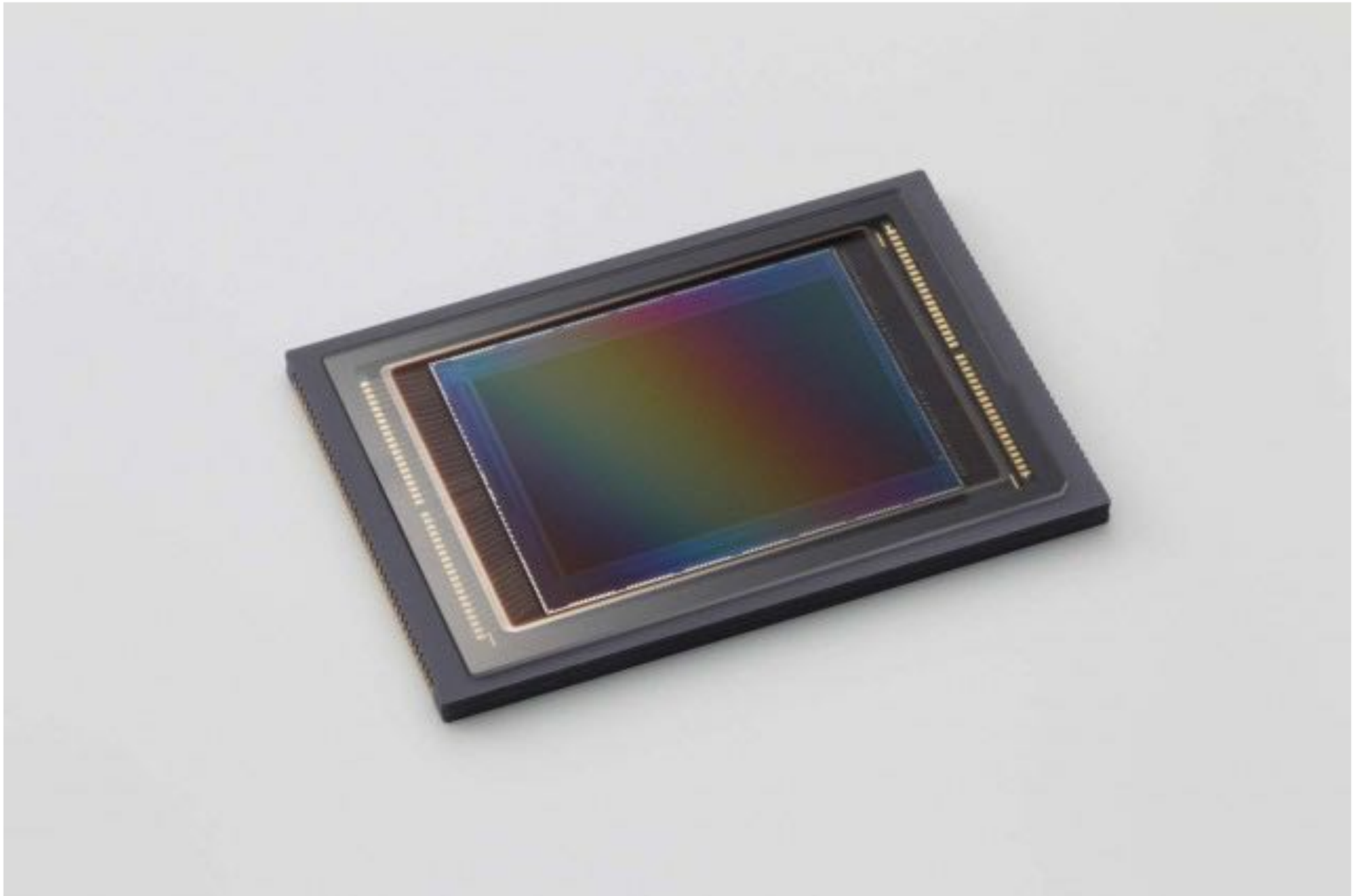
London and Upton

Shutter Exposes Sensor For Precise Duration



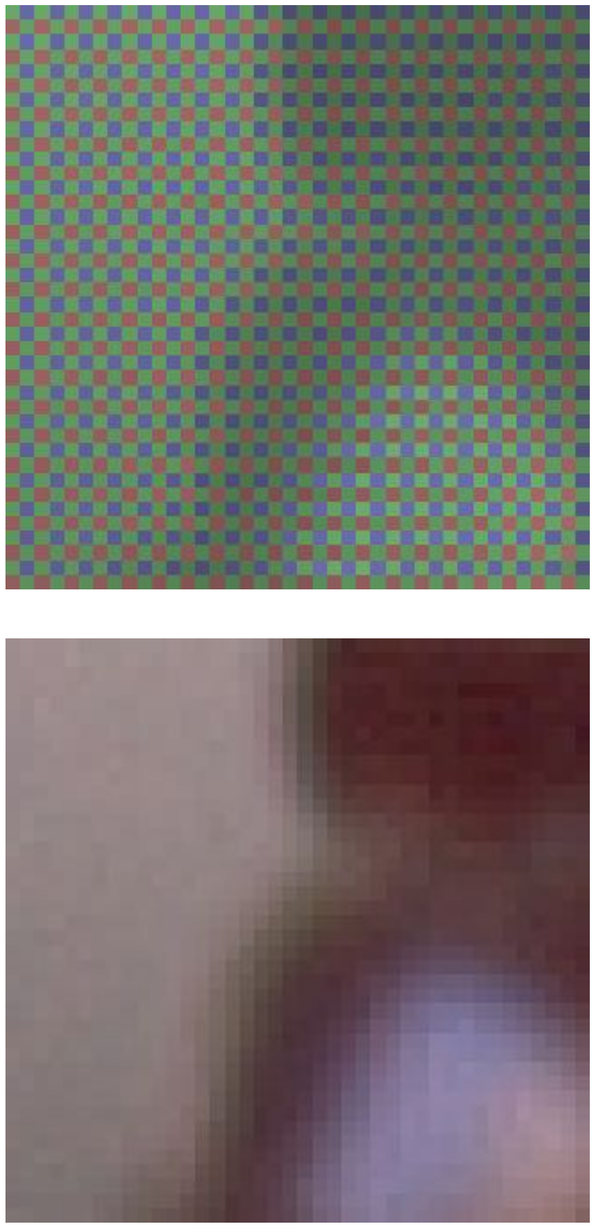
The Slow Mo Guys, https://youtu.be/CmjeCchGRQo

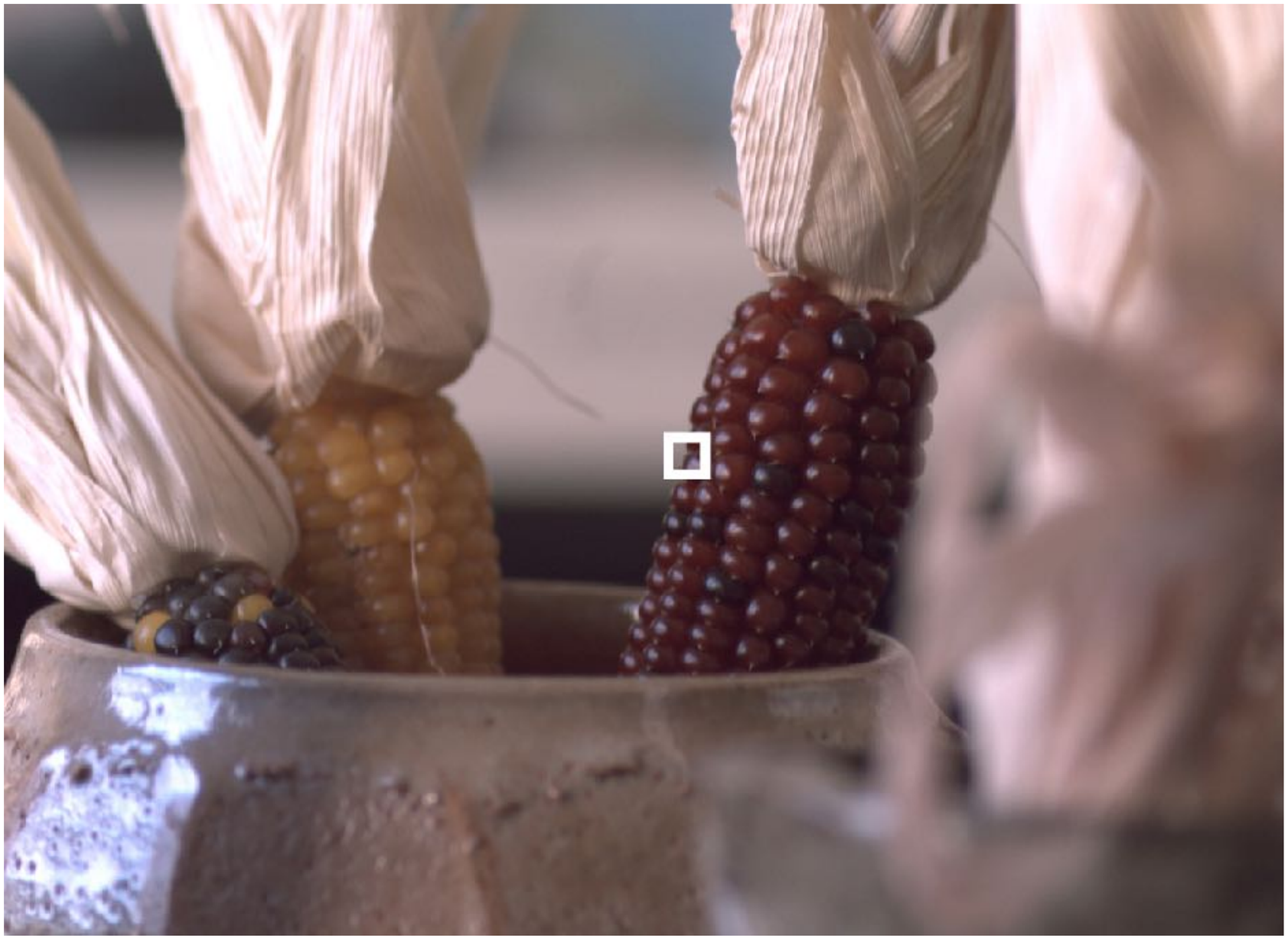
Sensor Accumulates Irradiance During Exposure



Ren Ng
Presented by David McAllister

Image Processing: From Sensor Values to Image





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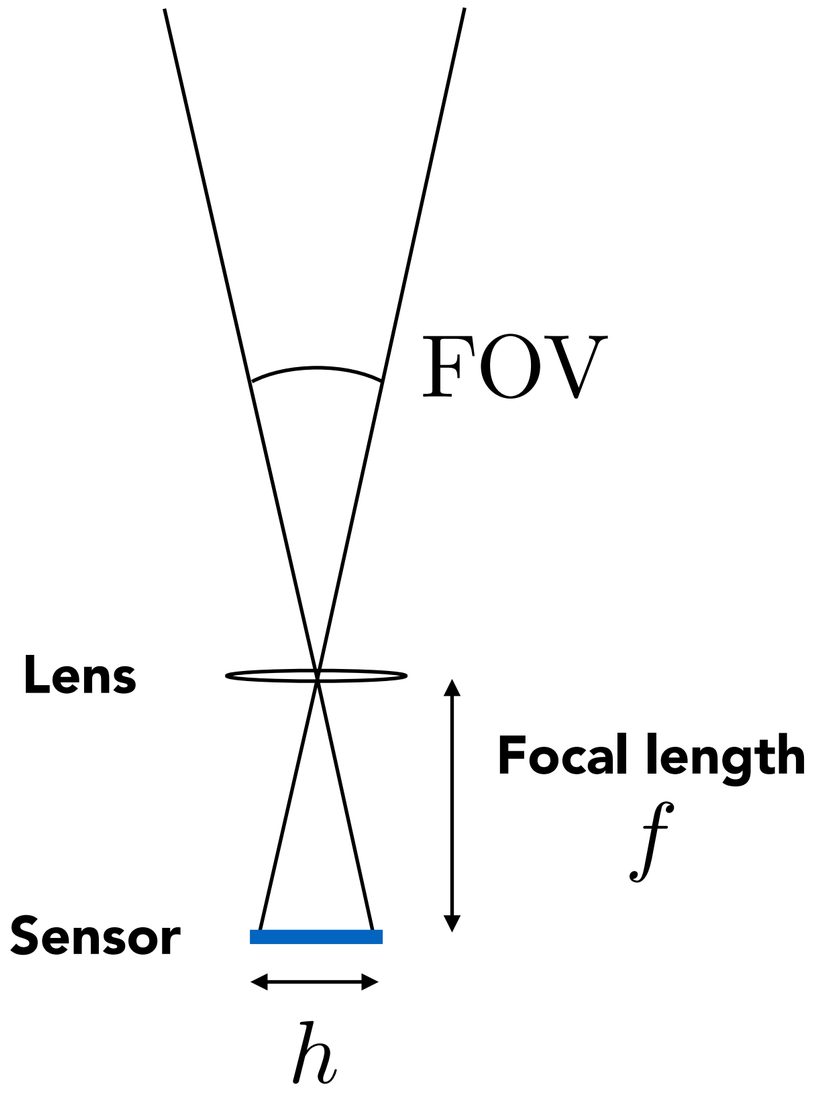
Ren Ng

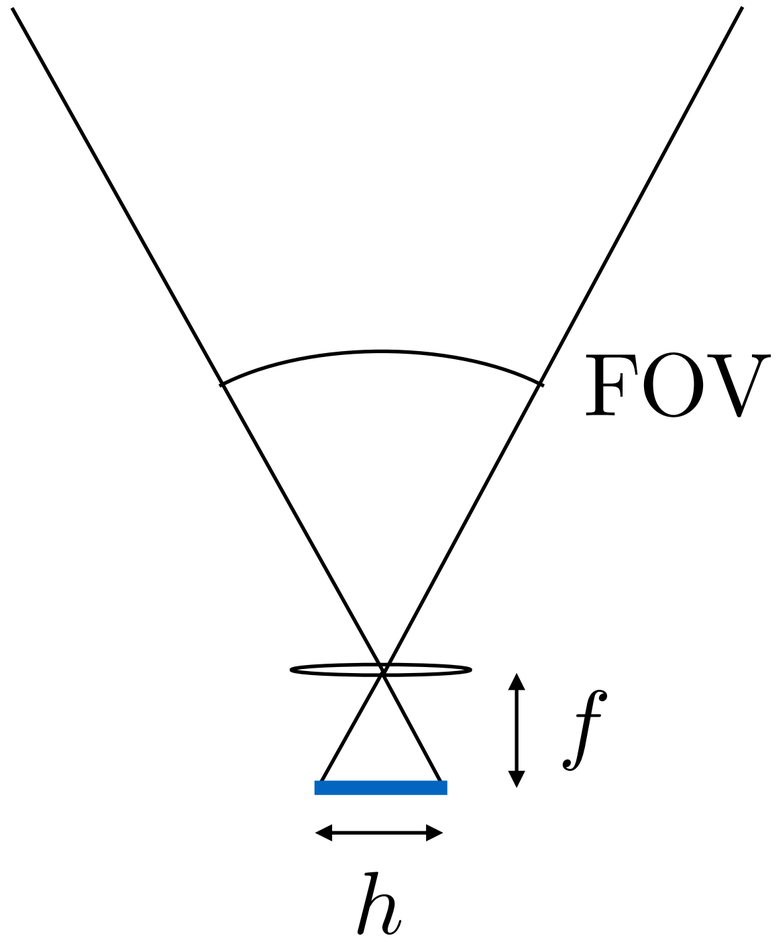
Presented by David McAllister

Optics of Image Formation:

Field of View

Effect of Focal Length on FOV

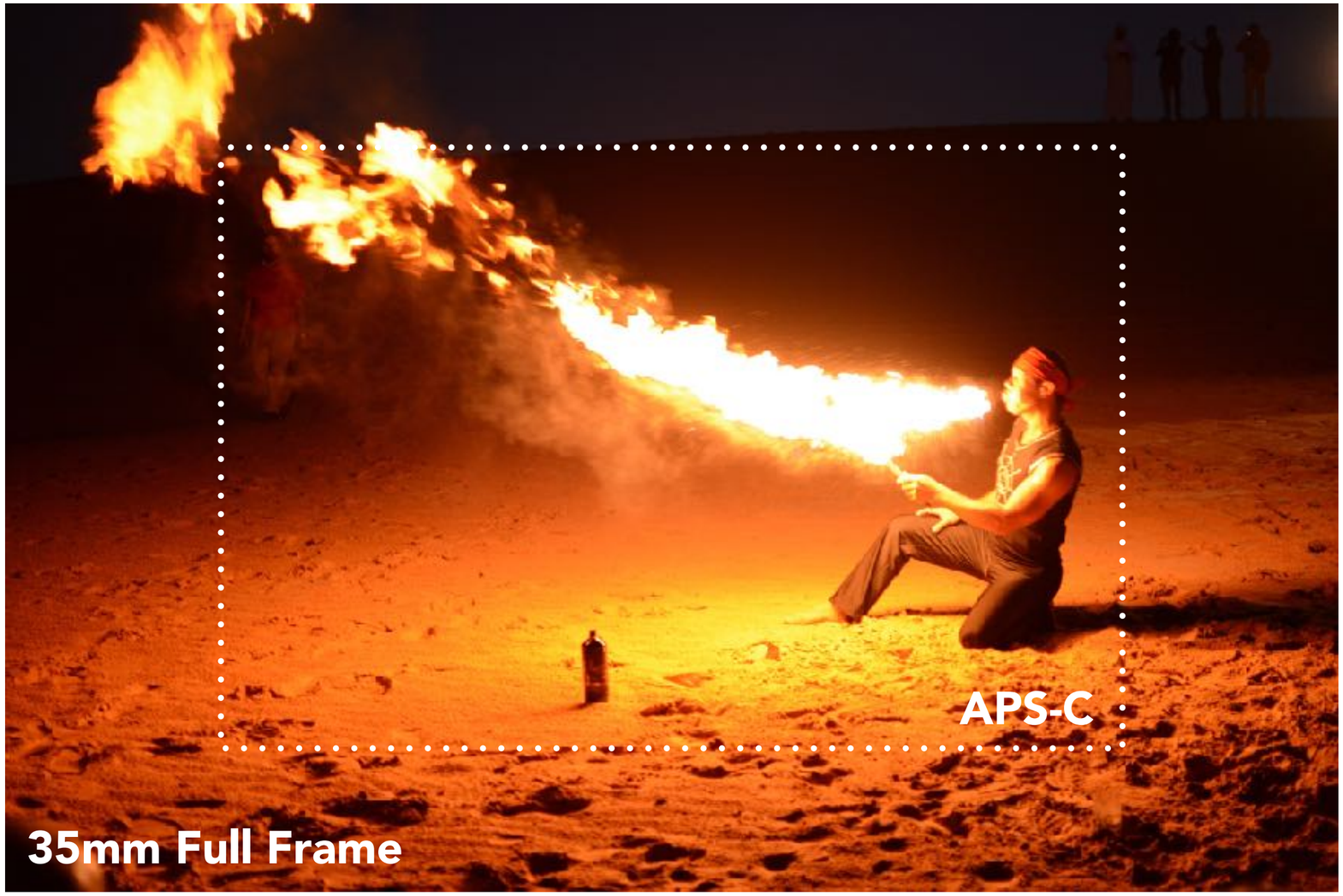


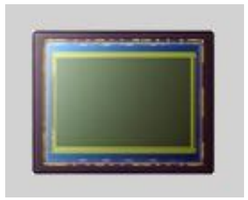


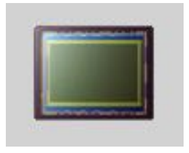
For a fixed sensor size, decreasing the focal length increases the field of view.

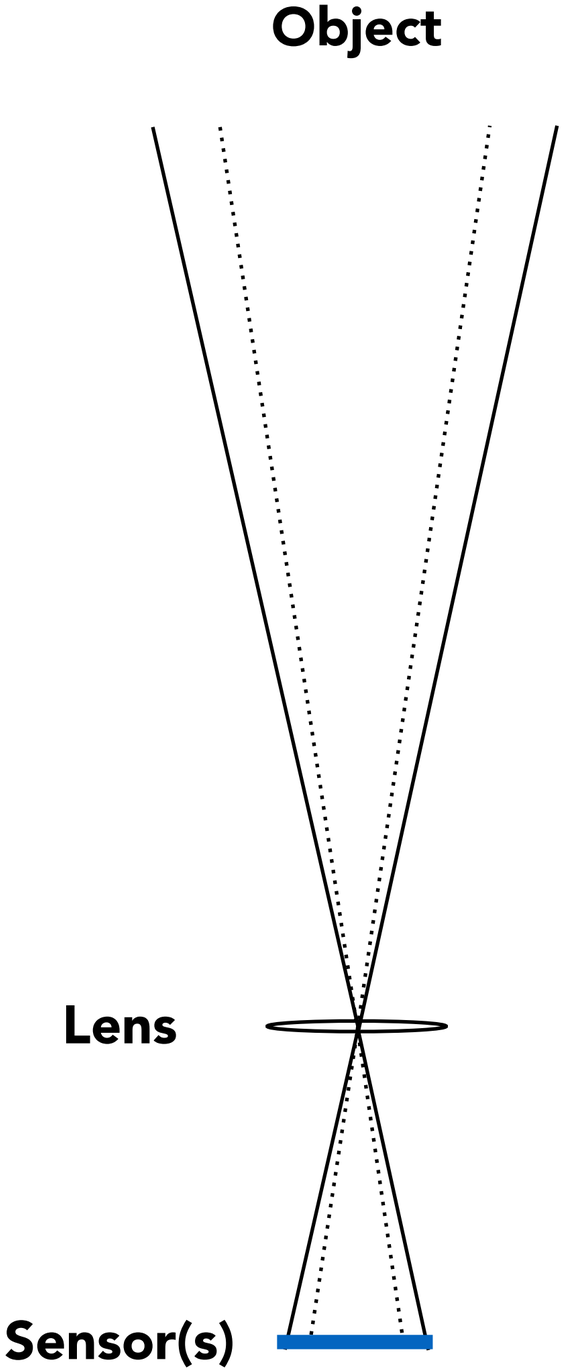
$$\text{FOV} = 2 \arctan\left(\frac{h}{2f}\right)$$

Effect of Sensor Size on FOV









Sensor Sizes

Sensor Name	Medium Format	Full Frame	APS-H	APS-C	4/3	1"	1/1.63"	1/2.3"	1/3.2"
Sensor Size	53.7 x 40.2mm	36 x 23.9mm	27.9x18.6mm	23.6x15.8mm	17.3x13mm	13.2x8.8mm	8.38x5.59mm	6.16x4.62mm	4.54x3.42mm
Sensor Area	21.59 cm²	8.6 cm²	5.19 cm²	3.73 cm²	2.25 cm²	1.16 cm²	0.47 cm²	0.28 cm²	0.15 cm²
Crop Factor	0.64	1.0	1.29	1.52	2.0	2.7	4.3	5.62	7.61
Image									
Example									

Credit: lensvid.com

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Maintain FOV on Smaller Sensor?







Lens

Large Sensor

Focal length



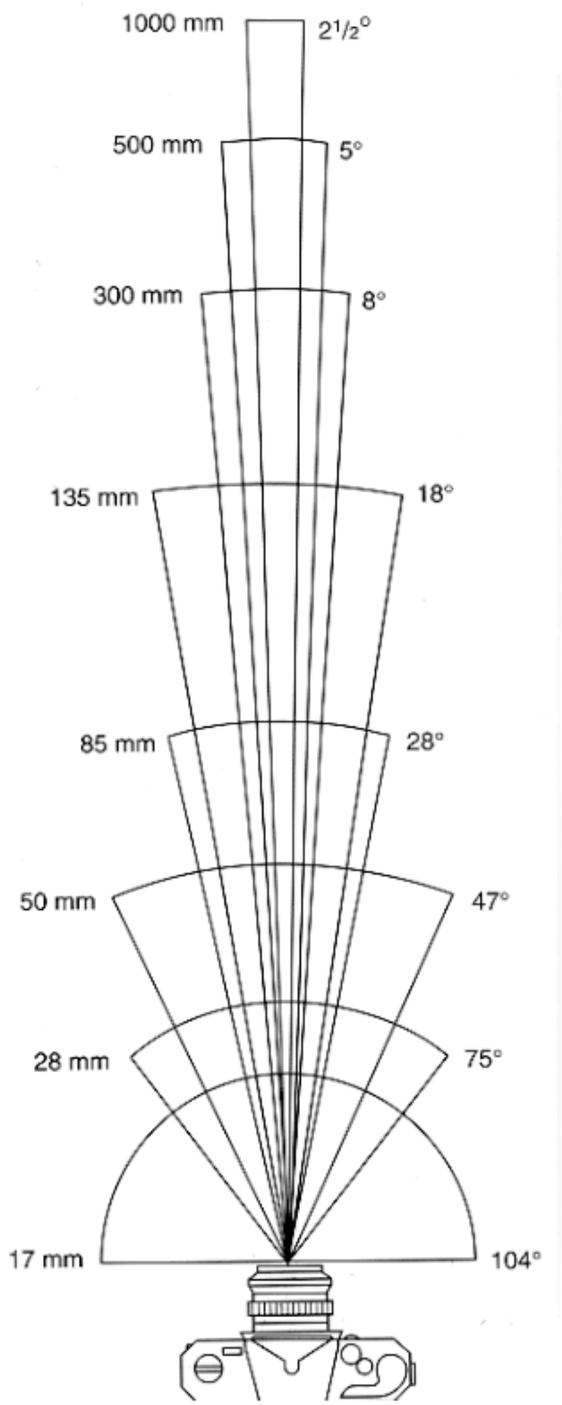
Lens

Small Sensor

Shorter focal length

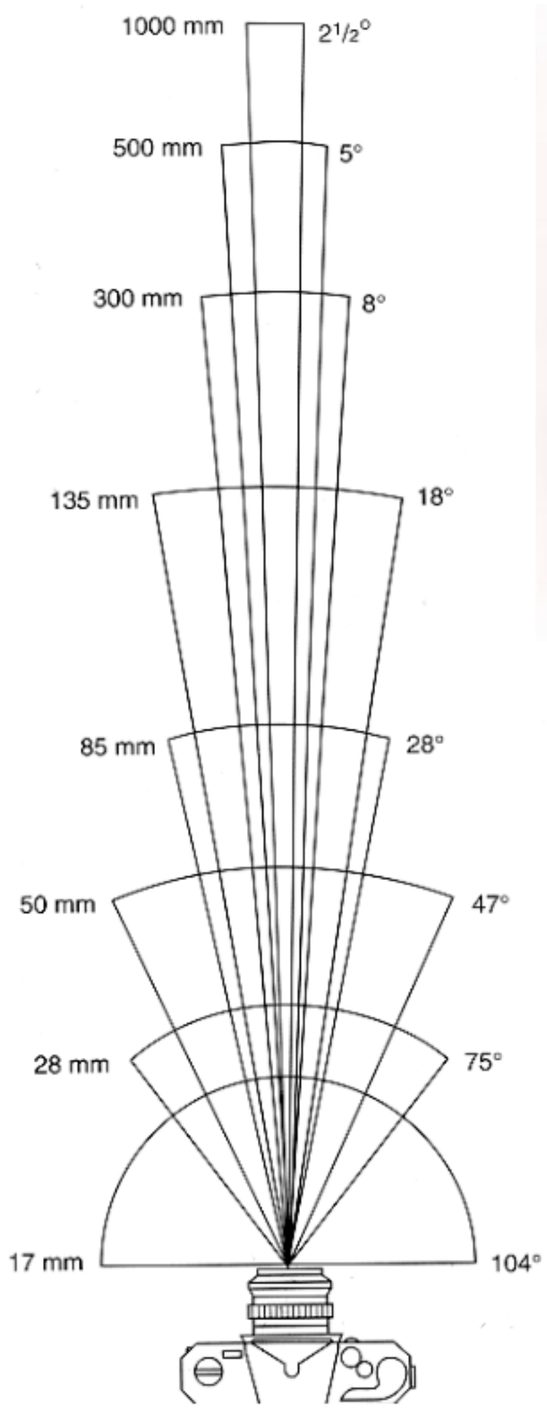
To maintain FOV, decrease focal length of lens in proportion to width/height of sensor

Focal Length v. Field of View



- For historical reasons, it is common to refer to angular field of view by focal length of a lens used on a 35mm-format film (36 x 24mm)
- Examples of focal lengths on 35mm format:
  - 17mm is wide angle 104°
  - 50mm is a “normal” lens 47°
  - 200mm is telephoto lens 12°
- Careful! When we say current cell phones have approximately 28mm “equivalent” focal length, this uses the above convention. The physical focal length is often 5-6 times shorter, because the sensor is correspondingly smaller

Focal Length v. Field of View







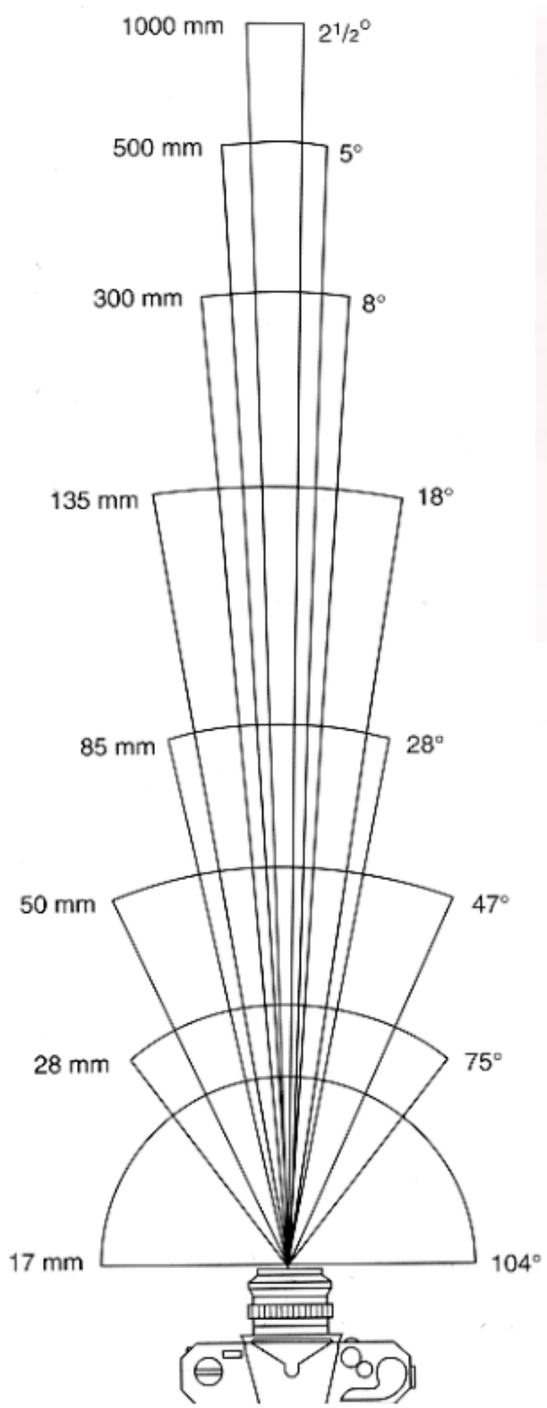
From London and Upton, and Canon EF Lens Work III

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Focal Length v. Field of View





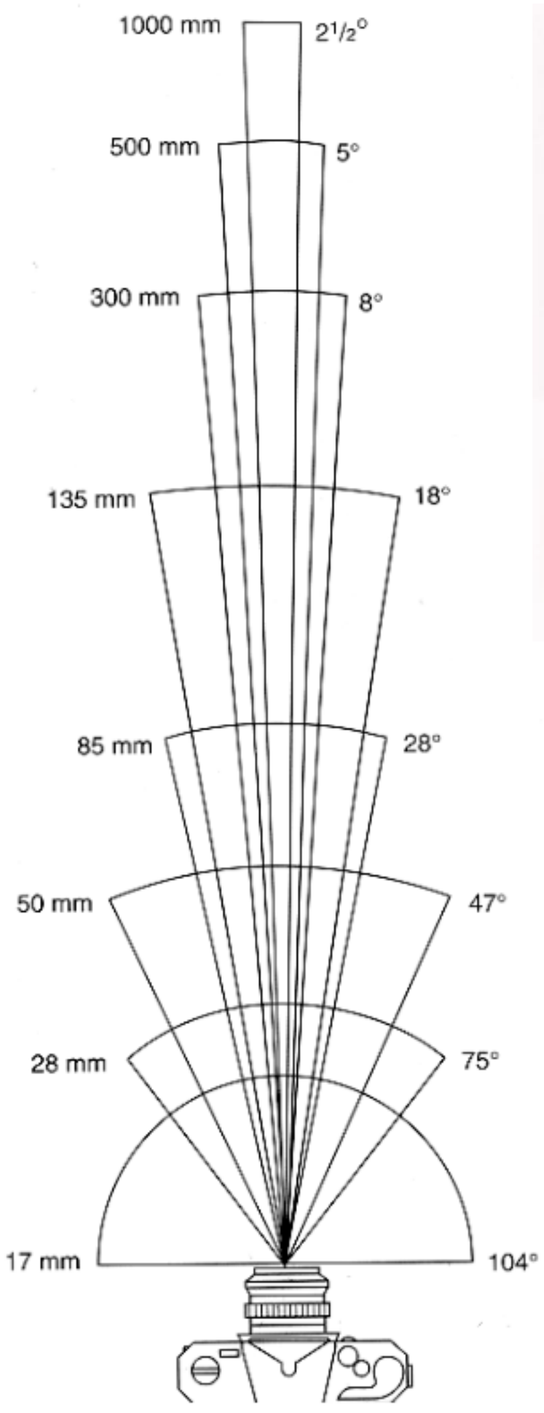
From London and Upton, and Canon EF Lens Work III

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Focal Length v. Field of View







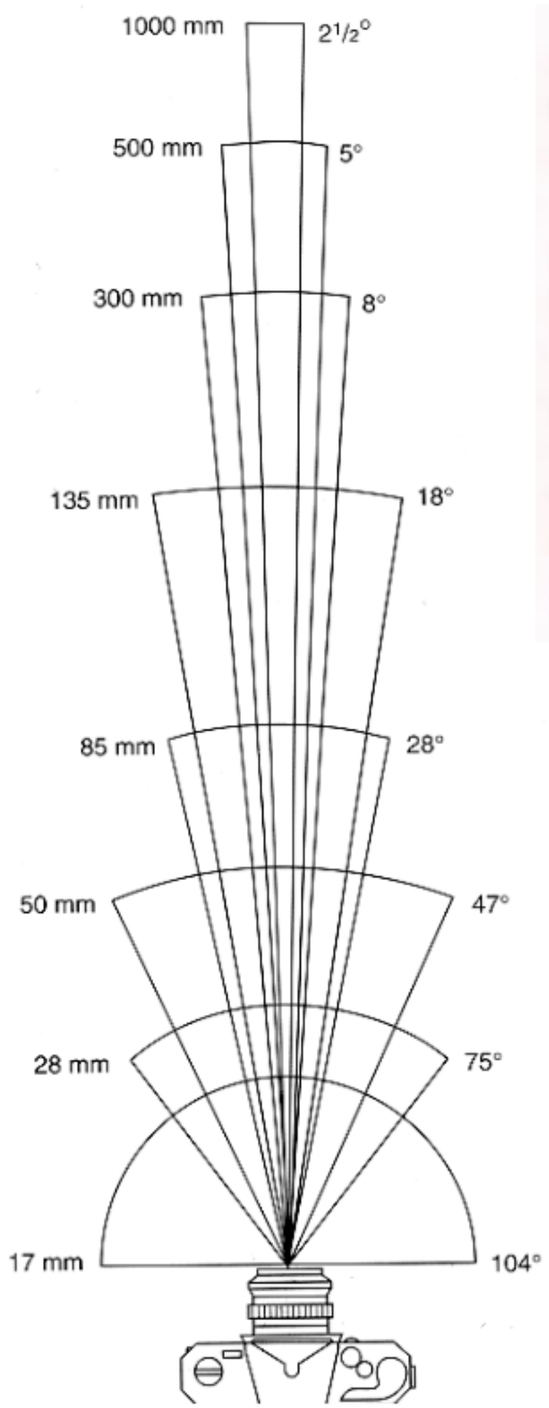
From London and Upton, and Canon EF Lens Work III

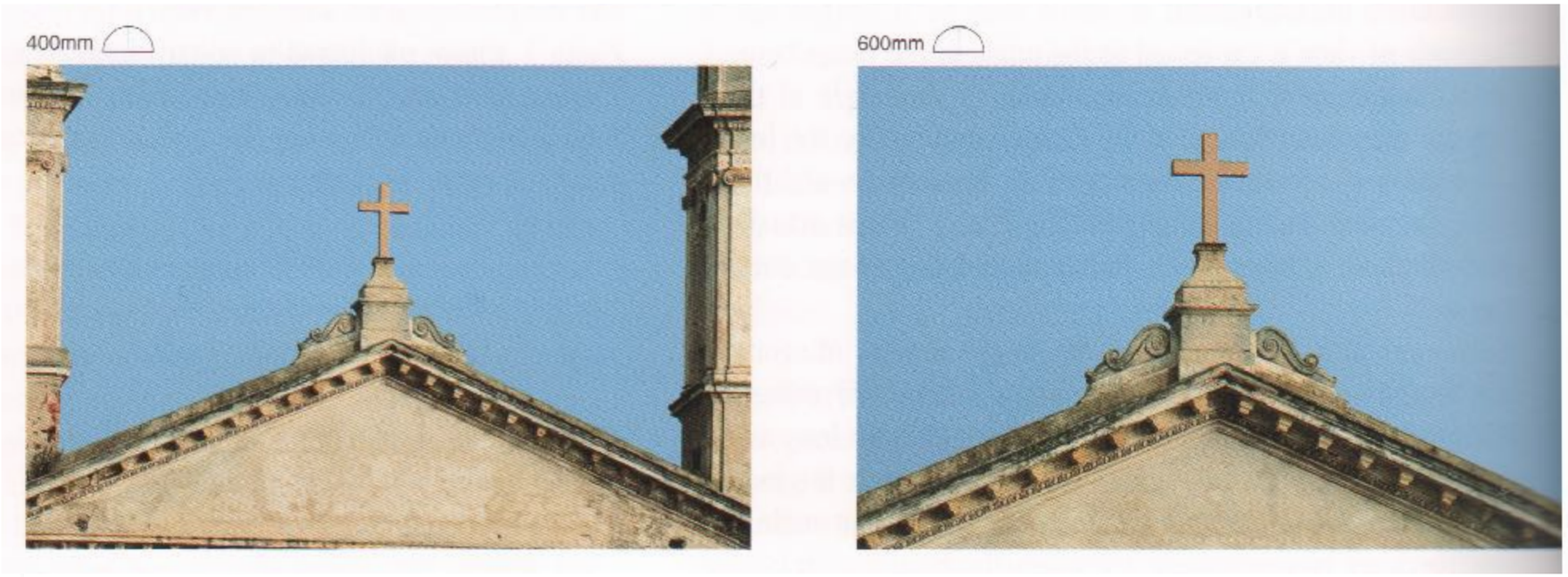
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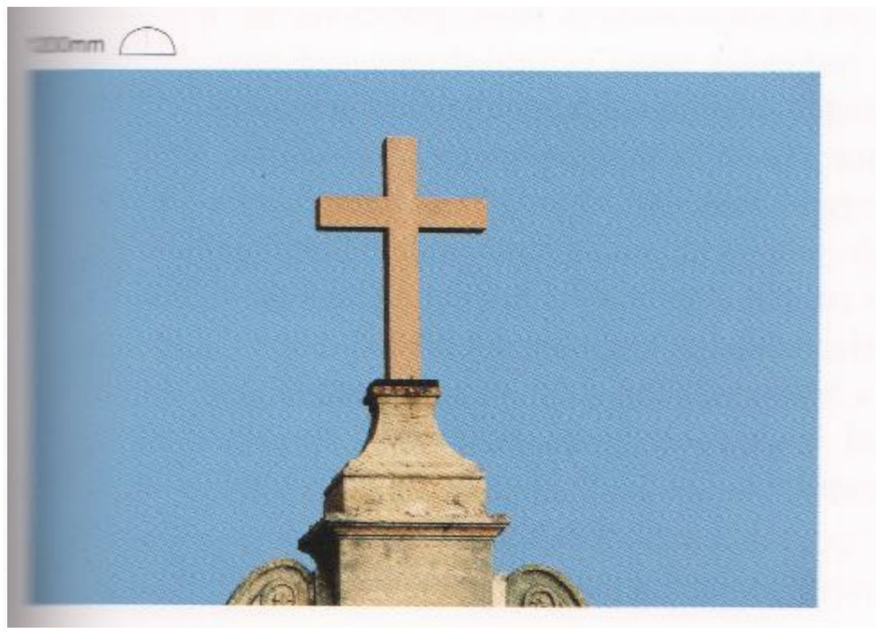
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Focal Length v. Field of View







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Wide angle: 15mm, f/2.8



Wide angle: 18mm, 1/750, f/8



Normal: 50mm, 1/80, f/1.4



Normal: 64mm, 1/3200, f/2.8

Telephoto: 150mm, 1/640, f/1.8



Telephoto: 200mm, 1/200, f/2.8



Telephoto: 420mm, 1/1600, f/4



Telephoto: 420mm, 1.0s, f/4

Perspective Composition (Photographer's Mindset)

Perspective Composition – Camera Position / Focal Length



16 mm

In this sequence, distance from subject increases with focal length to maintain image size of human subject.

Notice the dramatic change in background perspective.

From Canon EF Lens Work III

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Perspective Composition – Camera Position / Focal Length



24 mm

In this sequence, distance from subject increases with focal length to maintain image size of human subject.

Notice the dramatic change in background perspective.

From Canon EF Lens Work III

Perspective Composition – Camera Position / Focal Length



50 mm

In this sequence, distance from subject increases with focal length to maintain image size of human subject.

Notice the dramatic change in background perspective.

From Canon EF Lens Work III

Perspective Composition – Camera Position / Focal Length



135 mm

From Canon EF Lens Work III

In this sequence, distance from subject increases with focal length to maintain image size of human subject.

Notice the dramatic change in background perspective.

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Perspective Composition – Camera Position / Focal Length



200 mm

In this sequence, distance from subject increases with focal length to maintain image size of human subject.

Notice the dramatic change in background perspective.

From Canon EF Lens Work III

Perspective Composition – Camera Position / Focal Length

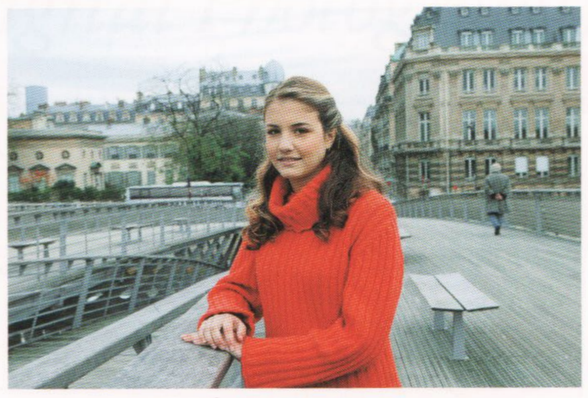
16mm



24mm



50mm



200mm



135mm



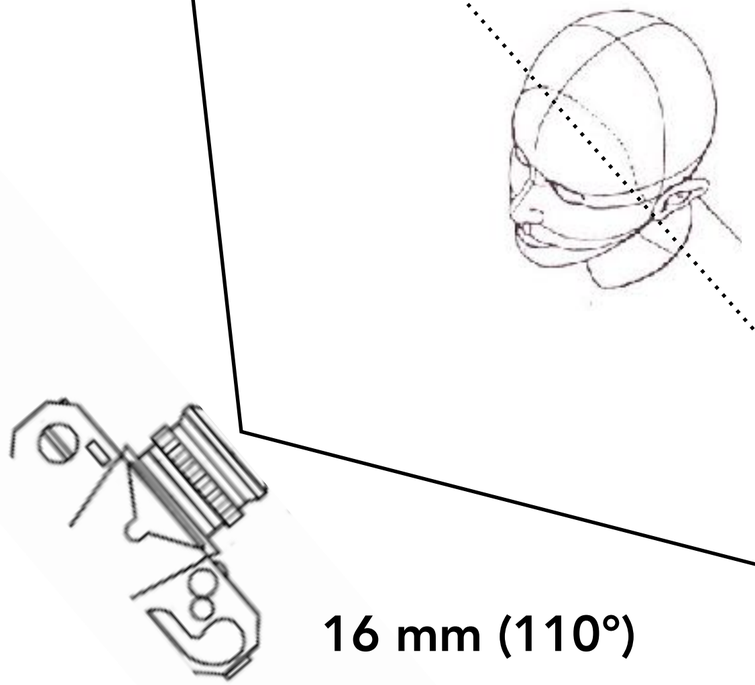
From Canon EF Lens Work III

In this sequence, distance from subject increases with focal length to maintain image size of human subject.

Notice the dramatic change in background perspective.

Perspective Composition





16 mm (110°)

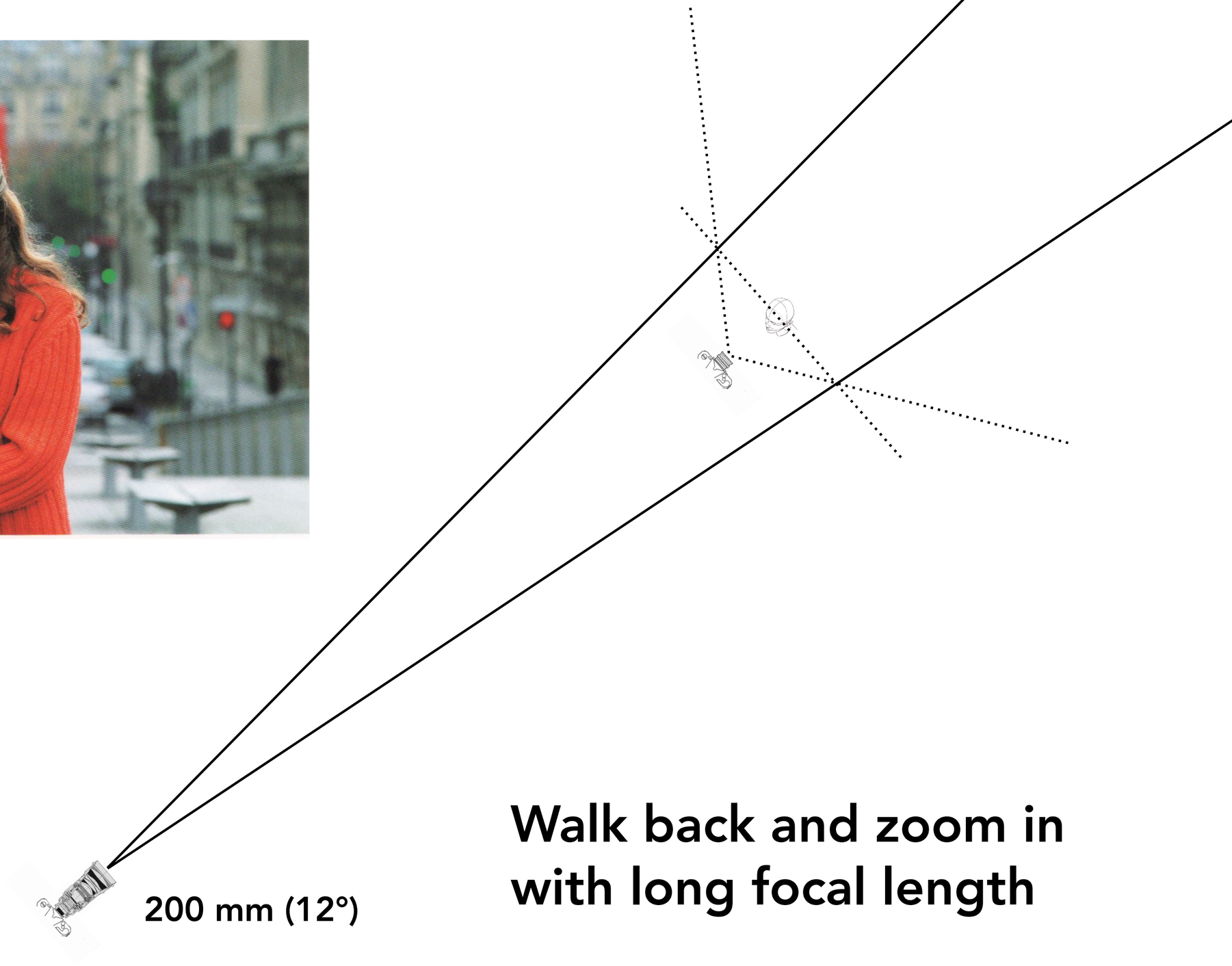
Up close and zoomed wide with short focal length

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Perspective Composition





Walk back and zoom in with long focal length

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Improve Your Own Photography

Tip 1: Make sure you have a strong subject

- Make it prominent, e.g. 1/3 of your image

Tip 2: Choose a good perspective relationship (relative size) between your subject and background (or foreground)

- Complement, don’t compete with the subject

Tip 3: Change the zoom and camera distance to your subject

- Implement: actively zoom, and move your camera in/out
- Even works with your smartphone!

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Dolly-Zoom Cinema Technique – a.k.a. “Vertigo Effect”



By Steven Spielberg in “Jaws” 1975

A Photographer's Mindset





"Choose your perspective before you choose your lens."

— Ming Thein, mingthein.com

Exposure

Exposure Levels - “Stops” are Logarithmic



- Here, different exposure levels with +/– 1 “stop” of exposure
- In photography, a “stop” = a doubling of exposure
- The natural, perceptual scale of exposure is logarithmic

https://www.dpmag.com/how-to/tip-of-the-week/how-and-why-to-use-auto-exposure-bracketing/

Exposure

- Exposure = irradiance x time x gain
- Irradiance
  - Power of light falling on image sensor pixel
  - Affected by scene brightness, pixel size, lens aperture…
- Exposure time
  - Duration that the image sensor exposed to light
  - Affected by shutter opening / closing
- Gain
  - Amplification of sensor pixel values
  - Affected by pixel-value amplifiers in image sensor

Exposure Controls: Aperture, Shutter, Gain (ISO)

Aperture

- Change the lens f-stop by opening / closing its physical aperture (if lens has iris control)

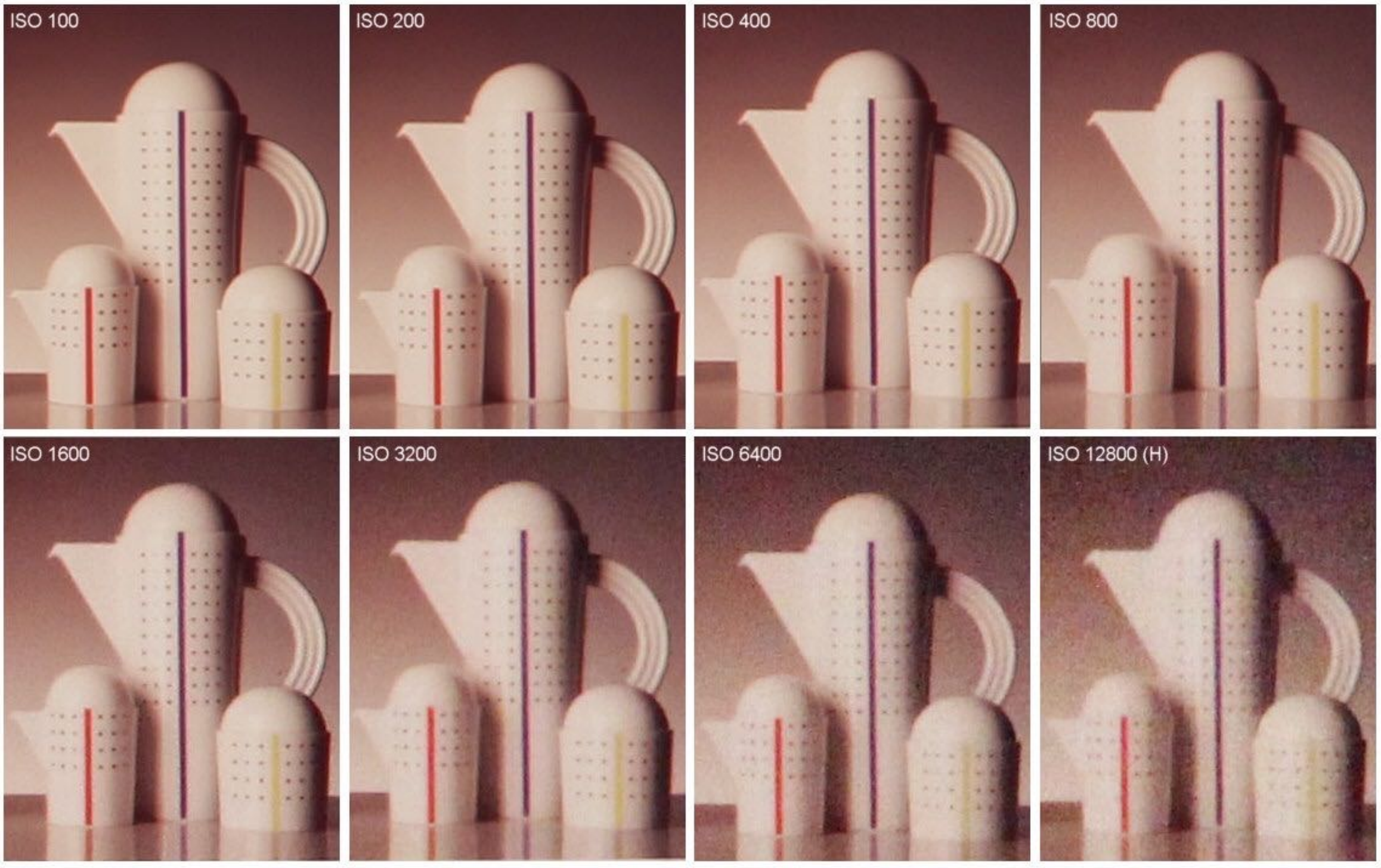
Shutter speed

- Change the duration that the sensor pixels are integrating light (physical or electronic shutter)

ISO gain

- Change the system amplification between sensor values and digital image values

Gain (ISO) — Noise Increases



Credit: bobatkins.com

ISO (Gain)

Image sensor: trade sensitivity for noise

- Multiply signal before analog-to-digital conversion
- Linear effect (ISO 200 needs half the light as ISO 100)
- Typically, set gain to lowest value that works for the scene light level, to minimize noise

Many Ways to Achieve the Same Exposure

Have multiple ways to adjust aperture, shutter, gain to achieve a desired exposure

Example: all the following pairs of aperture and shutter give equivalent exposure (not same image, though!)

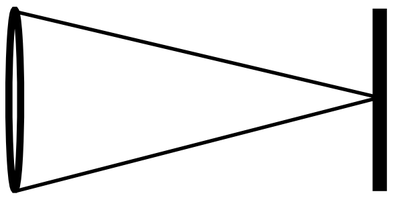
F-Stop	1.4	2.0	2.8	4.0	5.6	8.0	11.0	16.0	22.0	32.0
Shutter	1/500	1/250	1/125	1/60	1/30	1/15	1/8	1/4	1/2	1

If the exposure is too bright/dark, may need to adjust f-stop and/or shutter up/down.

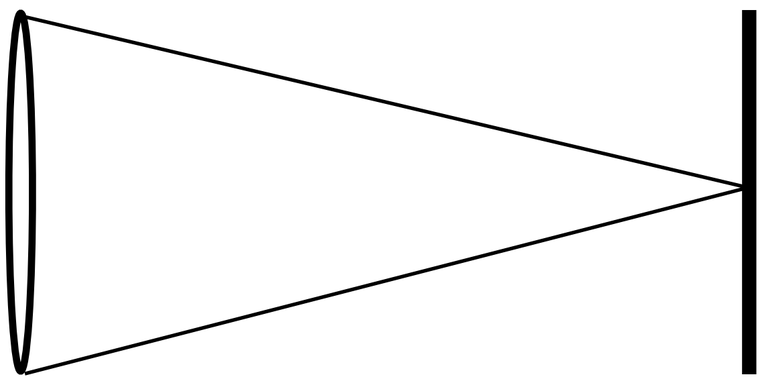
Definition: F-Number of a Lens

- The F-Number of a lens is defined as the focal length divided by the diameter of the aperture
- Common F-stops on real lenses: 1.4, 2, 2.8, 4.0, 5.6, 8, 11, 16, 22, 32
- 1 stop doubles exposure
- Notation: an f-stop of, e.g. 2 is sometimes written f/2, or F:2 or F2

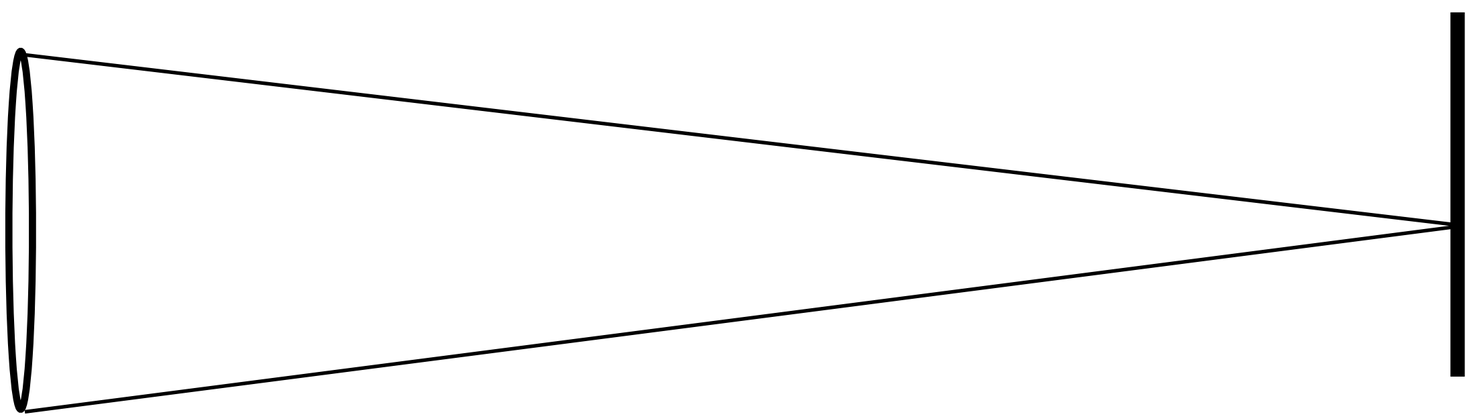
Example F-Number Calculations



$$D = 50\,\text{mm}$$
$$f = 100\,\text{mm}$$
$$N = f/D = 2$$



$$D = 100\,\text{mm}$$
$$f = 200\,\text{mm}$$
$$N = f/D = 2$$



$$D = 100\,\text{mm}$$
$$f = 400\,\text{mm}$$
$$N = f/D = 4$$

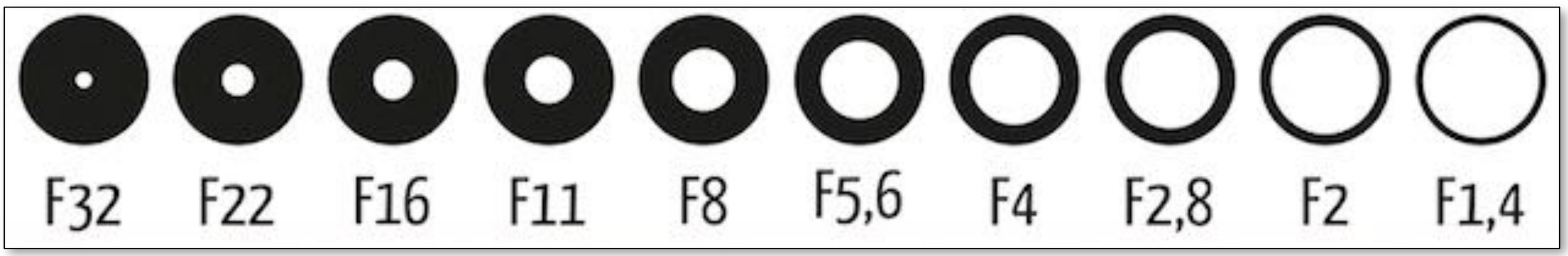
Lens's F-Number vs F-Number for Photo

A lens's F-Number is the maximum for that lens

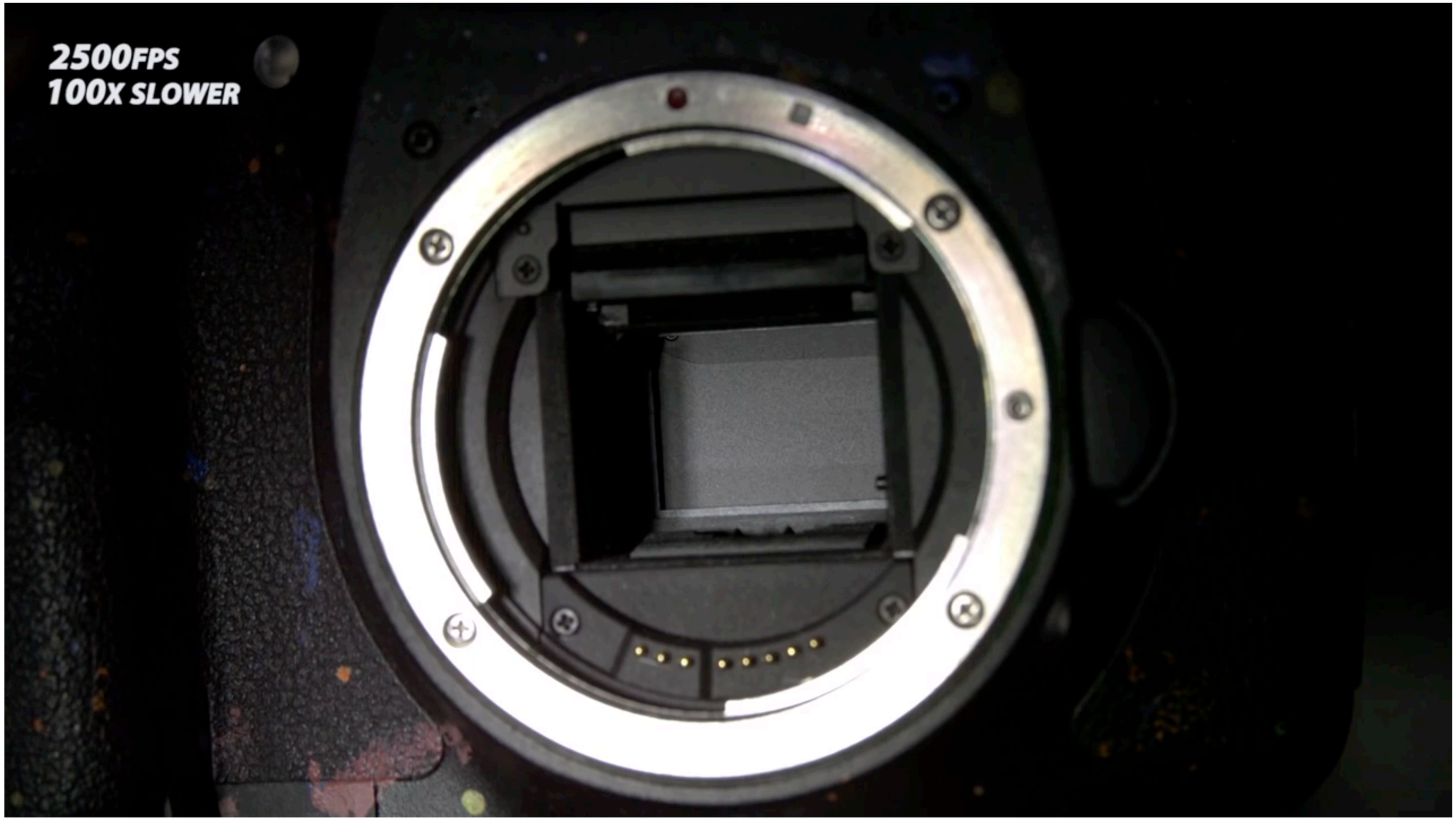
- E.g. 50 mm F/1.4 is a high-quality telephoto lens
  - Maximum aperture is 50/1.4 = 36 mm diameter

But for an individual photo, the lens aperture may be "stopped down" to a smaller size

- E.g. 50 mm F/1.4 lens stopped down to F/4
  - Aperture is closed down with an iris to 50/4 = 12.5 mm



Physical Shutter (1/25 Sec Exposure)



The Slow Mo Guys, https://youtu.be/CmjeCchGRQo

Main Side Effect of Shutter Speed

Motion blur: handshake, subject movement

Doubling shutter time doubles motion blur

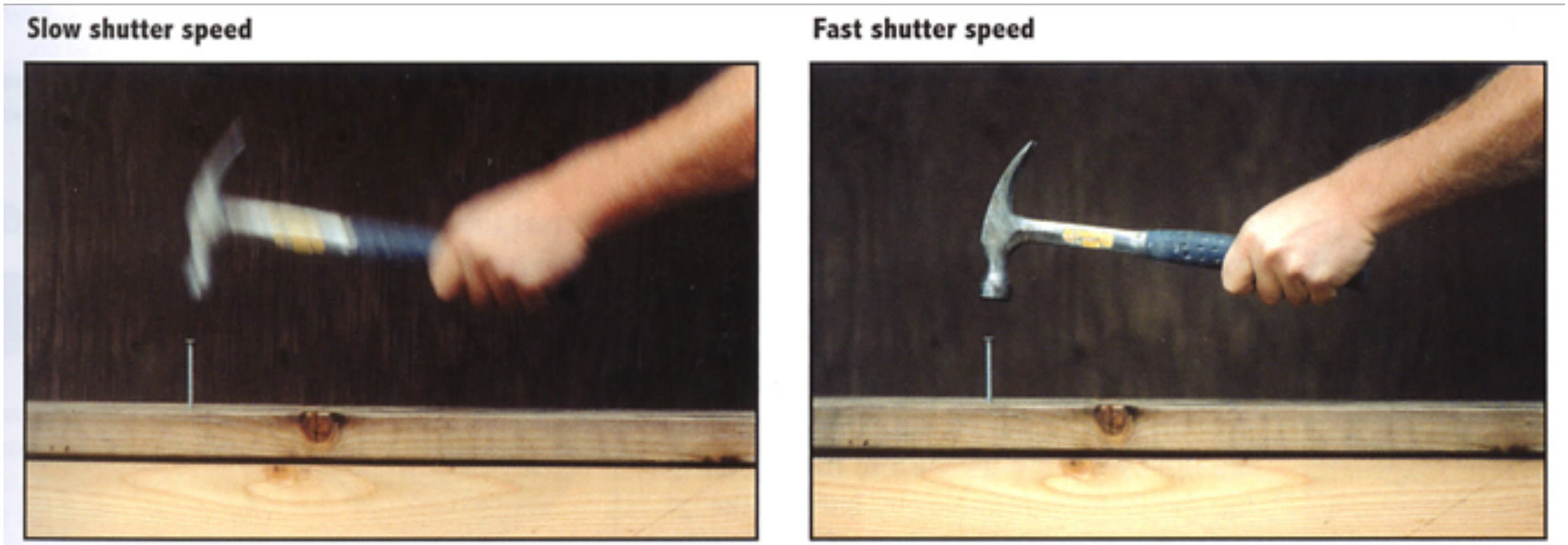


Gavin Hoey http://www.gavtrain.com/?p=3960

Main Side Effect of Shutter Speed

Motion blur: handshake, subject movement

Doubling shutter time doubles motion blur



London

Electronic Shutter

- Pixel is electronically reset to start exposure
- Fills with photoelectrons as light falls on sensor
- Reading out pixel electronically "ends" exposure
- Problem: most sensors read out pixels sequentially, takes time (e.g. 1/30 sec) to read entire sensor
  - If reset all pixels at the same time, last pixel read out will have longer exposure
  - So, usually stagger reset of pixels to ensure uniform exposure time
  - Problem: rolling shutter artifact

Electronic Rolling Shutter

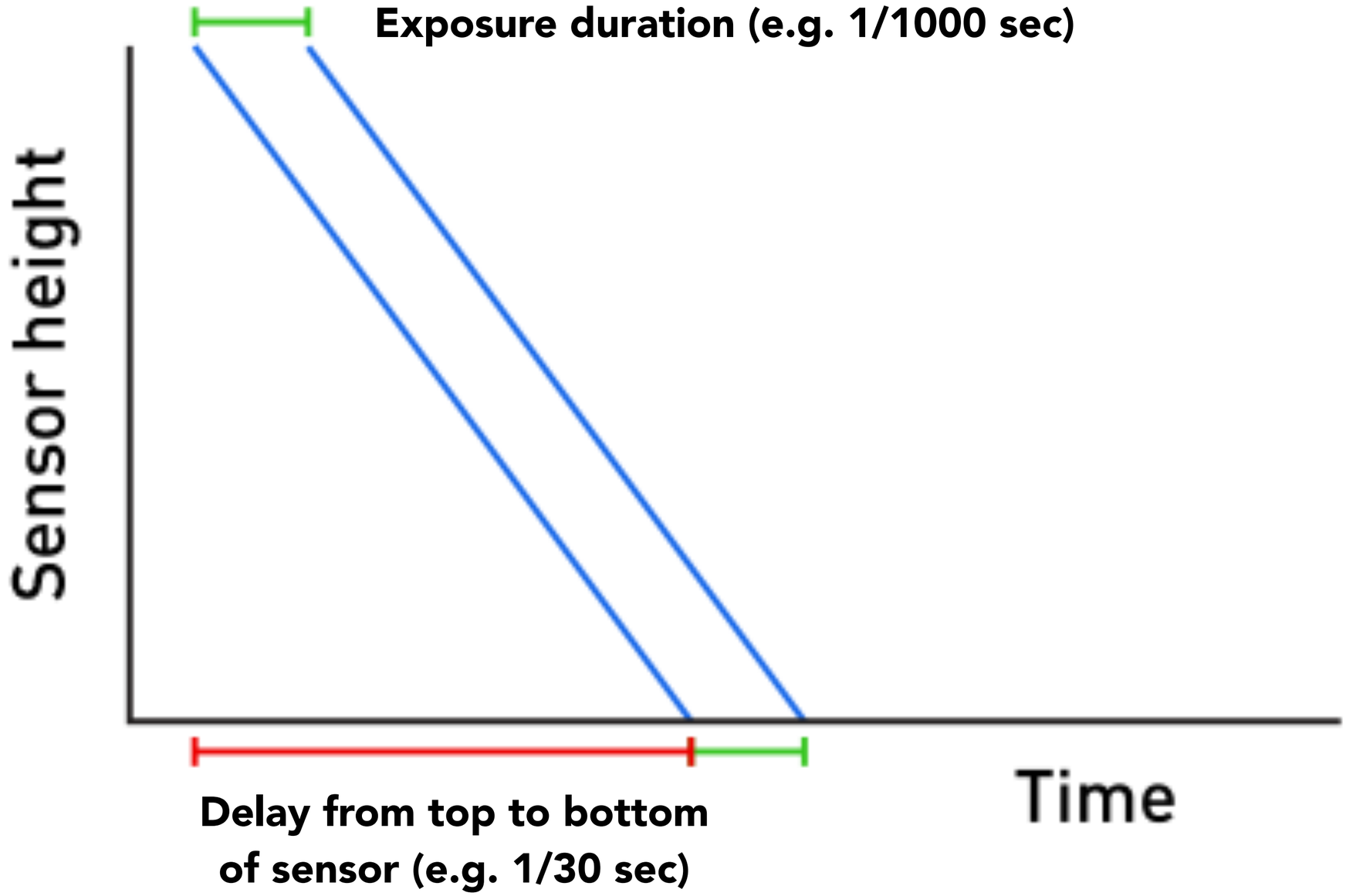


The Slow Mo Guys, https://youtu.be/CmjeCchGRQo

Ren Ng

Presented by David McAllister

Electronic Rolling Shutter

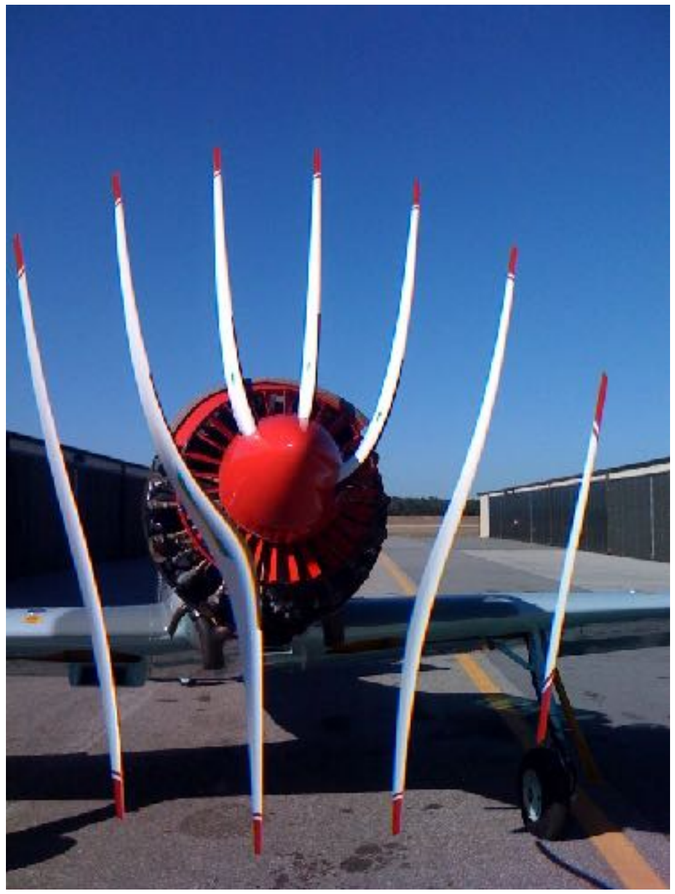


dpreview.com

Electronic Rolling Shutter

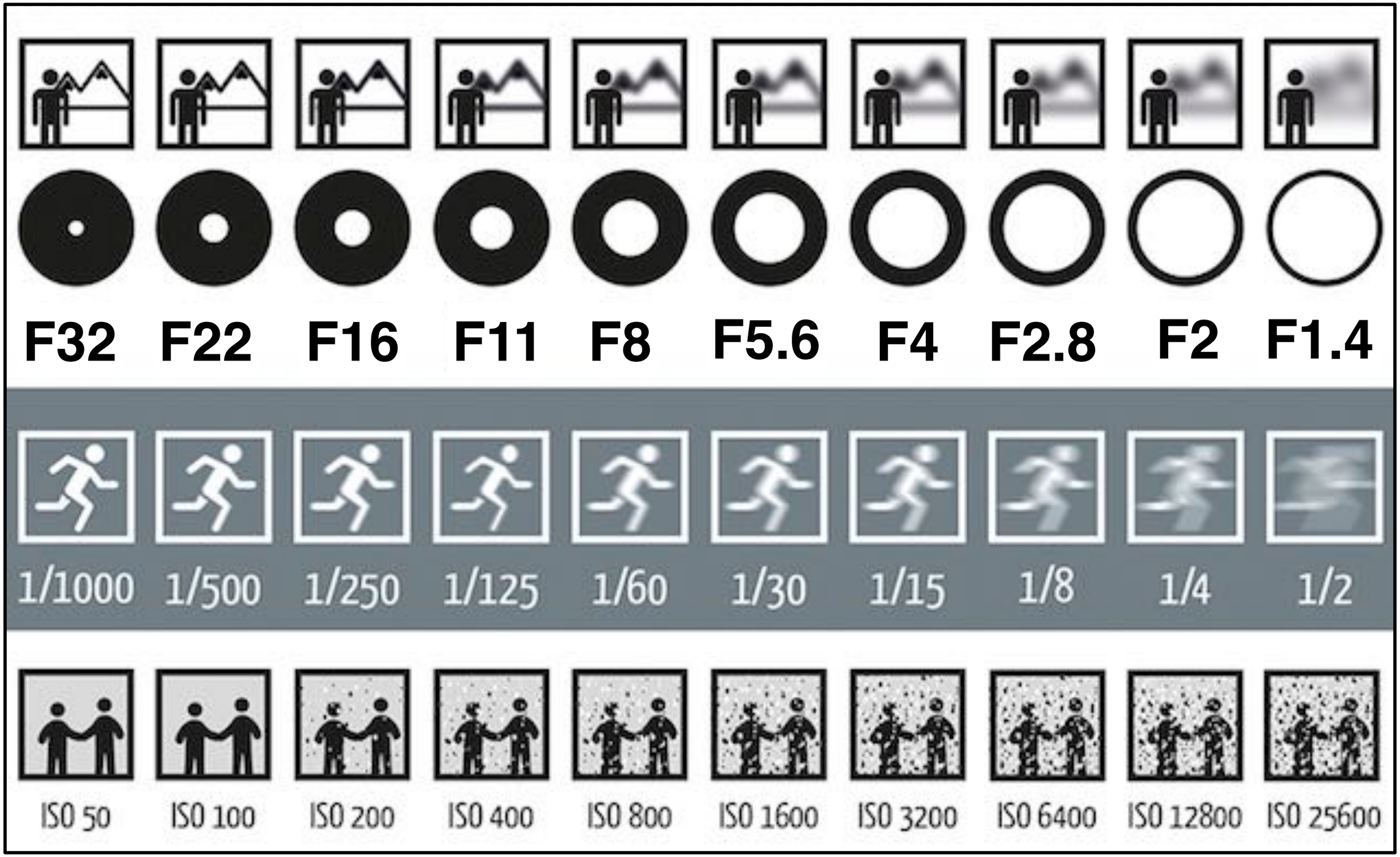


Credit: David Adler, B&H Photo Video
https://www.bhphotovideo.com/explora/video/tips-and-solutions/rolling-shutter-versus-global-shutter



Credit: Soren Ragsdale
https://flic.kr/p/5S6rKw

Exposure Controls: Aperture, Shutter, Gain (ISO)



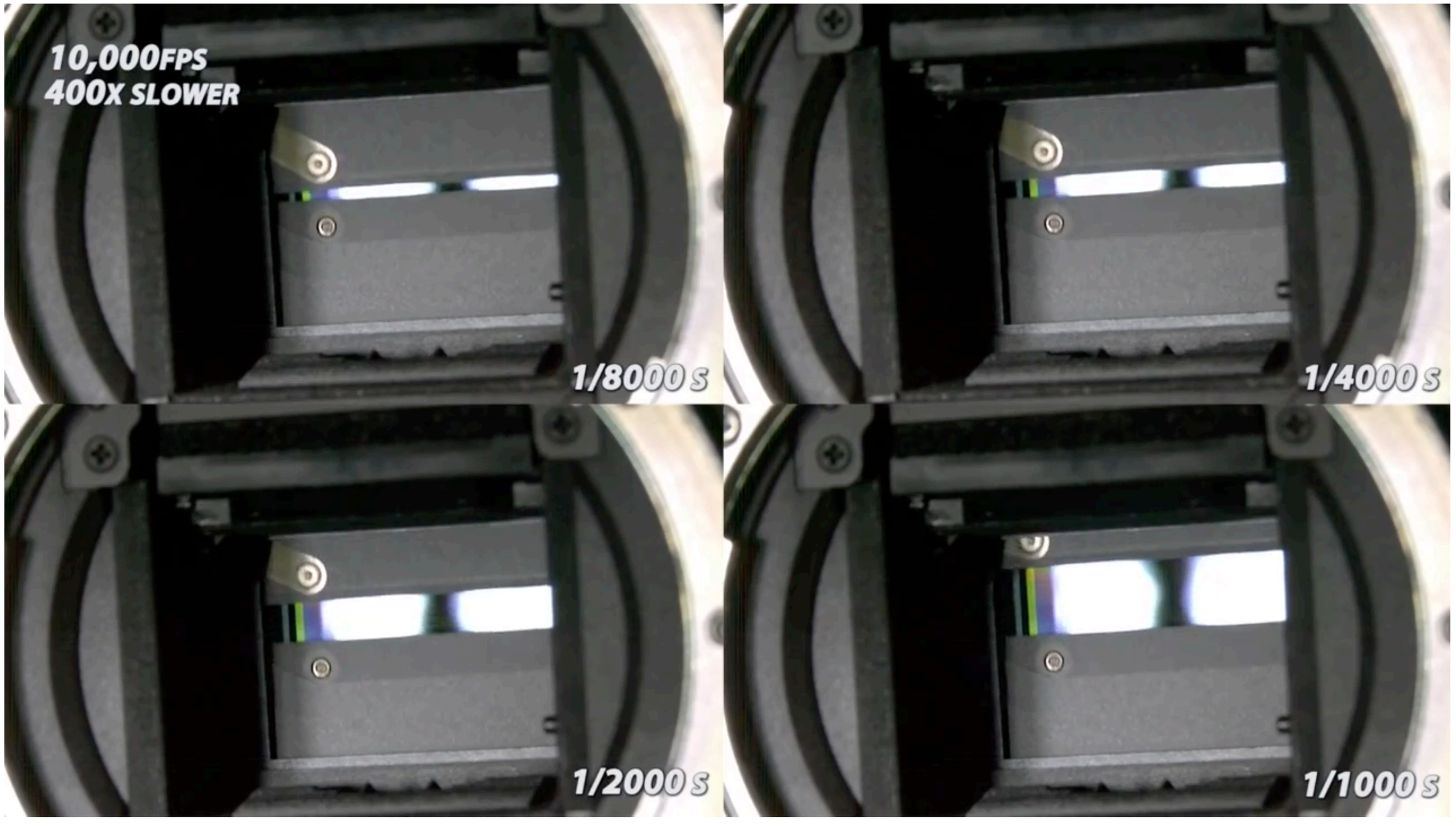
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Photoblog Hamburg

Physical Shutter (Fast Exposures)



The Slow Mo Guys, https://youtu.be/CmjeCchGRQo

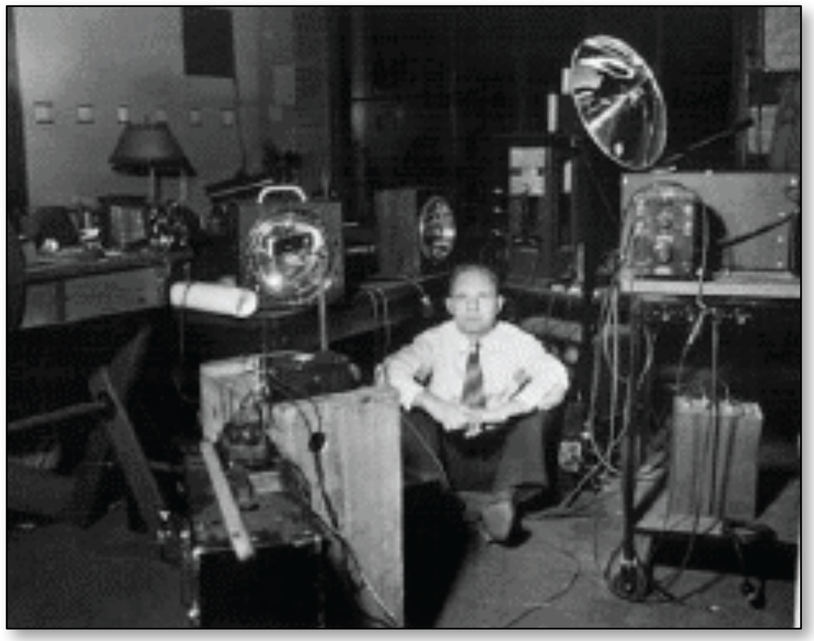
Exposure Duration:
Fast and Slow Photography

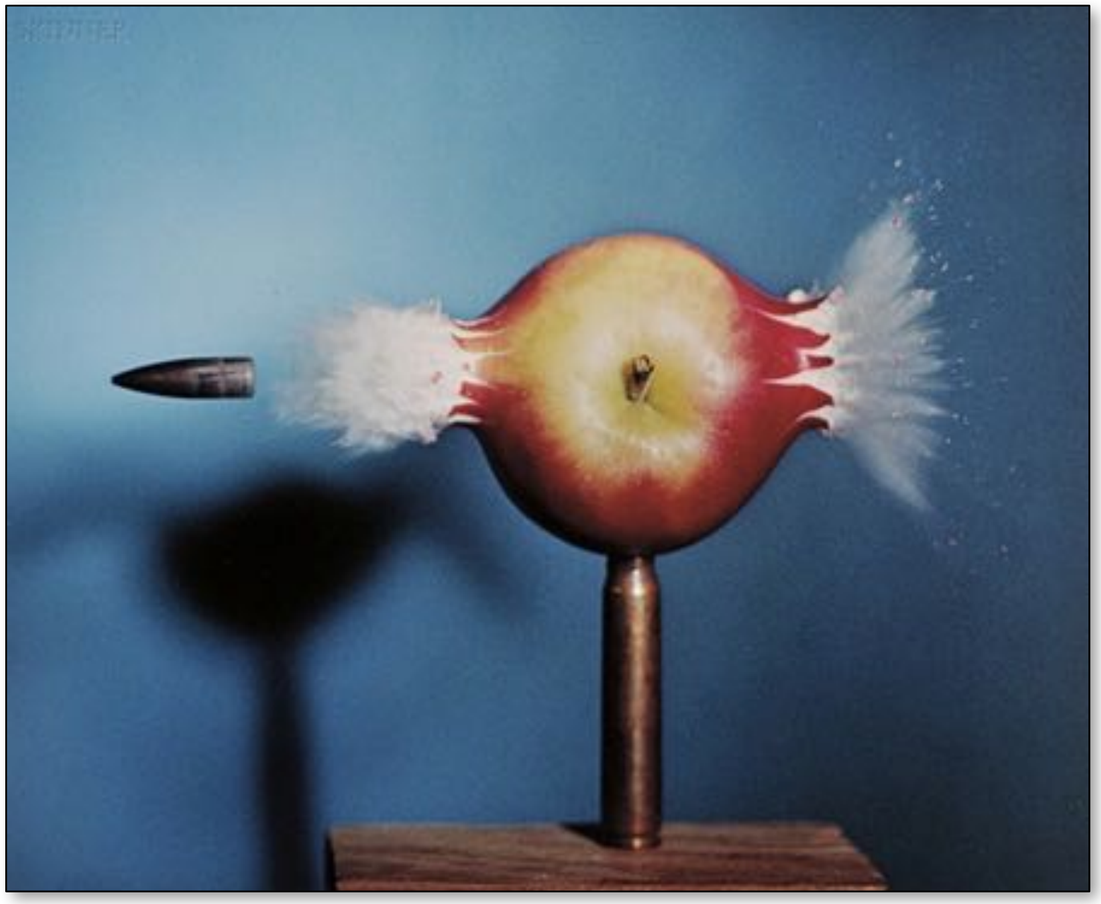
High-Speed Photography (Short Shutter)

long exposure

bright strobe illumination

gun synced to camera





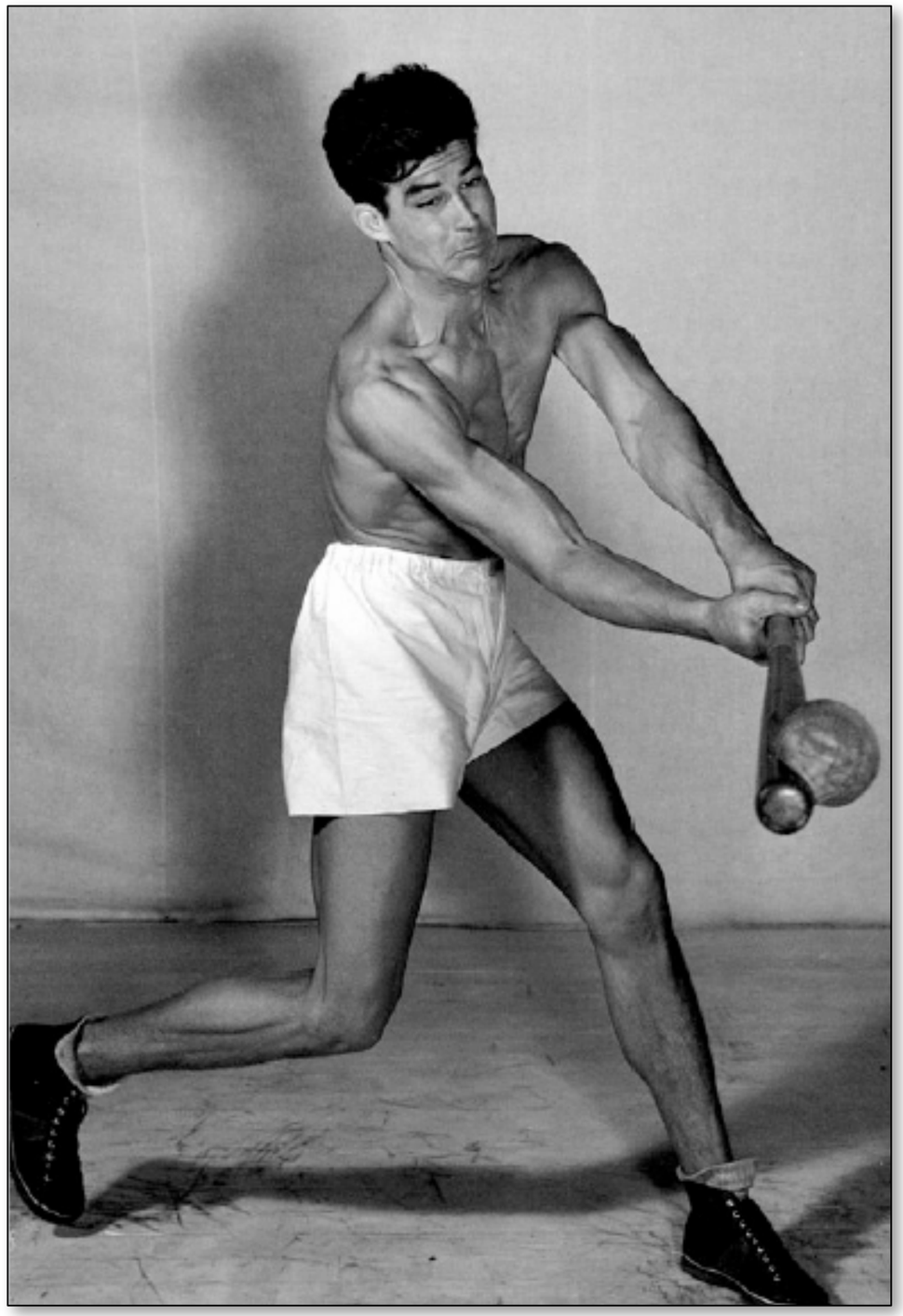
Harold Edgerton



Mark Watson

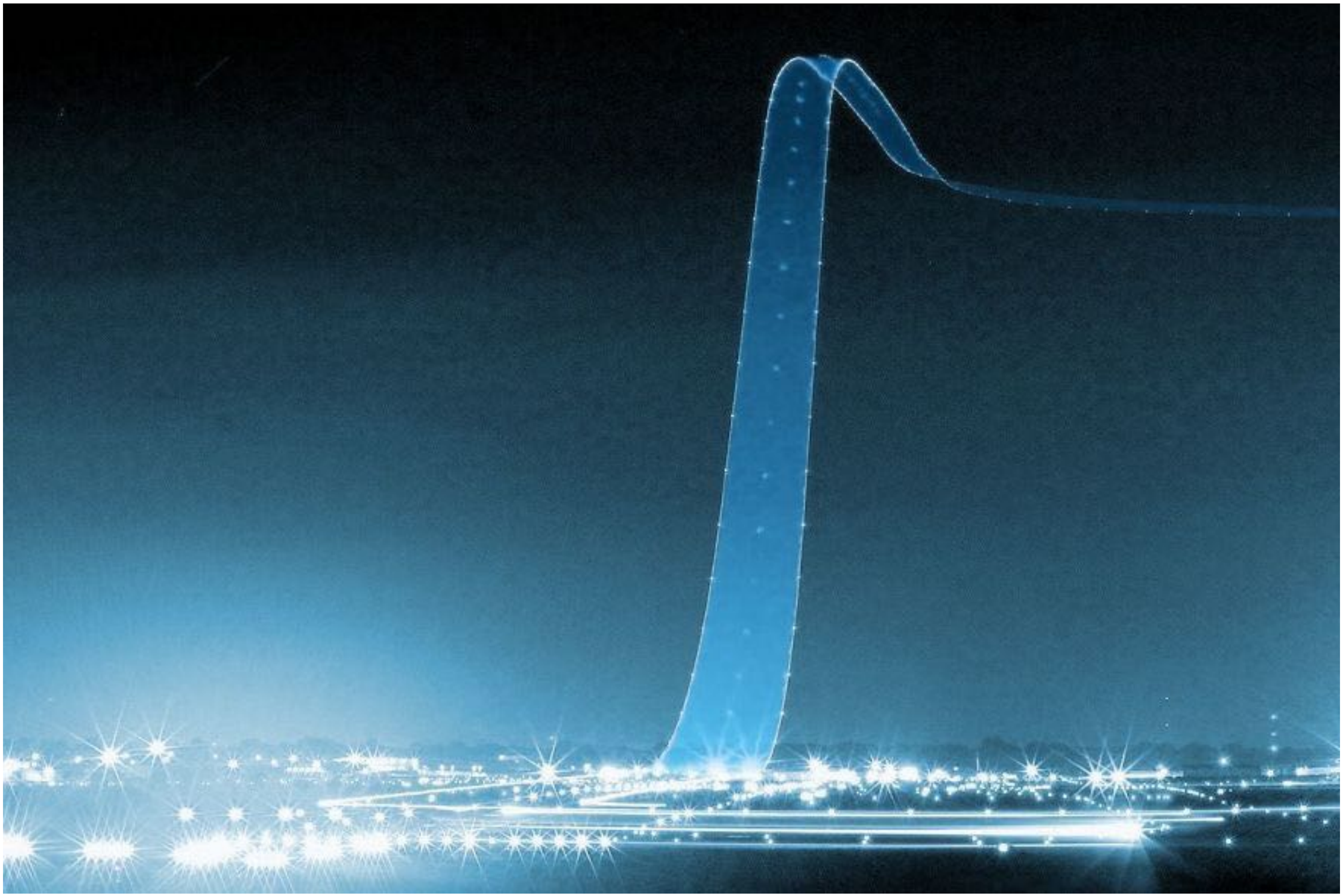
High-Speed Photography (Short Shutter)





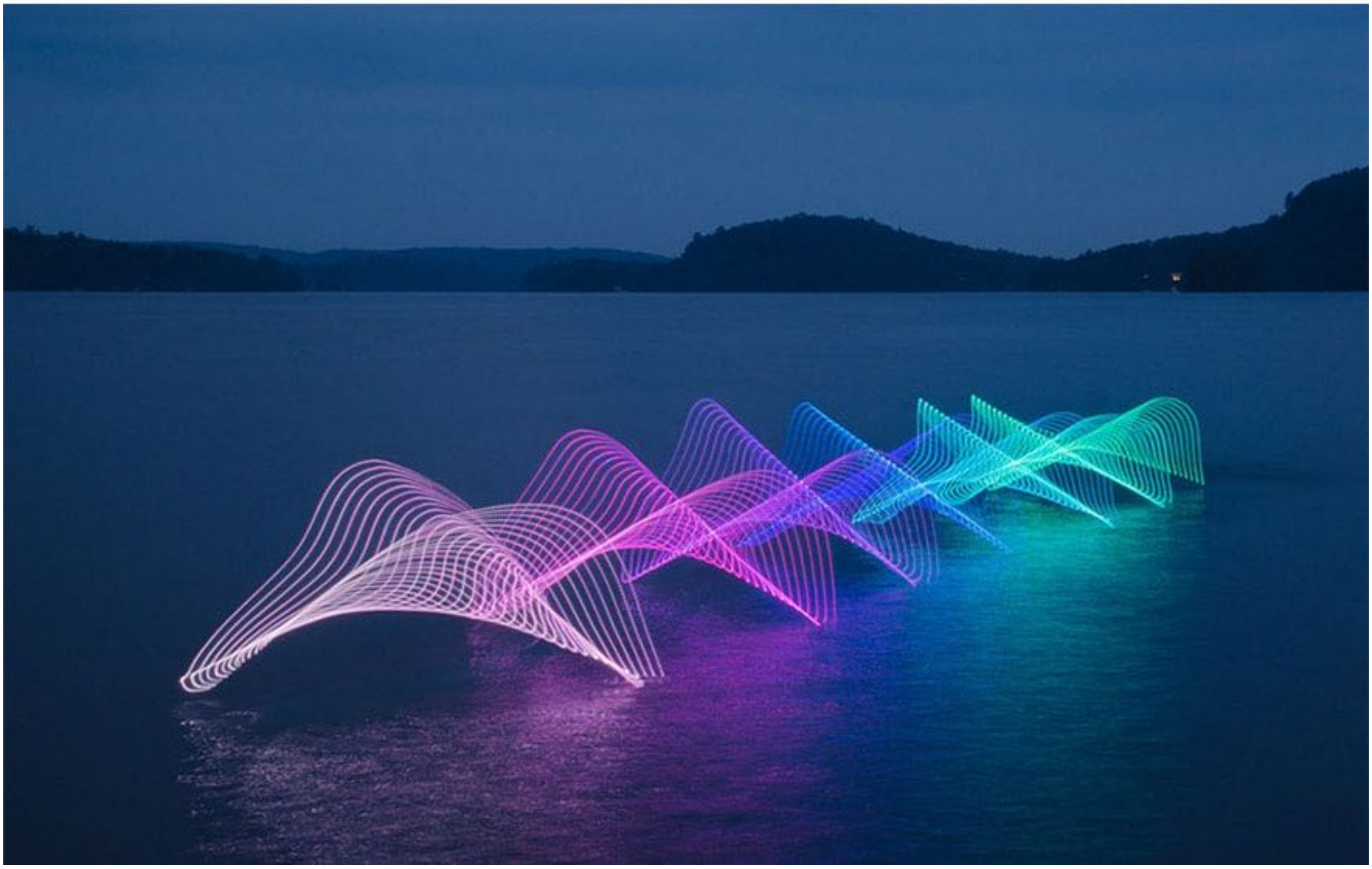
Harold Edgerton

Long-Exposure Photography



https://www.demilked.com/best-long-exposure-photos/

Long-Exposure Photography



https://www.demilked.com/best-long-exposure-photos/

Long-Exposure Photography



https://www.demilked.com/best-long-exposure-photos/

Optics of Lenses

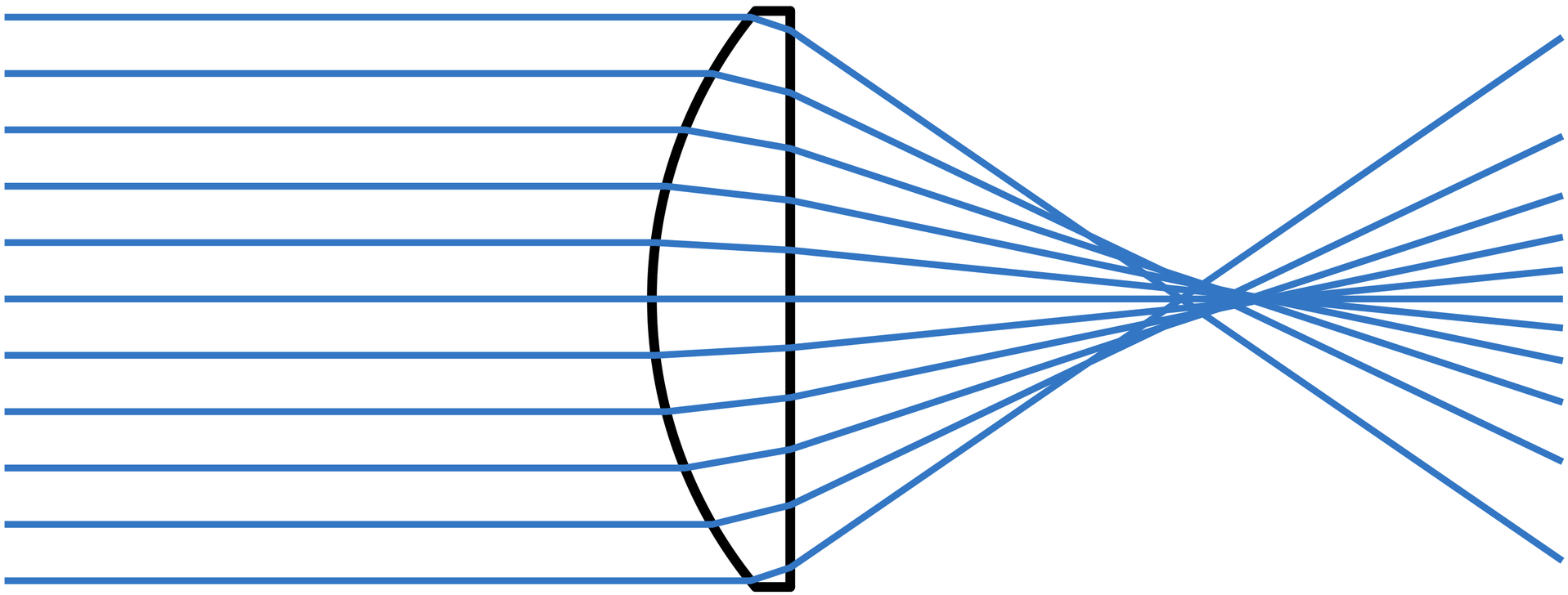
Real Lens Designs Are Highly Complex



[Apple]

Ren Ng
Presented by David McAllister

Real Lens Elements Are Not Ideal – Aberrations

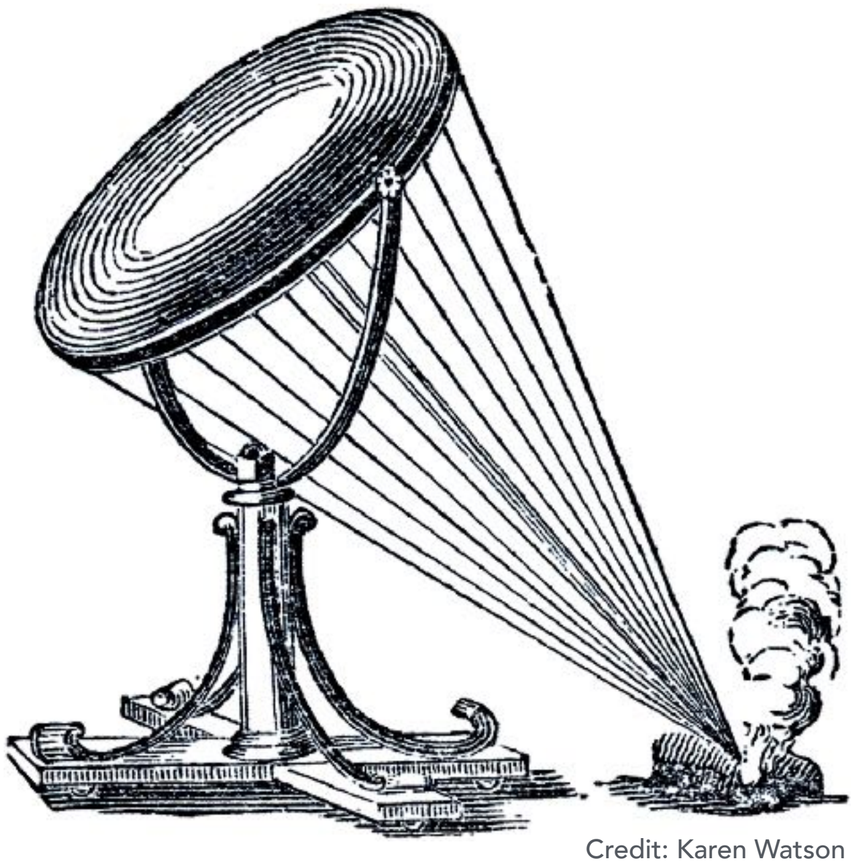


Real plano-convex lens (spherical surface shape).
Lens does not converge rays to a point anywhere.

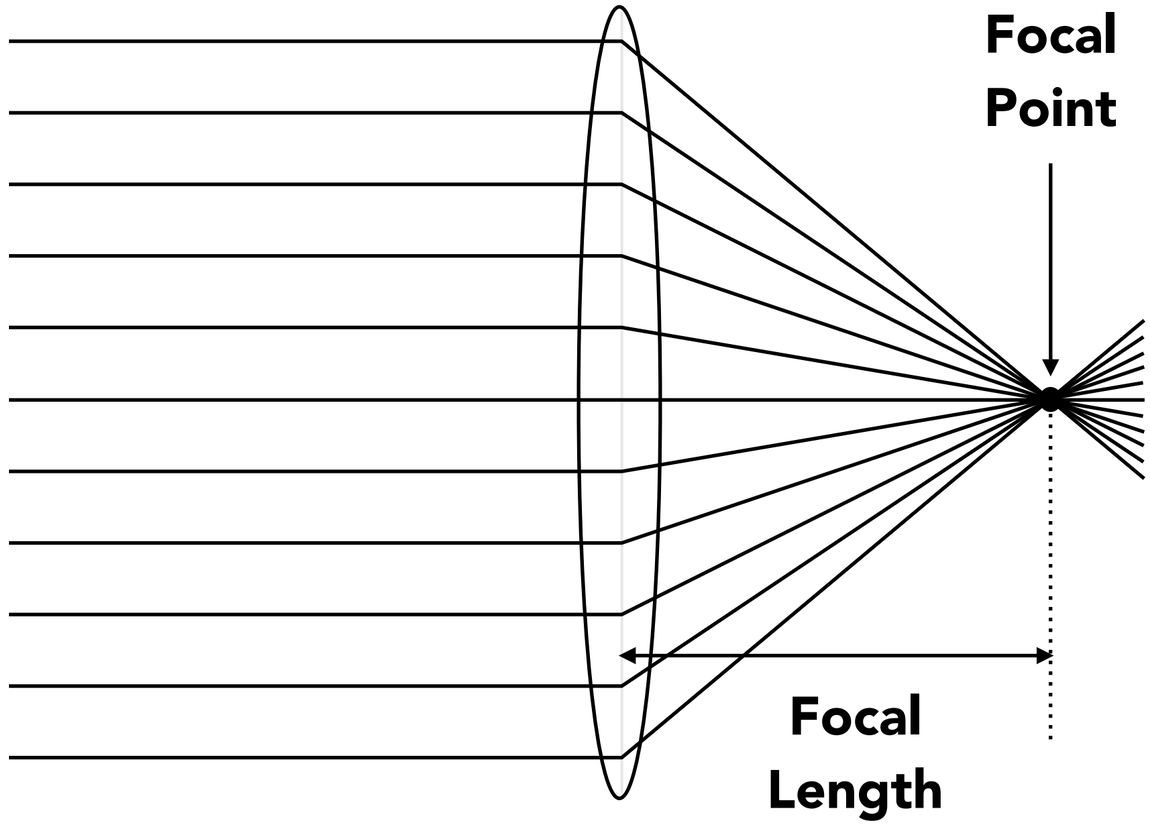
Ren Ng
Presented by David McAllister

First: Thin Lens Approximation

Ideal Thin Lens – Focal Point



Credit: Karen Watson



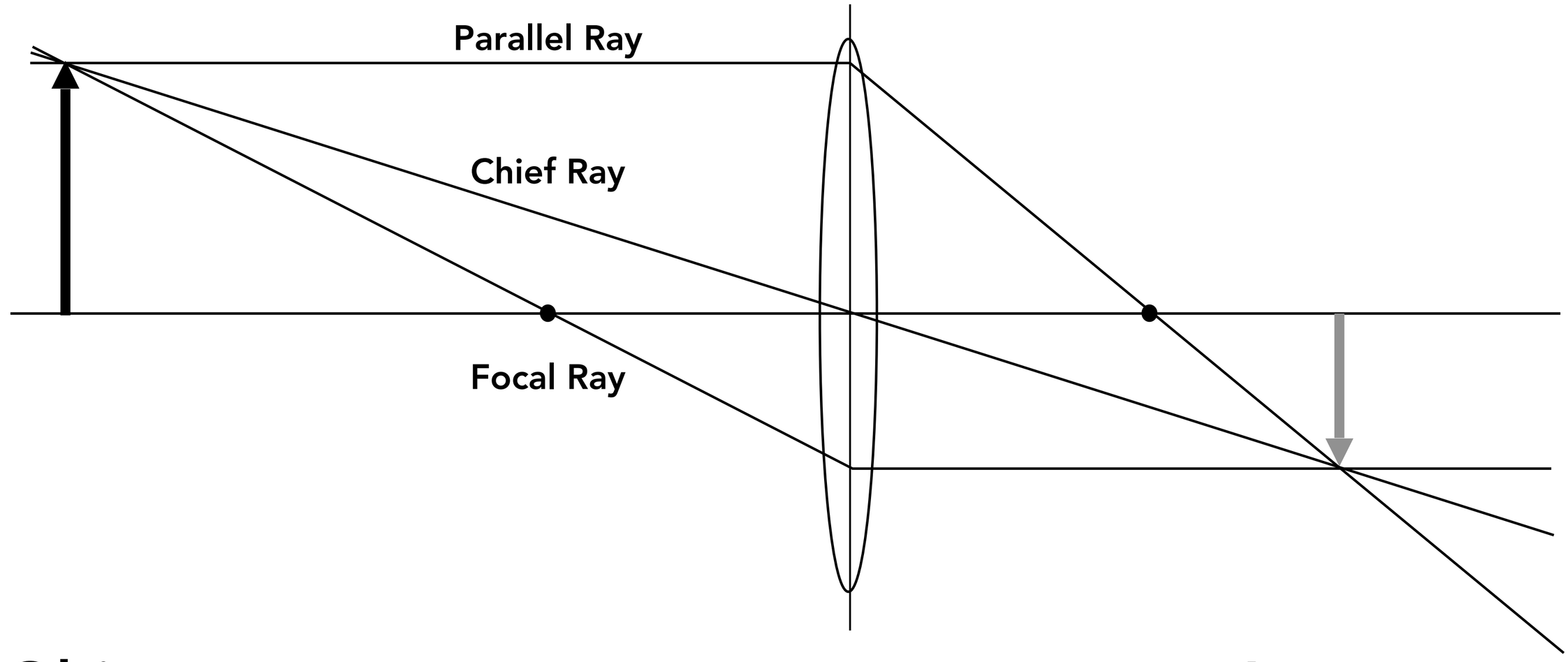
Assume all parallel rays entering a lens pass through its focal point.

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Gauss' Ray Diagrams

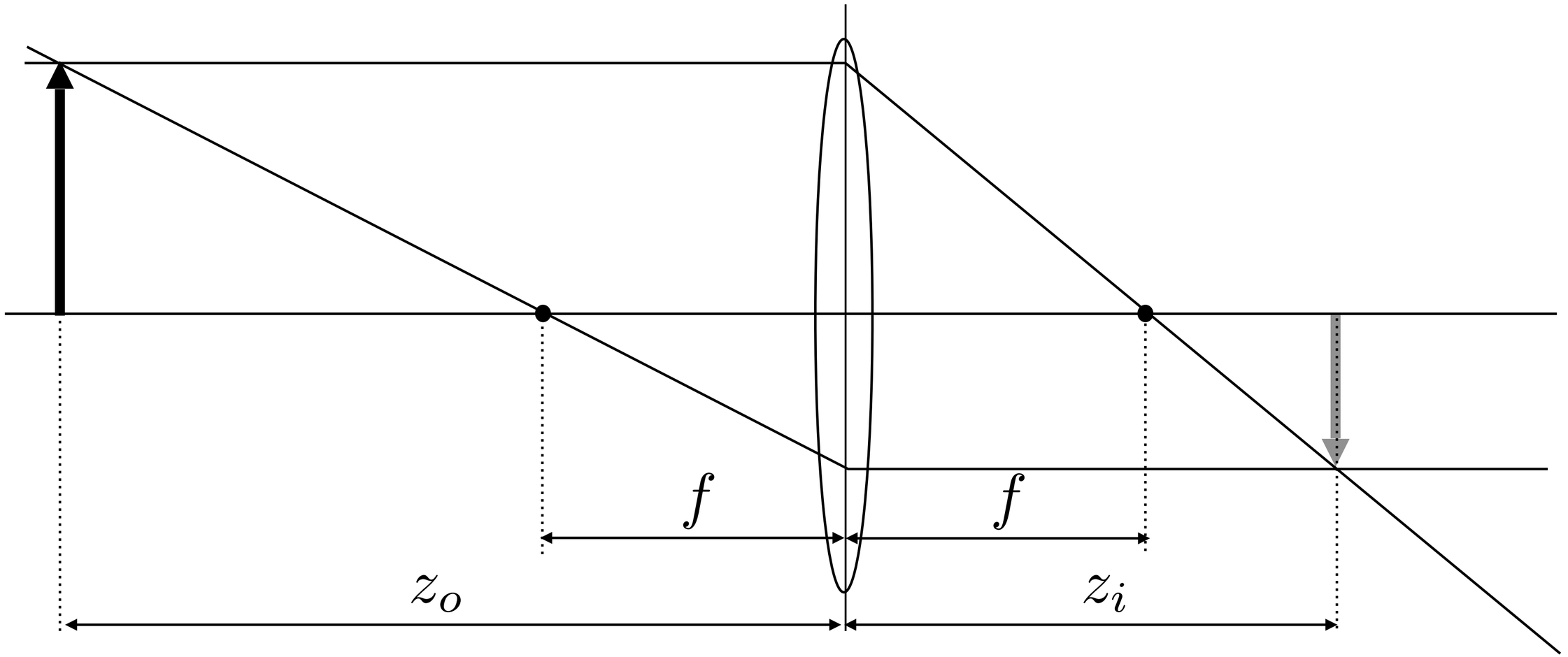
Gauss' Ray Tracing Construction



CS184

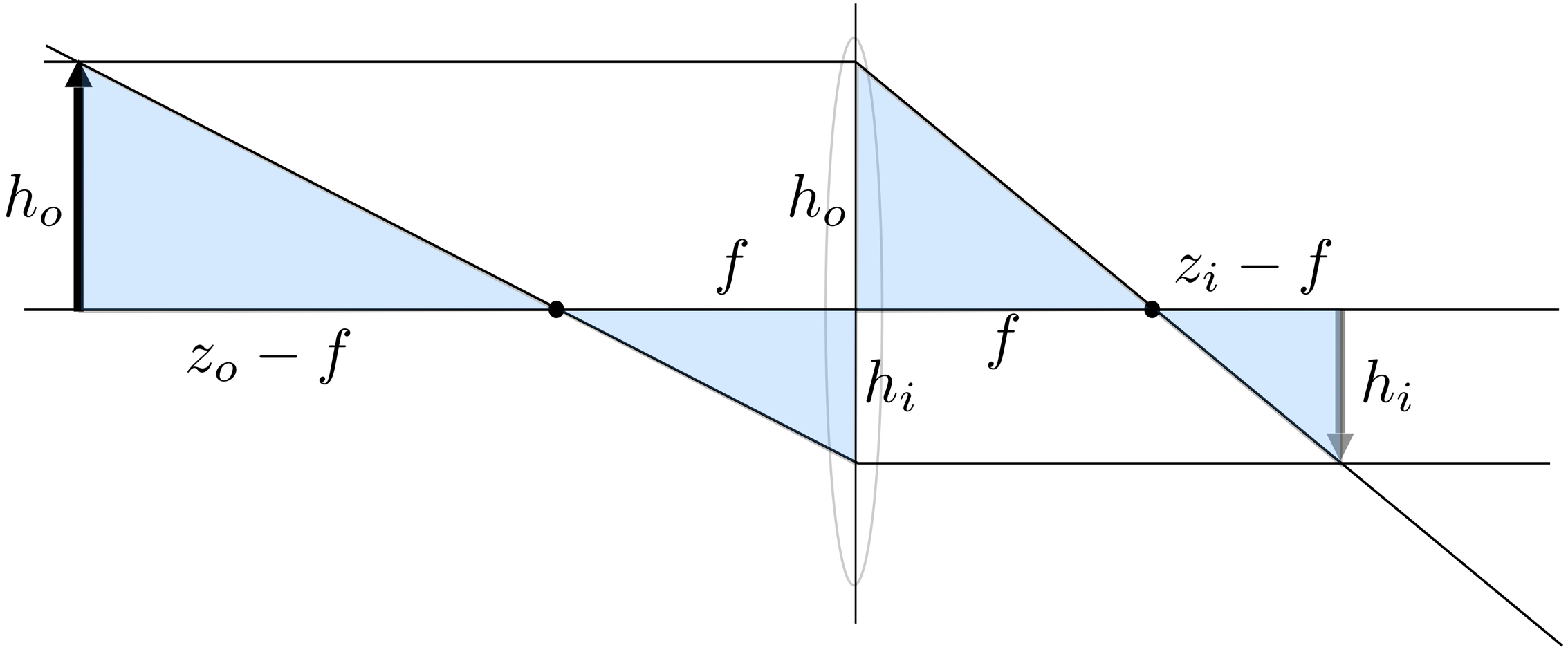
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Gauss' Ray Tracing Construction



What is the relationship between conjugate depths z_o, z_i?

Gauss' Ray Tracing Construction



$$\frac{h_o}{z_o - f} = \frac{h_i}{f}$$

$$\frac{h_o}{f} = \frac{h_i}{z_i - f}$$

Gauss' Ray Tracing Construction

$$\frac{h_o}{z_o - f} = \frac{h_i}{f} \qquad \frac{h_o}{f} = \frac{h_i}{z_i - f}$$

$$\frac{h_o}{h_i} = \frac{z_o - f}{f} \qquad \frac{h_o}{h_i} = \frac{f}{z_i - f}$$

$$\frac{z_o - f}{f} = \frac{f}{z_i - f}$$

Object / image heights

factor out - applies to all rays

$$(z_o - f)(z_i - f) = f^2$$

Newtonian Thin Lens Equation

$$z_o z_i - (z_o + z_i) f + f^2 = f^2$$

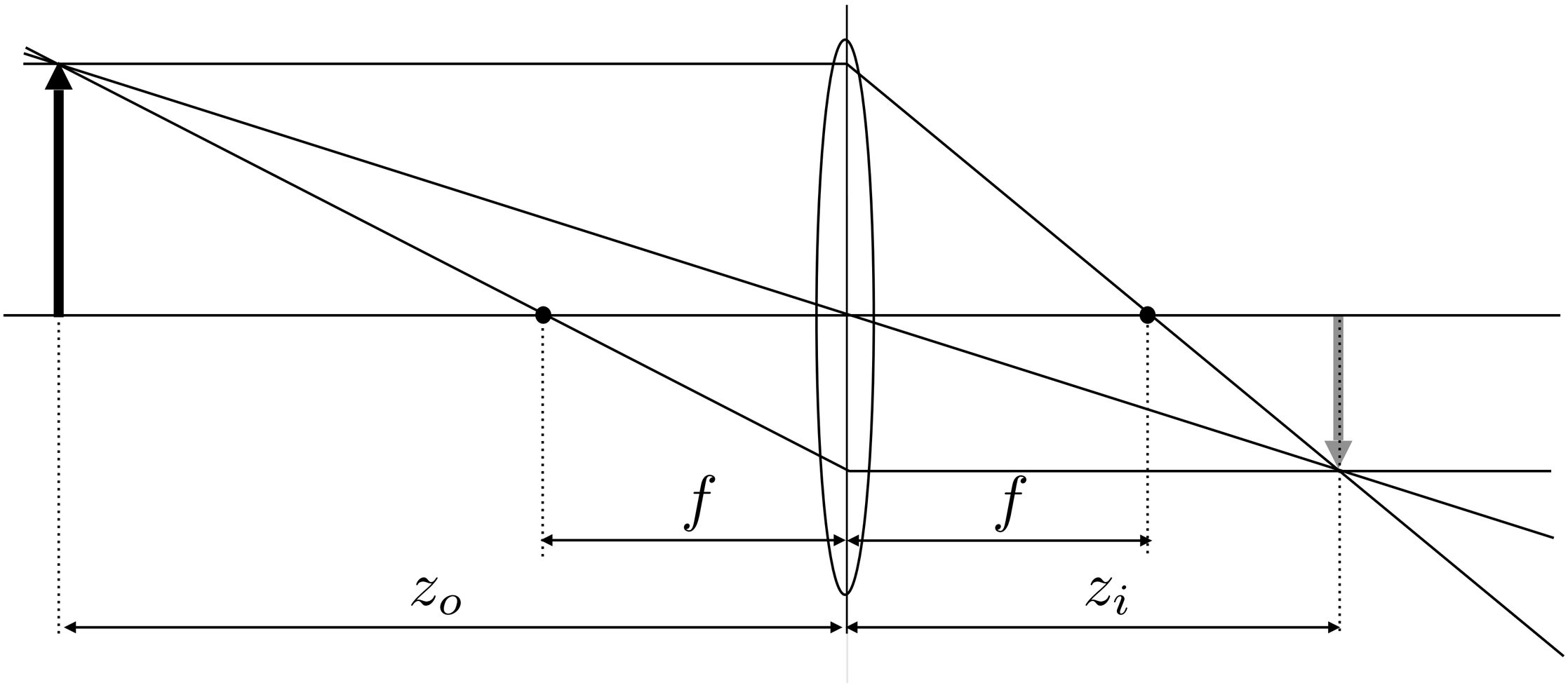
$$z_o z_i = (z_o + z_i) f$$

$$\frac{1}{f} = \frac{1}{z_i} + \frac{1}{z_o}$$

Gaussian Thin Lens Equation

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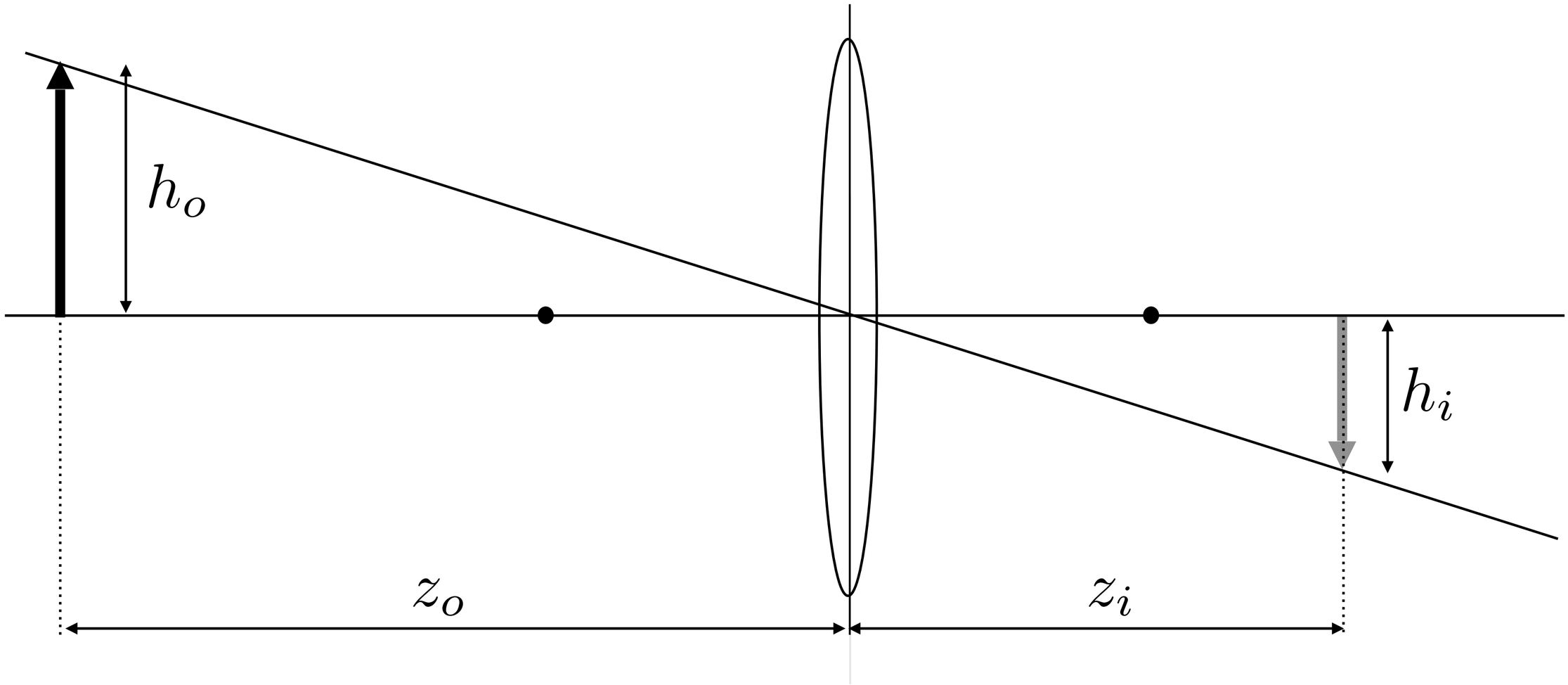
The Thin Lens Equation



$$\frac{1}{f} = \frac{1}{z_i} + \frac{1}{z_o}$$

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Magnification



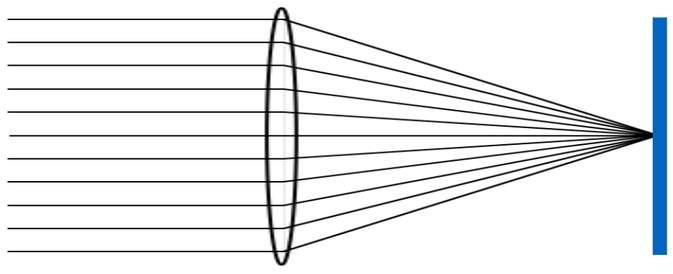
$$m = \frac{h_i}{h_o} = \frac{z_i}{z_o}$$

Magnification Example – Focus at Infinity

$$\frac{1}{f} = \frac{1}{z_i} + \frac{1}{z_o} \qquad m = \frac{z_i}{z_o}$$

If focused on a distant mountain

- $z_o \approx \infty$, so $z_i = f$
- sensor at focal point
- magnification ≈ 0



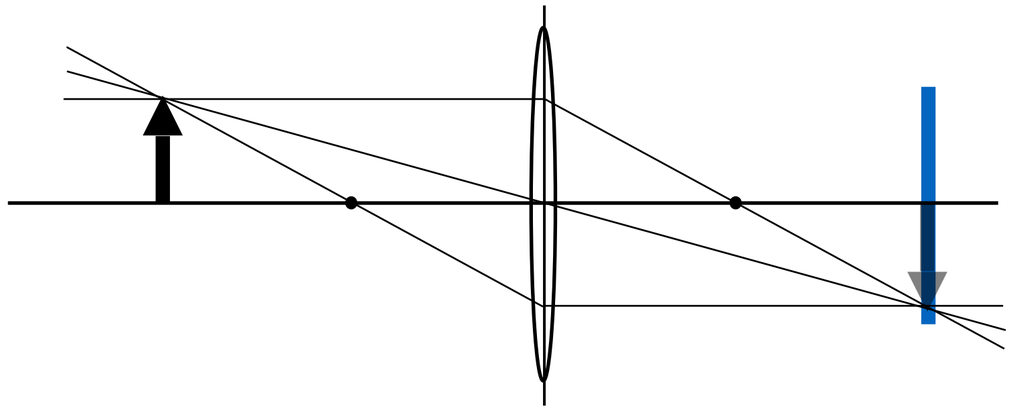


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Magnification Example – Focus at 1:1 Macro



$$\frac{1}{f} = \frac{1}{z_i} + \frac{1}{z_o} \qquad m = \frac{z_i}{z_o}$$



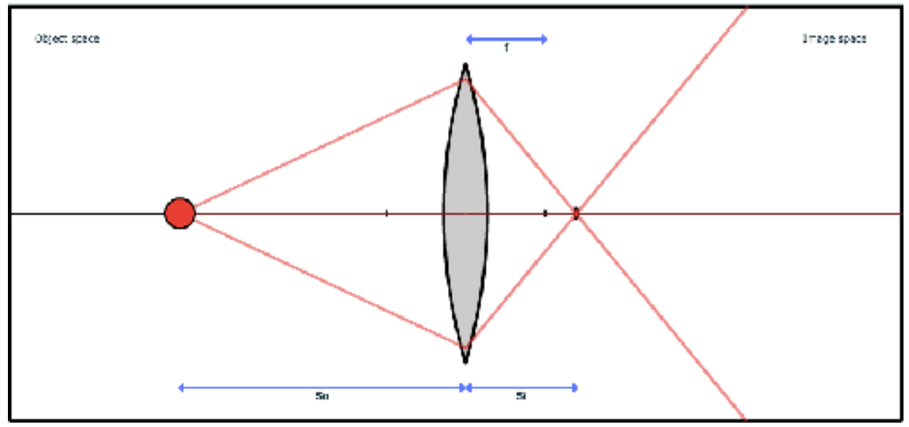
What configuration do we need to achieve a magnification of 1 (i.e. image and object the same size, a.k.a. 1:1 macro)?

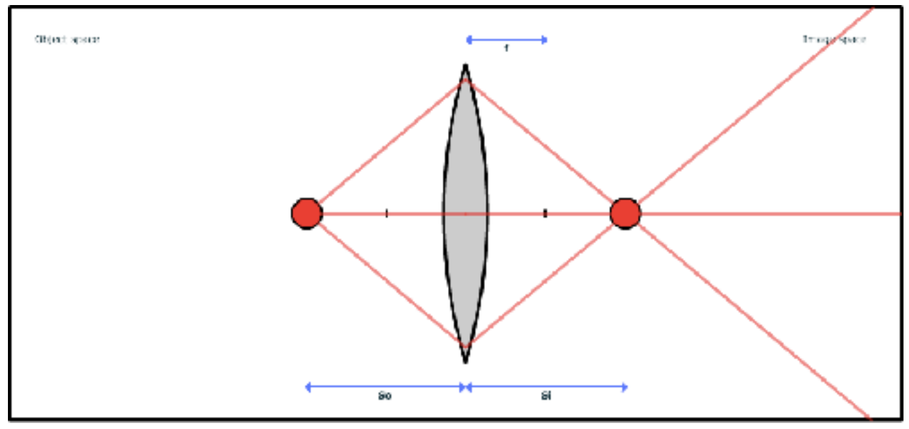
- Need $z_i = z_o$, so $z_i = z_o = 2f$ —- sensor at twice focal length
- In 1:1 imaging, if the sensor is 36 mm wide, an object 36 mm wide will fill the frame

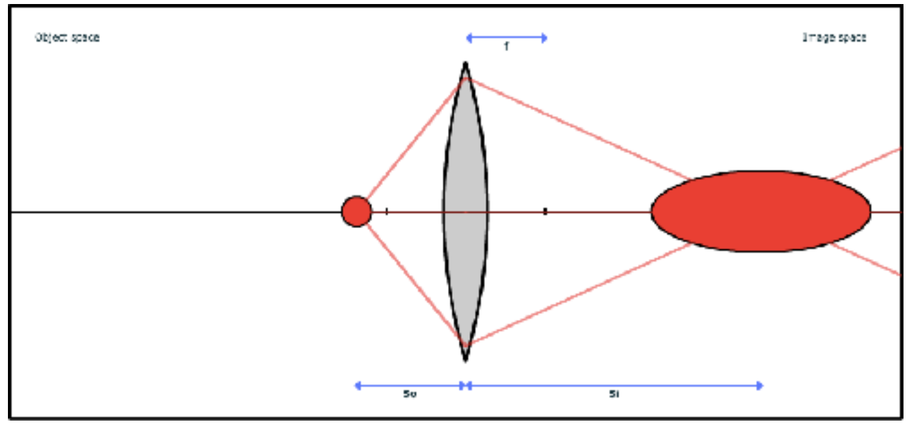
Thin Lens Effects — Observations in 3D

3D image of object is:

- Compressed in depth for low magnification
- 1:1 in 3D for unit magnification
- Stretched in depth for high magnification







Credit: Stanford CS 178

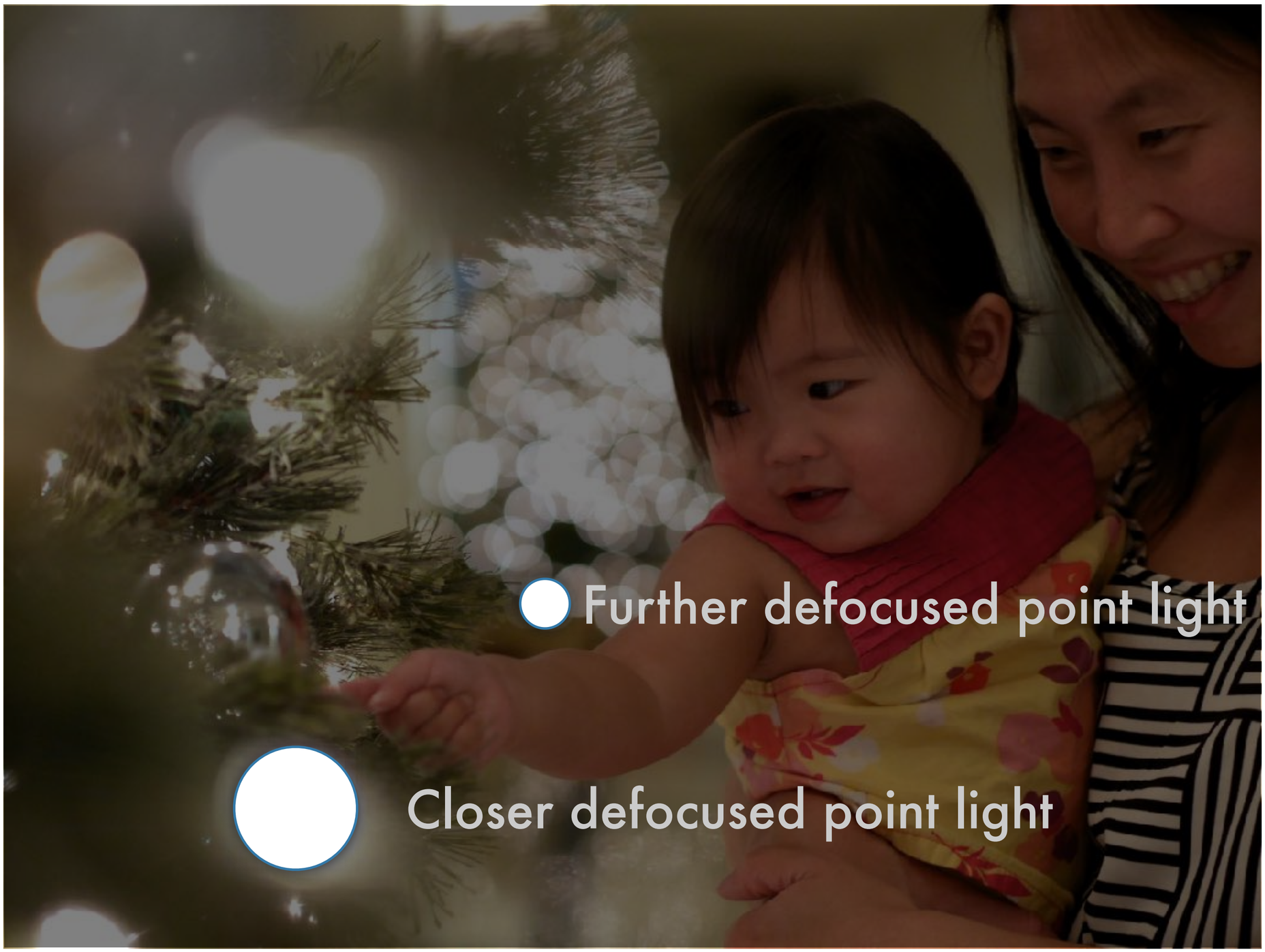
Defocus Blur

Circle of Confusion



Ren Ng
Presented by David McAllister

Circle of Confusion



Ren Ng
Presented by David McAllister

Circle of Confusion



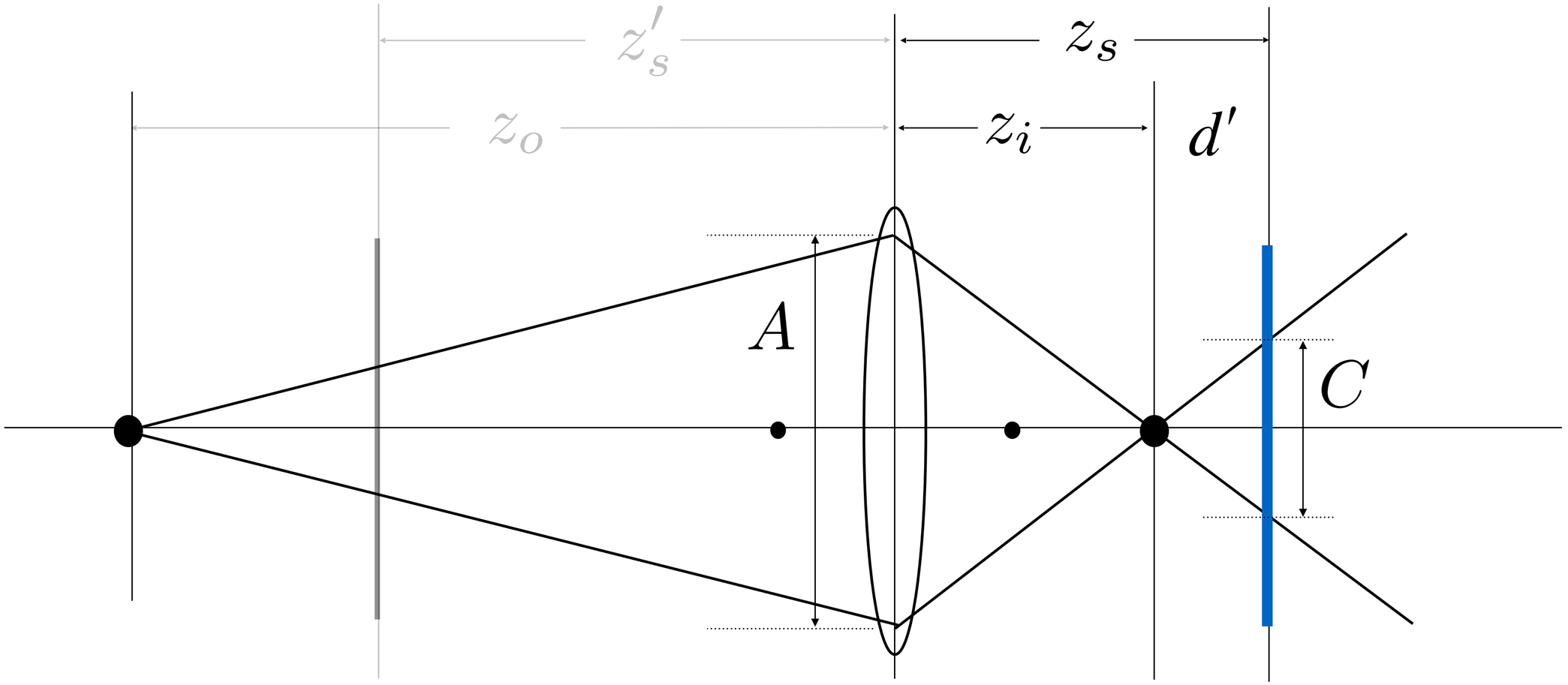
Size of blur kernel depends on depth from focal plane.
Only see the blur kernel itself if you have a point light. Why?

Circle of Confusion



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Computing Circle of Confusion Diameter (C)

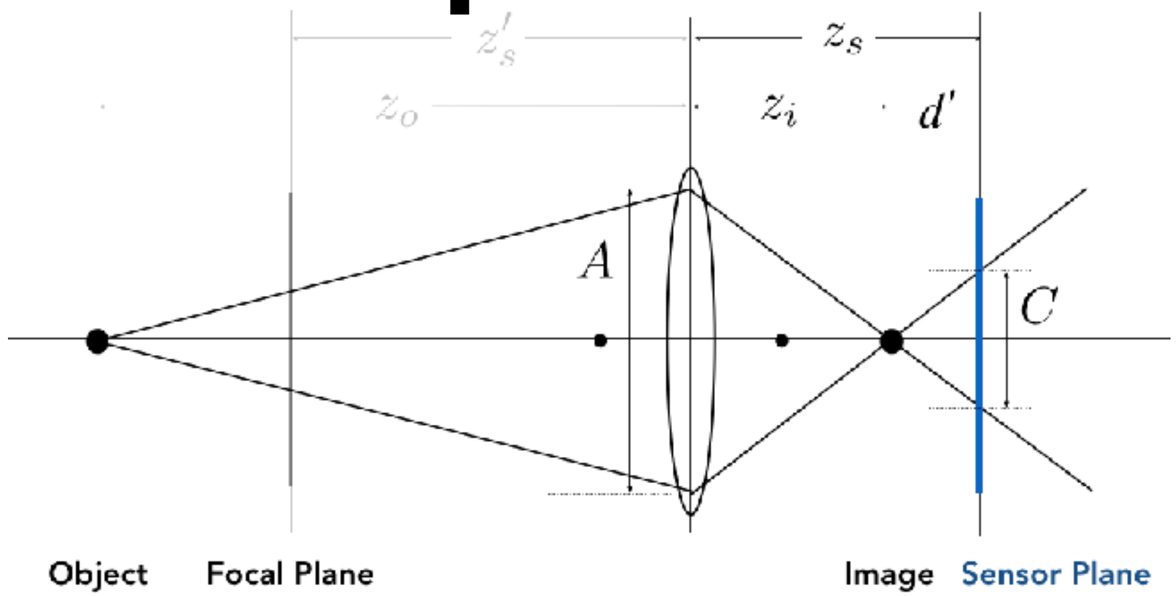


Circle of confusion is proportional to the size of the aperture

$$\frac{C}{A} = \frac{d'}{z_i} = \frac{|z_s - z_i|}{z_i}$$

Circle of Confusion – Example

50mm f/2 lens
Full frame sensor (36x24mm)
Focus: 1 meter
Background: 10 meter
Foreground: 0.3 meter



$$A = 50\text{mm}/2 = 25\text{mm}$$

$$z_s = \frac{1}{1/50 - 1/1000} \approx 52.63\text{mm}$$

$$C = A\frac{|z_s - z_i|}{z_i}$$



$$\text{Background: } z_i = \frac{1}{1/50 - 1/10{,}000} \approx 50.25\text{mm}$$

$$C = A|z_s - z_i|/z_i = 1.18\text{mm}$$ **~130 pixels on 4K TV**

$$\text{Foreground: } z_i = \frac{1}{1/50 - 1/300} \approx 55.56\text{mm}$$

$$C = A|z_s - z_i|/z_i = 3.07\text{mm}$$ **~338 pixels on 4K TV**

Size of Circle of Confusion is Inversely Proportional to F-Number for Photo



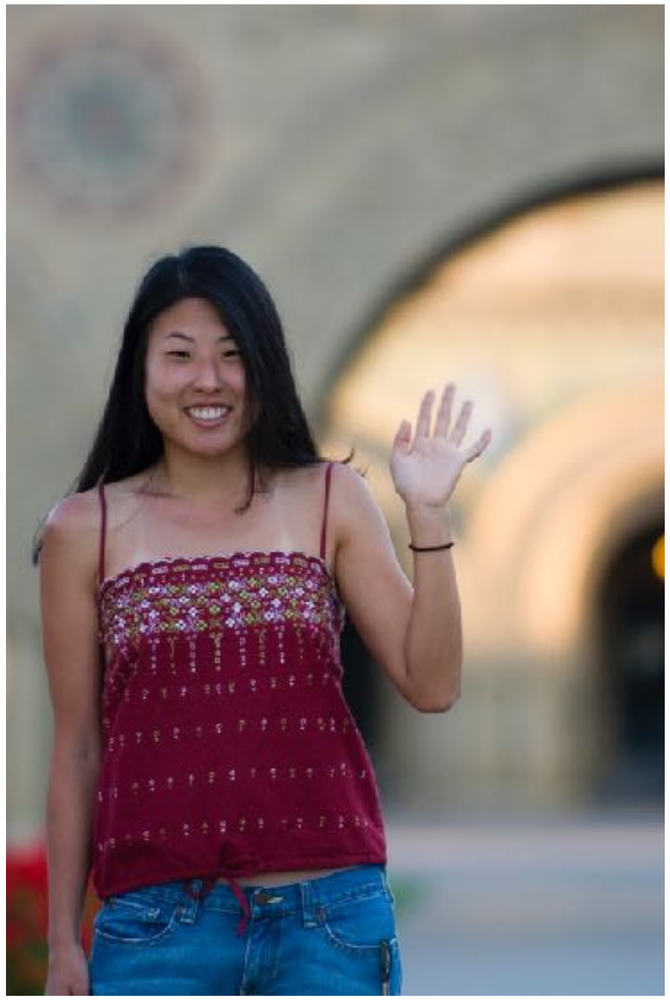
R. Berdan, canadiannaturephotographer.com

$$C = A\frac{|z_s - z_i|}{z_i} = \frac{f}{N}\frac{|z_s - z_i|}{z_i}$$

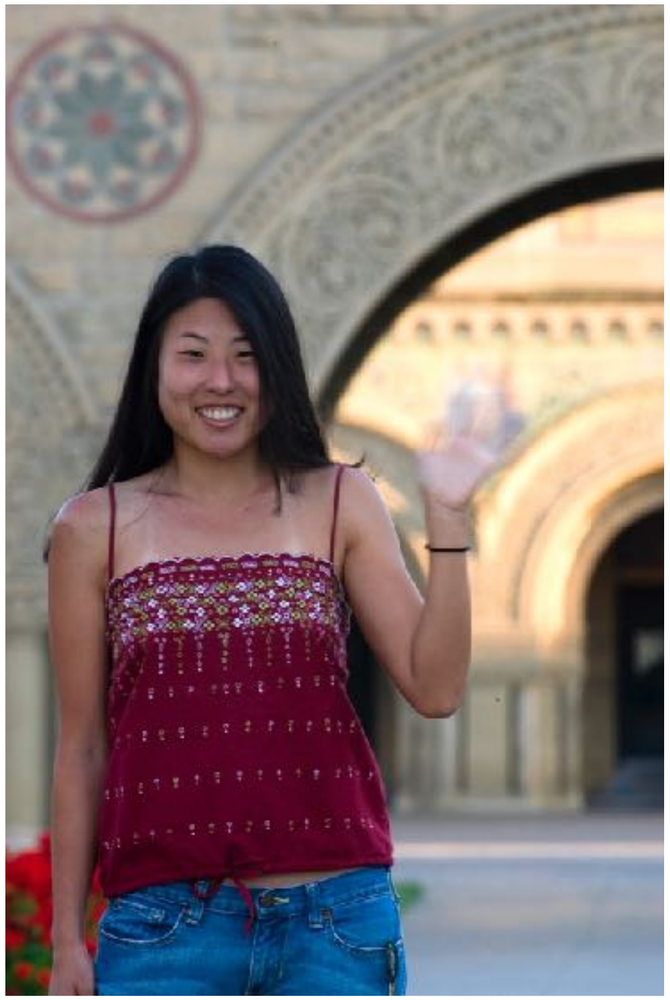
Exposure Tradeoffs

Depth of Field vs Motion Blur

Same Exposure: Depth of Field vs Motion Blur



f / 4
1/125 sec



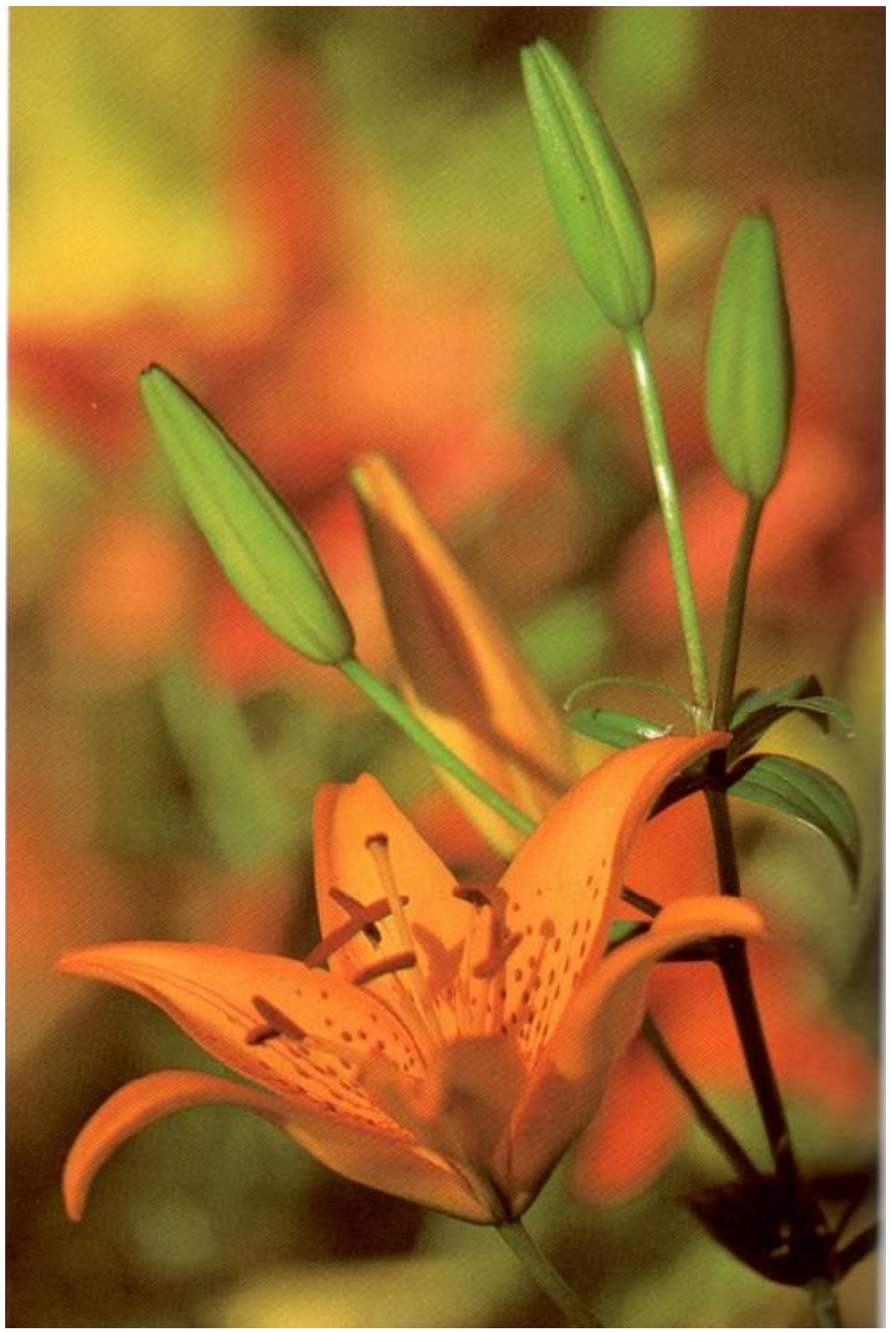
f / 11
1/15 sec



f / 32
1/2 sec

- Photographers must trade off depth of field and motion blur for moving subjects

Shallow Depth of Field Can Create a Stronger Image





From Peterson, Understanding Exposure
200mm, f/4, 1/1000 (left) and f/11, 1/125 (right)

Motion Blur Can Help Tell The Story



From Peterson, Understanding Exposure
1/60, f/5.6, 180mm

Attendance Time

If you are seated in class, go to this form and sign in:

- https://bit.ly/184attendance

Notes:

- Time-stamp will be taken when you submit form.
  Do it now, won't count later.
- Don't tell friends outside class to fill it out now, because we will audit at some point in semester.
- Failing audit will have large negative consequence. You don't need to, because you have an alternative!

CS184

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Presented by David McAllister

Depth of Field

Depth of Field (DOF)

Large aperture opening



Small aperture opening

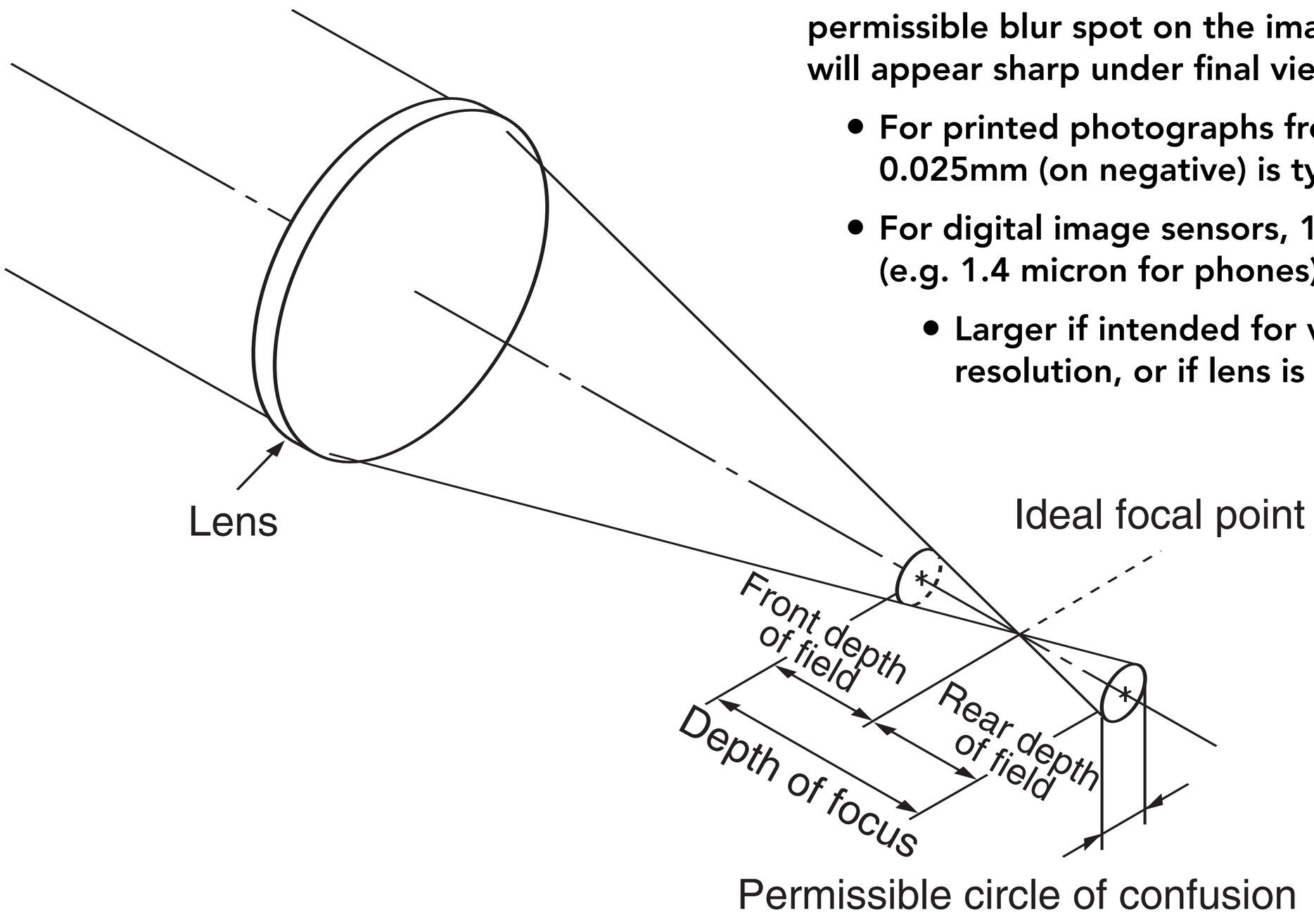


From London and Upton

- Depth of field is the range of object depths that are rendered with acceptable sharpness in an image

Ren Ng
Presented by David McAllister

Circle of Confusion for Depth of Field

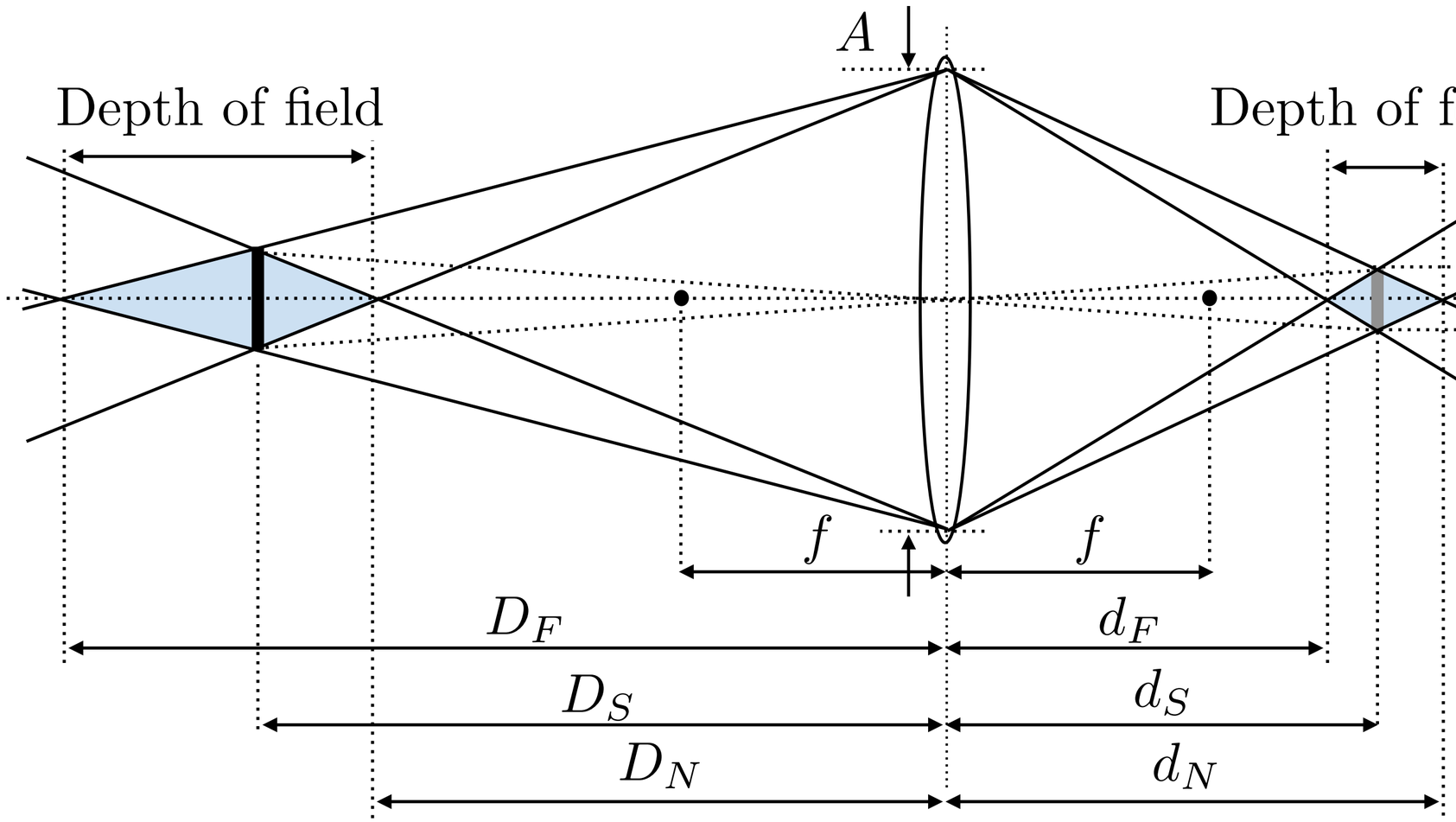


Set circle of confusion as the maximum permissible blur spot on the image plane that will appear sharp under final viewing conditions

- For printed photographs from 35mm film, 0.025mm (on negative) is typical
- For digital image sensors, 1 pixel is typical (e.g. 1.4 micron for phones)
    - Larger if intended for viewing at web resolution, or if lens is poor

[Canon, EF Lens Work III]

Depth of Field



$$\frac{d_N - d_S}{d_N} = \frac{C}{A}$$

$$\frac{d_S - d_F}{d_F} = \frac{C}{A}$$

$$N = \frac{f}{A}$$

$$\frac{1}{D_F} + \frac{1}{d_F} = \frac{1}{f}$$

$$\frac{1}{D_S} + \frac{1}{d_S} = \frac{1}{f}$$

$$\frac{1}{D_N} + \frac{1}{d_N} = \frac{1}{f}$$

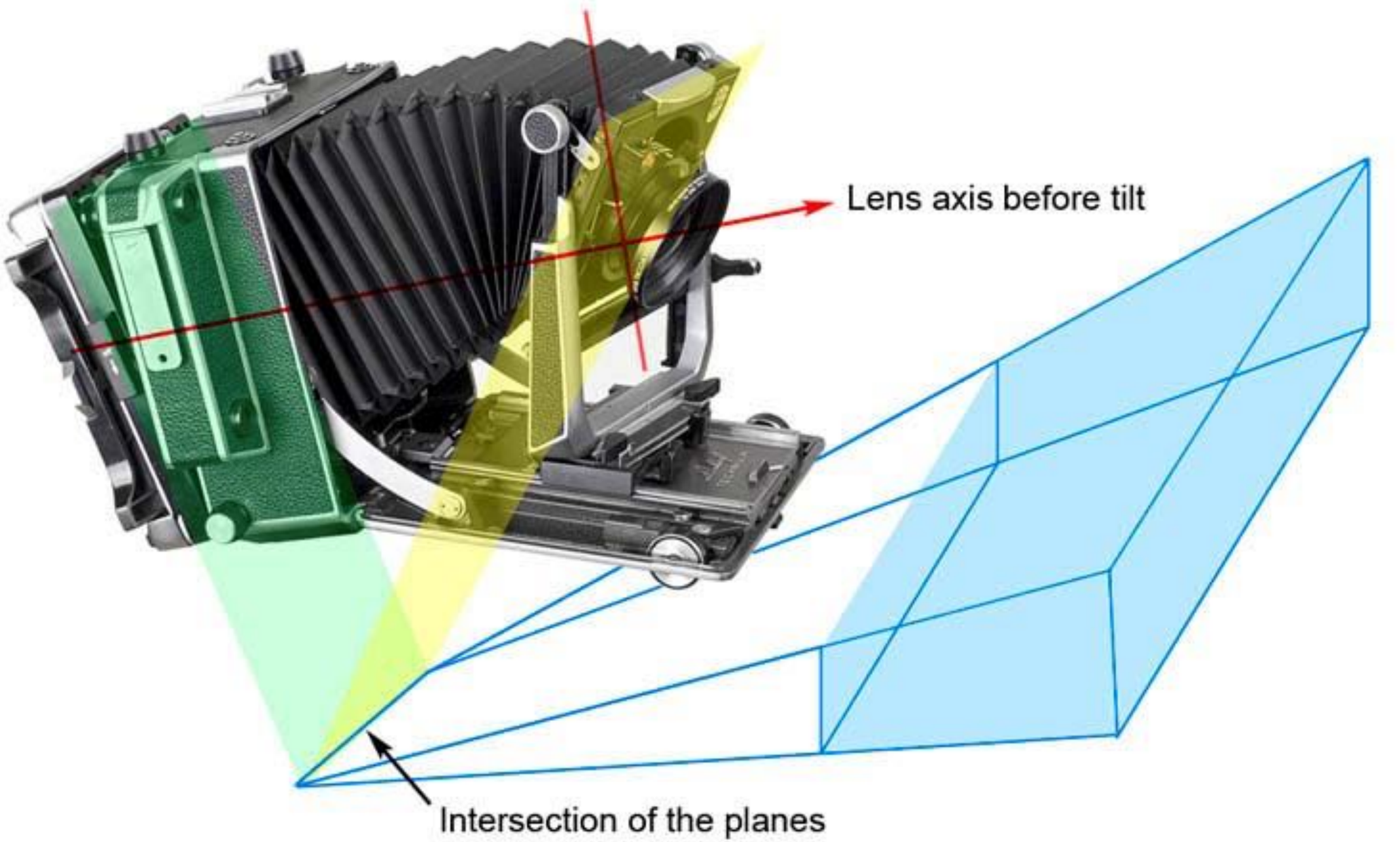
$$\text{DOF} = D_F - D_N$$

$$D_F = \frac{D_S f^2}{f^2 - NC(D_S - f)} \qquad D_N = \frac{D_S f^2}{f^2 + NC(D_S - f)}$$

Other Focus / DOF Situations to Consider

- How does sensor size affect defocus blur and DOF?
  - E.g. consider cell phone vs 35mm format sensors
- For a given lens & f-stop, how does moving closer/further from the subject (and adjusting focus onto subject) affect defocus / DOF of other objects?
- In 1:1 macro, does focal length affect DOF?
- What is the lens-sensor separation for hyperfocal condition (largest DOF possible), for full-resolution viewing vs web-resolution viewing?

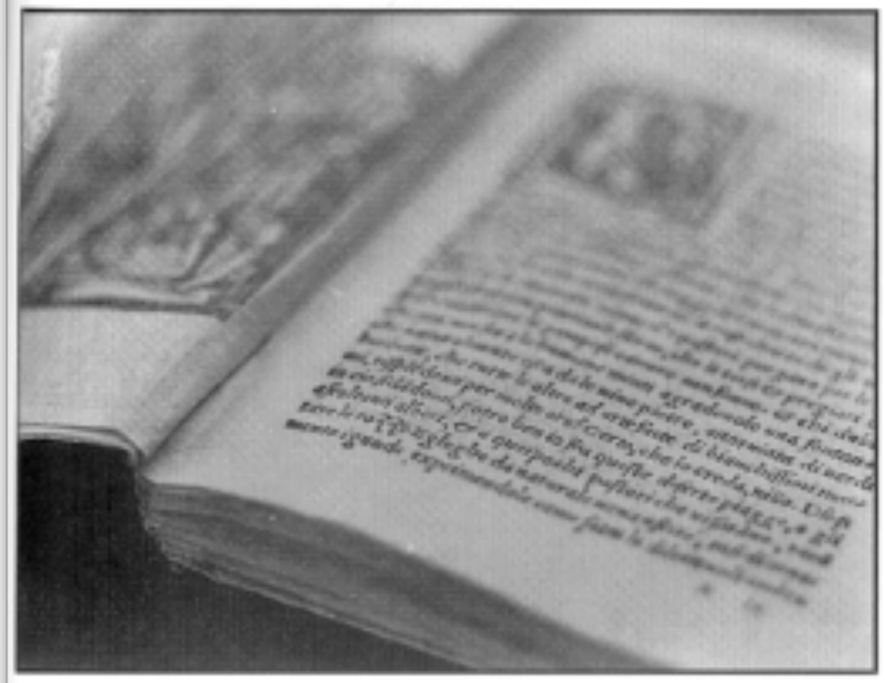
View Camera, Scheimpflug Rule



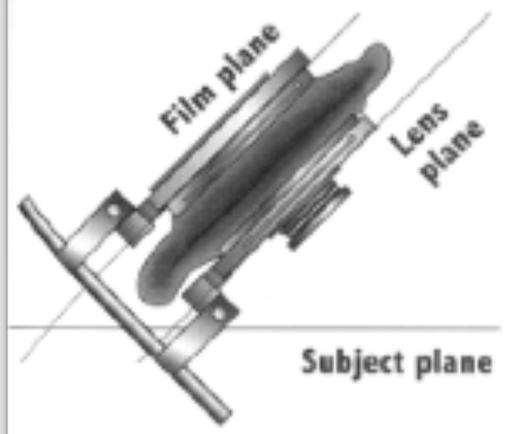
Source: David Summerhayes, http://www.luminous-landscape.com/tutorials/focusing-ts.shtml

View Camera, Scheimpflug Rule

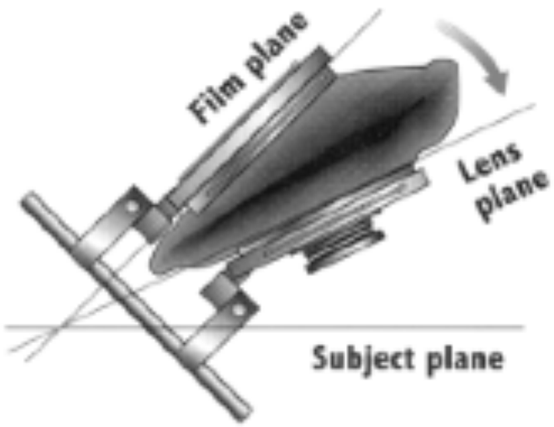
ADJUSTING THE PLANE OF FOCUS TO MAKE THE ENTIRE SCENE SHARP







The book is partly out of focus because the lens plane and the film plane are not parallel to the subject plane. Instead of a regular accordion bellows, the diagrams show a bag bellows that can bring camera front and back closer together for use with a short focal-length lens.



Tilting the front of the camera forward brings the entire page into sharp focus. The camera diagram illustrates the Scheimpflug principle, explained at right.

[London]

Ray Tracing Ideal Thin Lenses

Example of Rendering with Lens Focus



Credit: Bertrand Benoit. “Sweet Feast,” 2009. [Blender /VRay]

Example of Rendering with Lens Focus



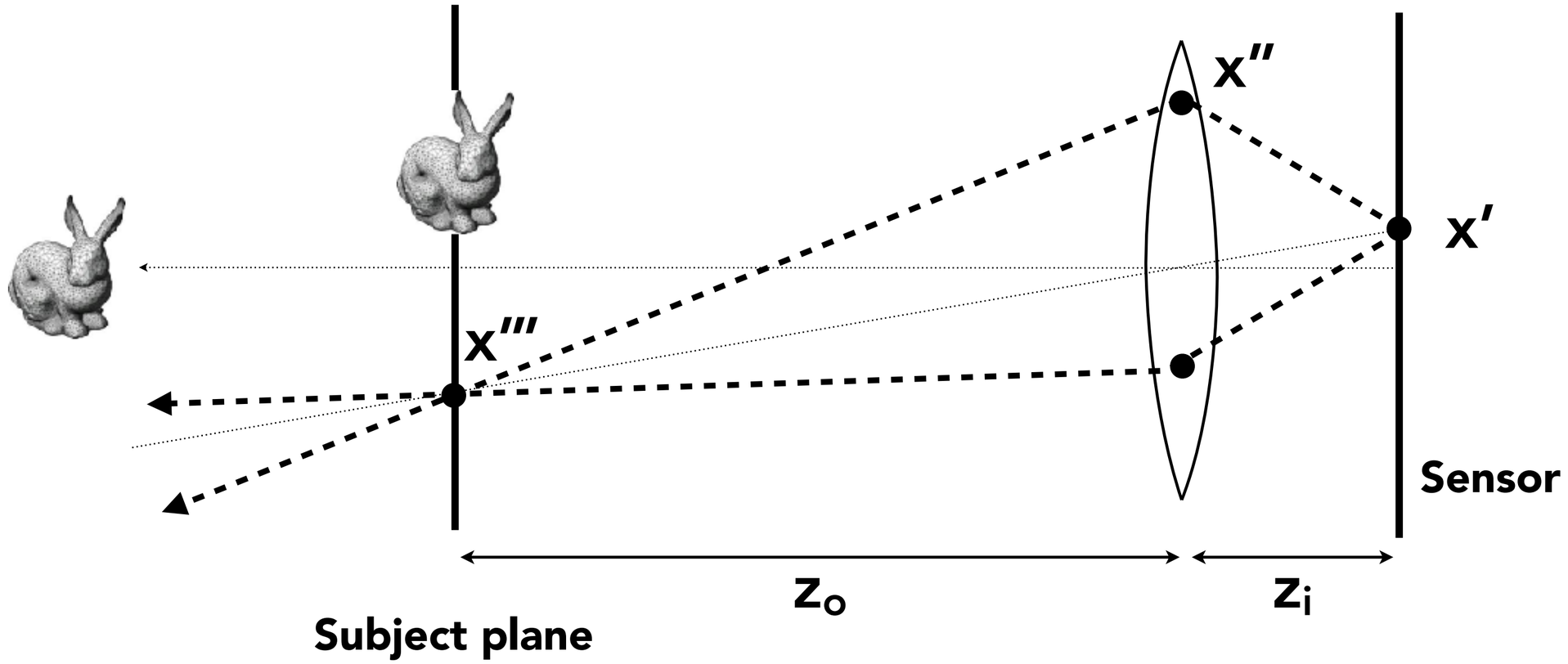
Credit: Giuseppe Albergo. “Colibri” [Blender]

Example of Rendering with Lens Focus



Pharr and Humphreys

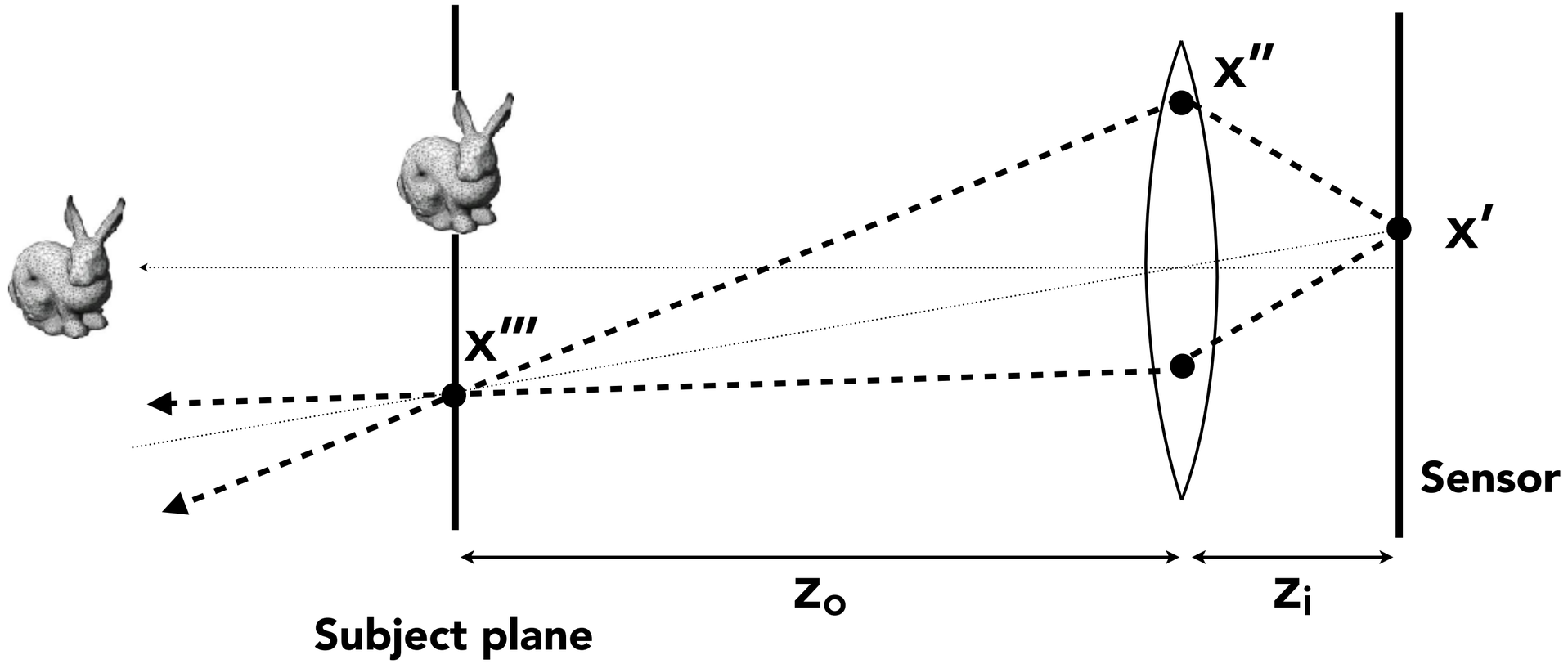
Ray Tracing for Defocus Blur (Thin Lens)



Setup (photography composition principles)

- Choose sensor size, lens focal length and aperture size
- Choose depth of subject of interest z_o
  - Calculate corresponding depth of sensor z_i from thin lens equation (focusing)

Ray Tracing for Defocus Blur (Thin Lens)



To compute value of pixel at position x′ by Monte Carlo integration:

- Select random points x″ on lens plane
- Rays pass from point x′ on image plane z_i through points x″ on lens
- Each ray passes through conjugate point x‴ on the plane of focus z_o
  - Can determine x‴ from Gauss′ ray diagram
  - So just trace ray from x″ to x‴
- Estimate radiance on rays using path-tracing, and sum over all points x″

Ren Ng

Presented by David McAllister

Example of Rendering with Lens Focus



Pharr and Humphreys

Bokeh

Bokeh

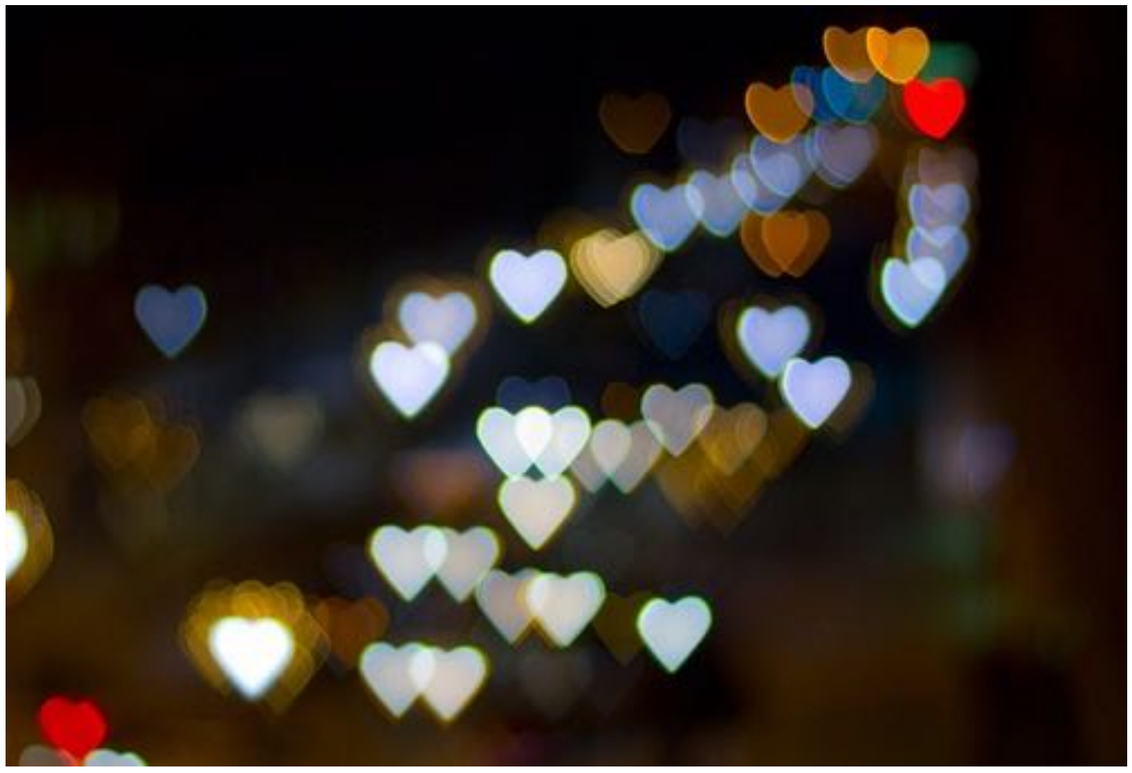
Bokeh is the shape and quality of out-of-focus blur

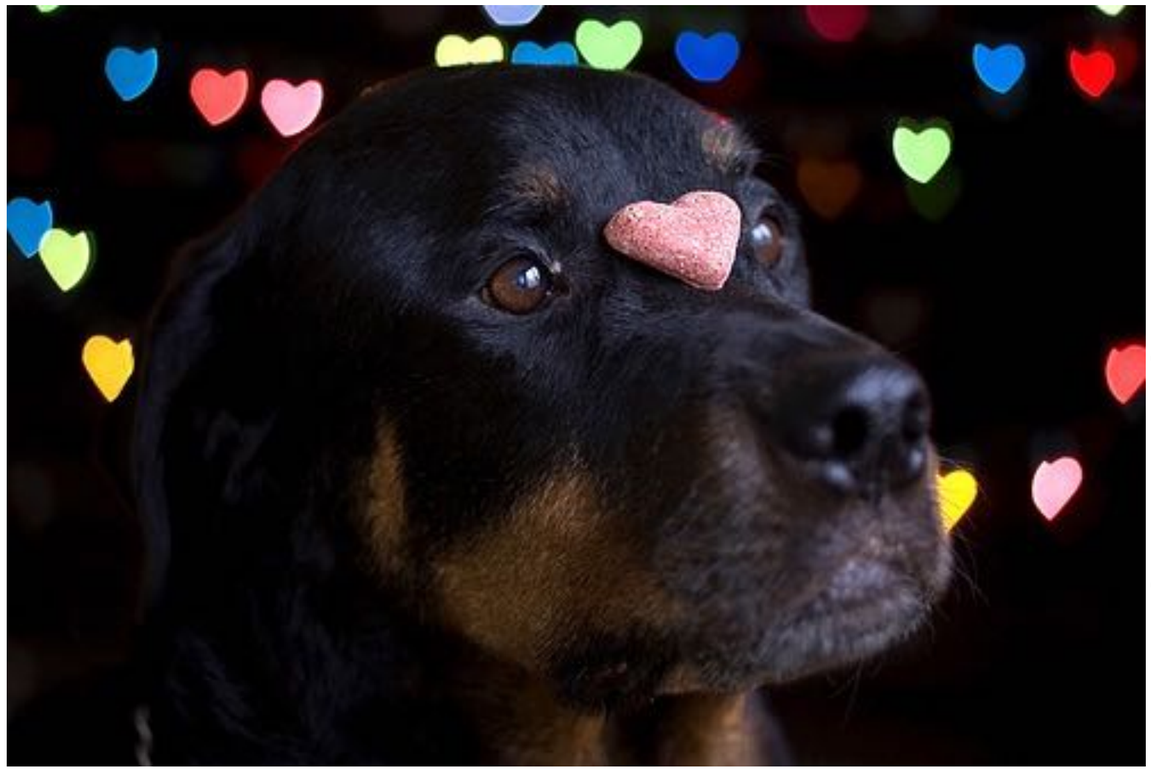
- For small, out-of-focus lights, bokeh takes on the shape of the lens aperture



M Yashna, flickr, 40mm f/3.0

Bokeh





diyphotography.net

Heart-shaped bokeh?



CS184

Ren Ng

Presented by David McAllister

Bokeh



Dino Quinzani, Leica Noctilux 50mm, f/0.95

Why does the bokeh vary across the image?

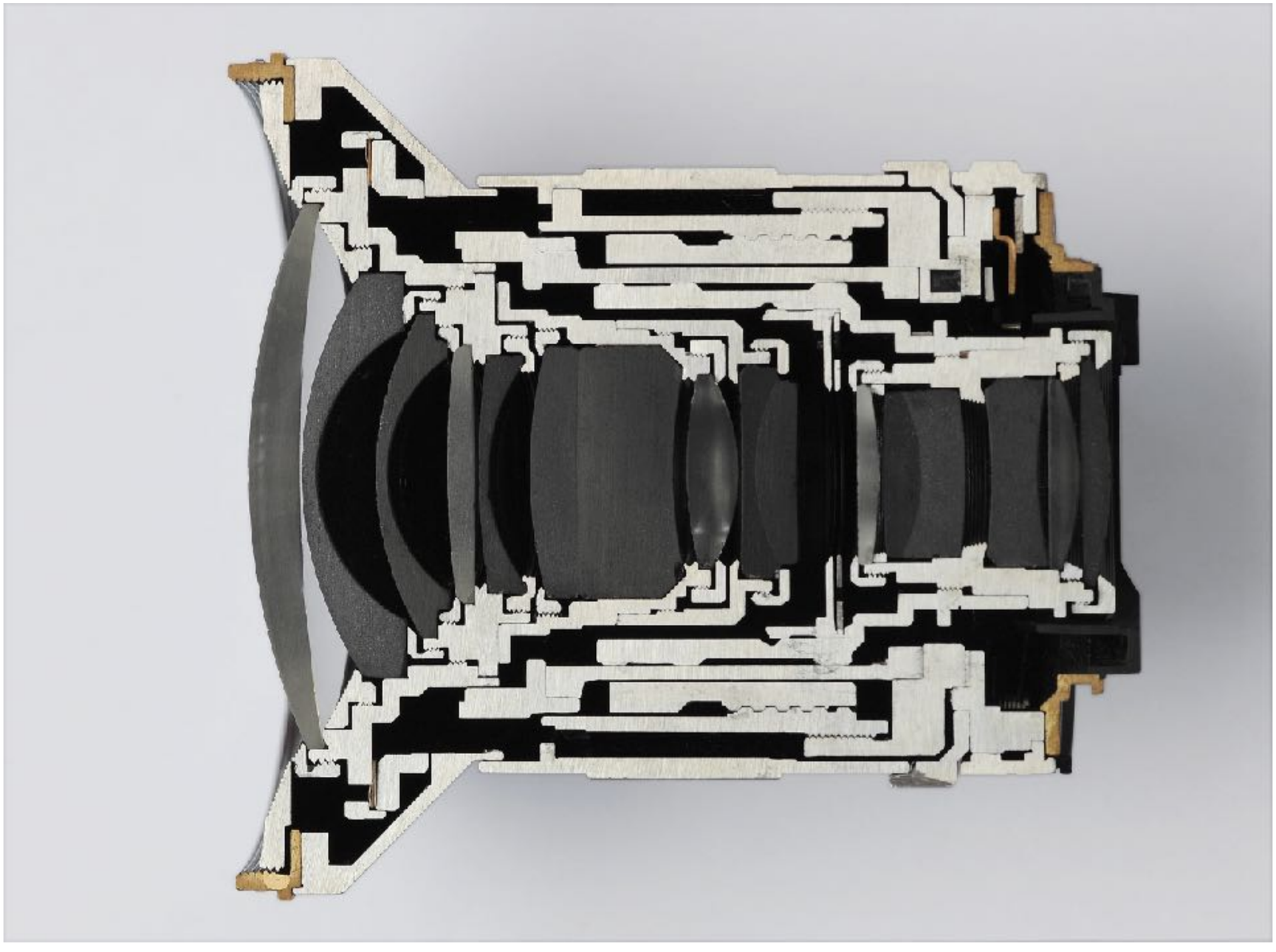
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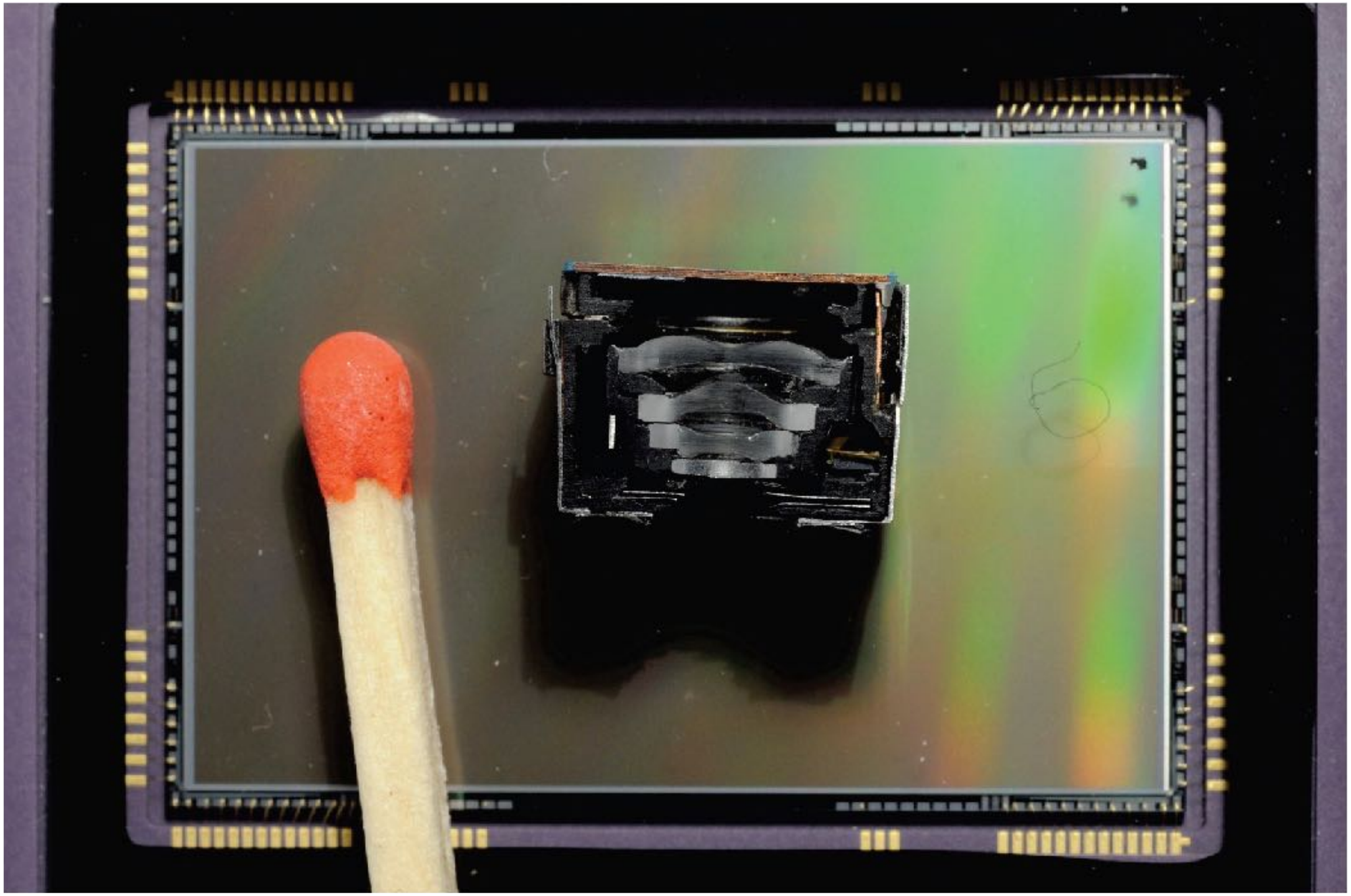
Real Compound Lenses

Modern Lens Designs Are Highly Complex



Photographic lens cross section

Modern Lens Designs Are Highly Complex



4 element mobile phone lens (on 24x36mm sensor)

CS184

Ren Ng

Presented by David McAllister

Modern Lens Designs Are Highly Complex



[Apple]

Ren Ng
Presented by David McAllister

Modern Lens Designs Are Highly Complex

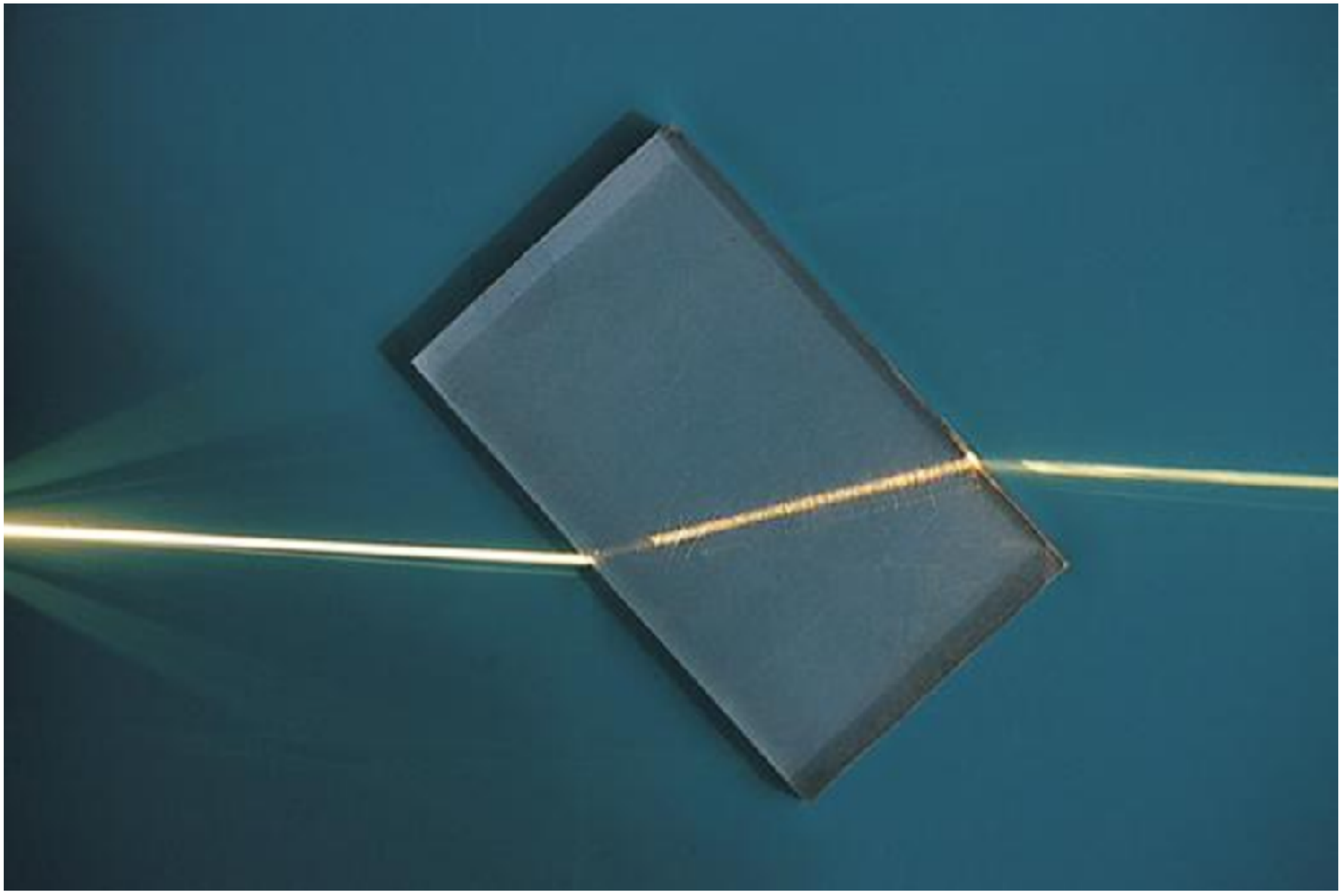




Zeiss flickr.com account

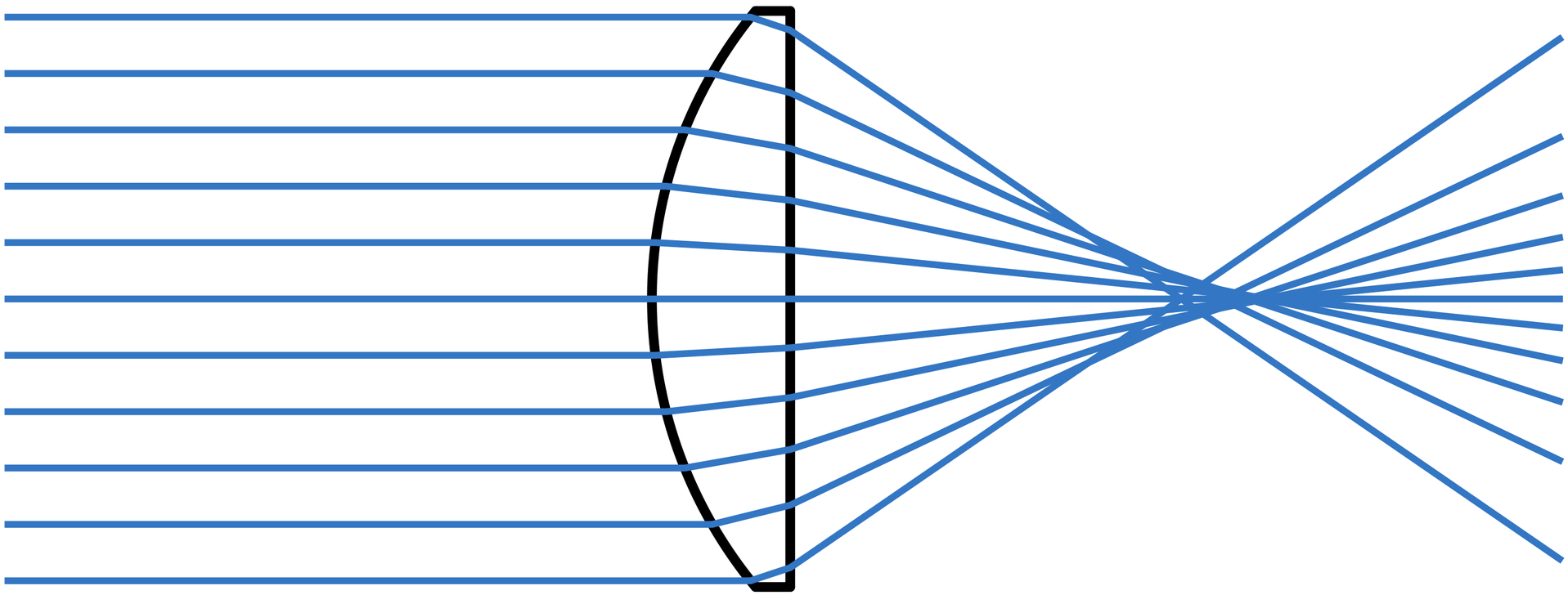
Microscope objective

Recall: Snell's Law of Refraction



$$\eta_i \sin\theta_i = \eta_t \sin\theta_t$$

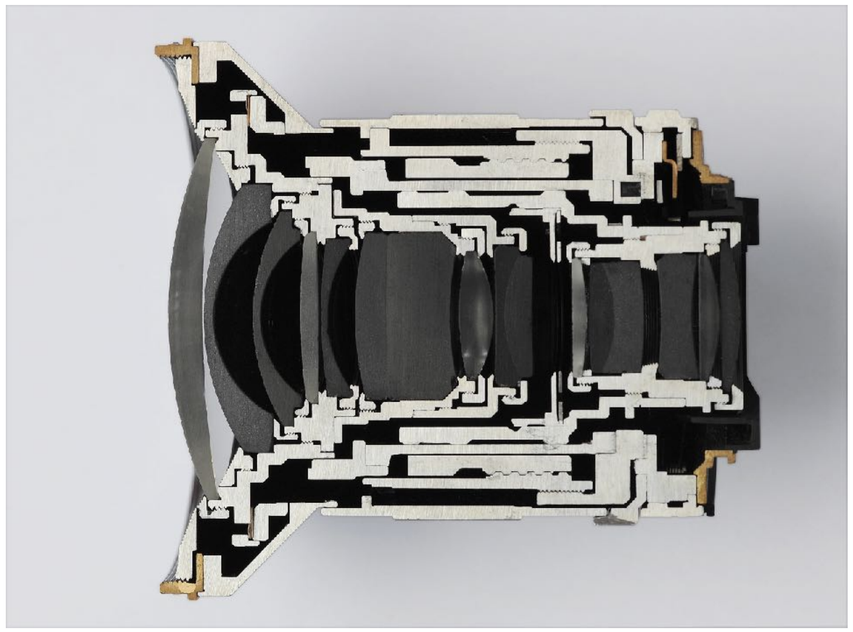
Real Refraction Through A Lens Is Not Ideal – Aberrations



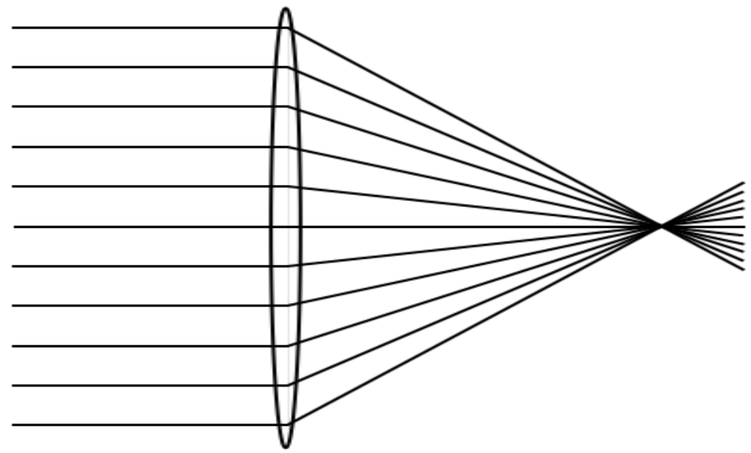
Real plano-convex lens (spherical surface shape).
Lens does not converge rays to a point anywhere.

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Real Lenses vs Ideal Thin Lenses



ilovephotography.com



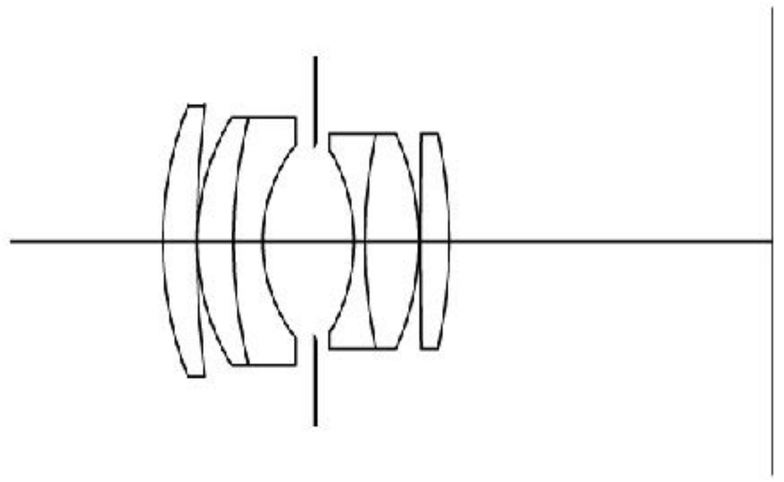
- Real optical system
- Multiple physical elements in compound design
- Optical aberrations prevent rays from converging perfectly

- Theoretical abstraction
- Assume all rays refract at a plane & converge to a point
- Quick and intuitive calculation of main imaging effects

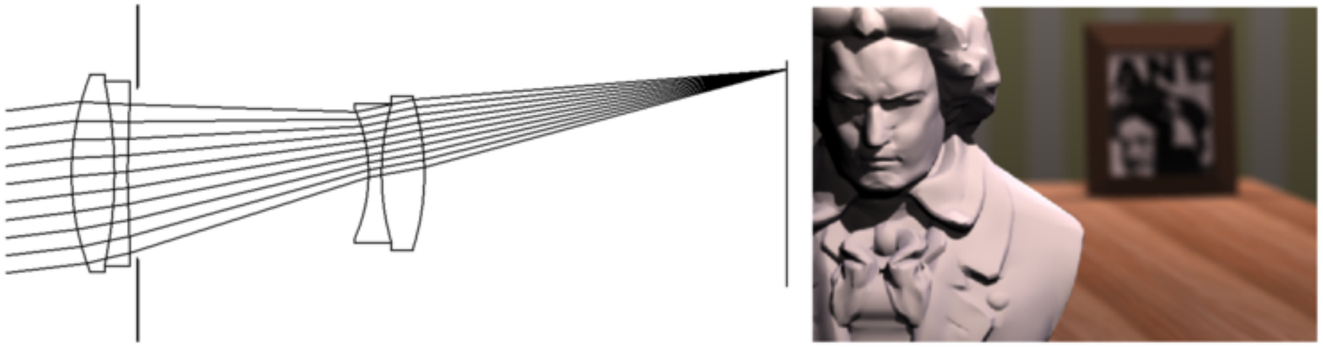
Example Lens Formula: Double Gauss

Data from W. Smith, Modern Lens Design, p 312

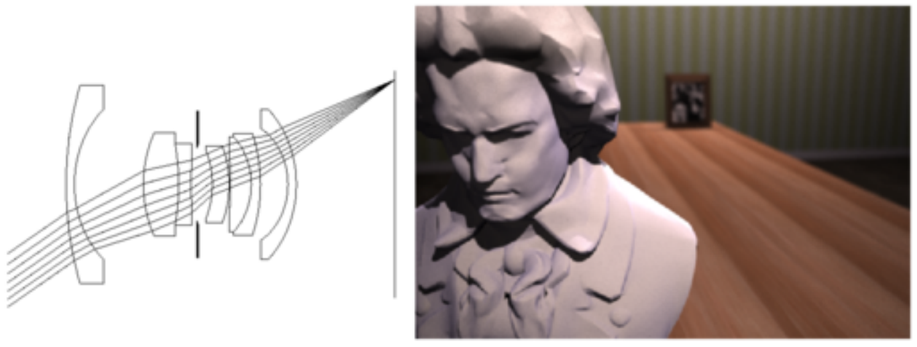
Radius (mm)	Thick (mm)	n_d	V-no	Aperture (mm)
58.950	7.520	1.670	47.1	50.4
169.660	0.240			50.4
38.550	8.050	1.670	47.1	46.0
81.540	6.550	1.699	30.1	46.0
25.500	11.410			36.0
	9.000			34.2
–28.990	2.360	1.603	38.0	34.0
81.540	12.130	1.658	57.3	40.0
–40.770	0.380			40.0
874.130	6.440	1.717	48.0	40.0
–79.460	72.228			40.0



Ray Tracing Through Real Lens Designs



200 mm telephoto



35 mm wide-angle



50 mm double-gauss



16 mm fisheye

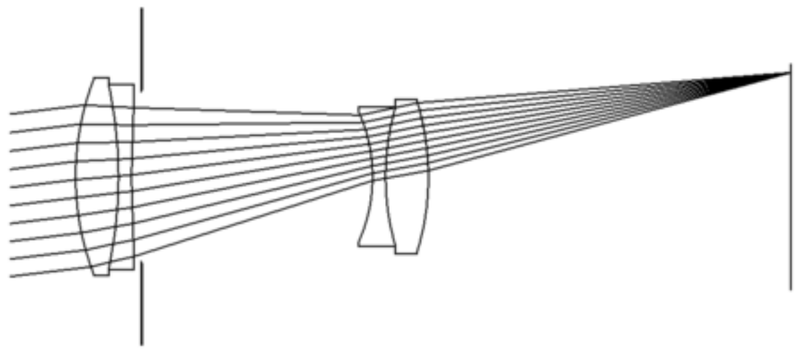
From Kolb, Mitchell and Hanrahan (1995)

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Presented by David McAllister

Ray Tracing Through Real Lens Designs



200 mm telephoto



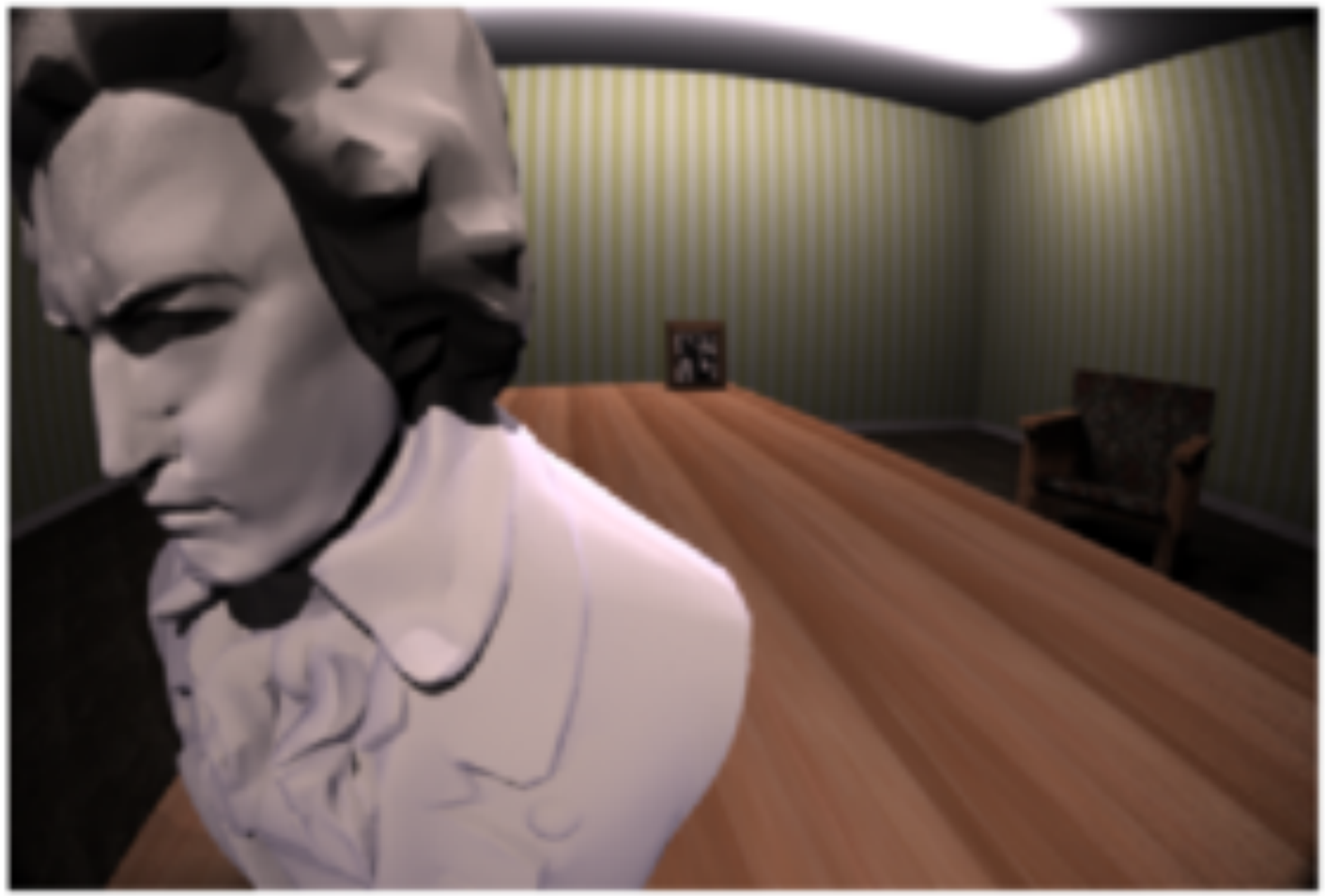
Notice shallow depth of field (out of focus background)

CS184

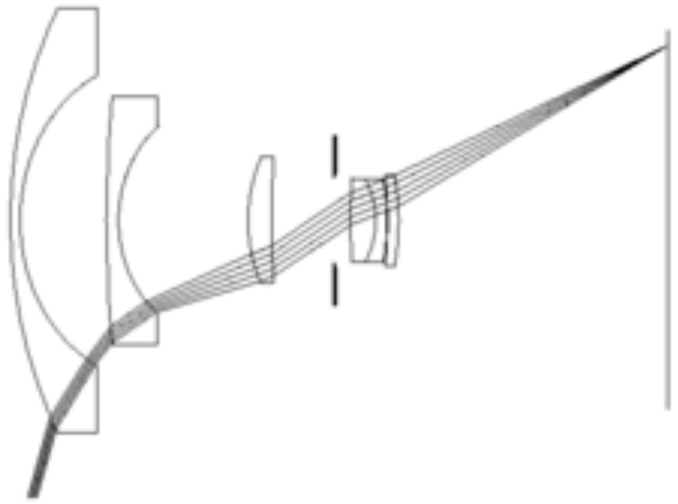
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Ray Tracing Through Real Lens Designs



16 mm fisheye



Notice distortion in the corners (straight lines become curved)

CS184

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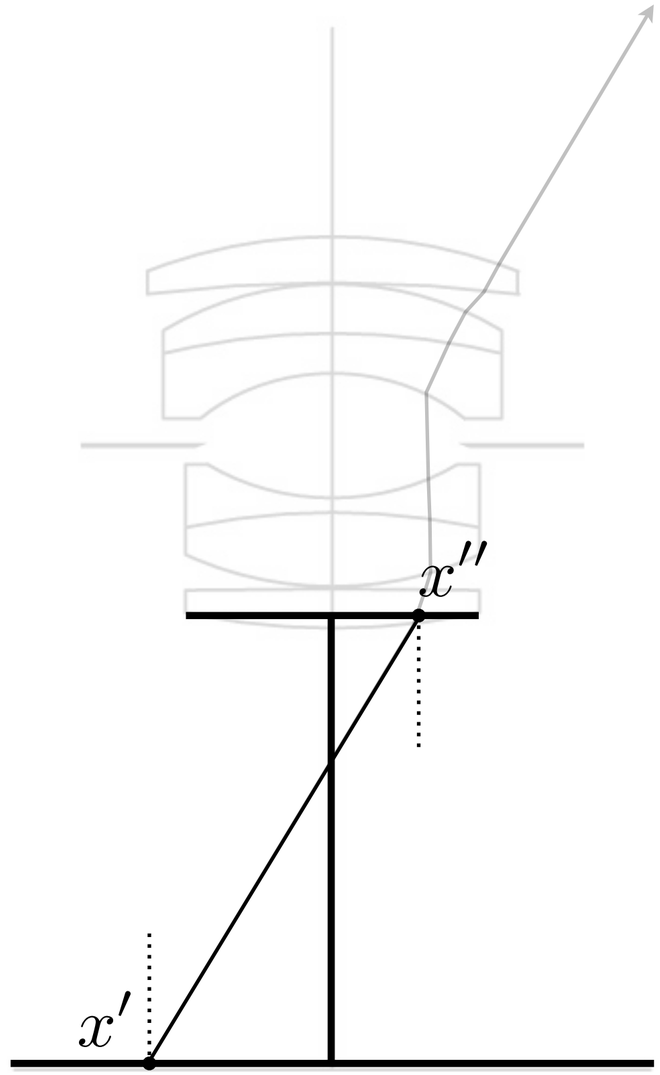
Ray Tracing Real Lens Designs

Monte Carlo approach

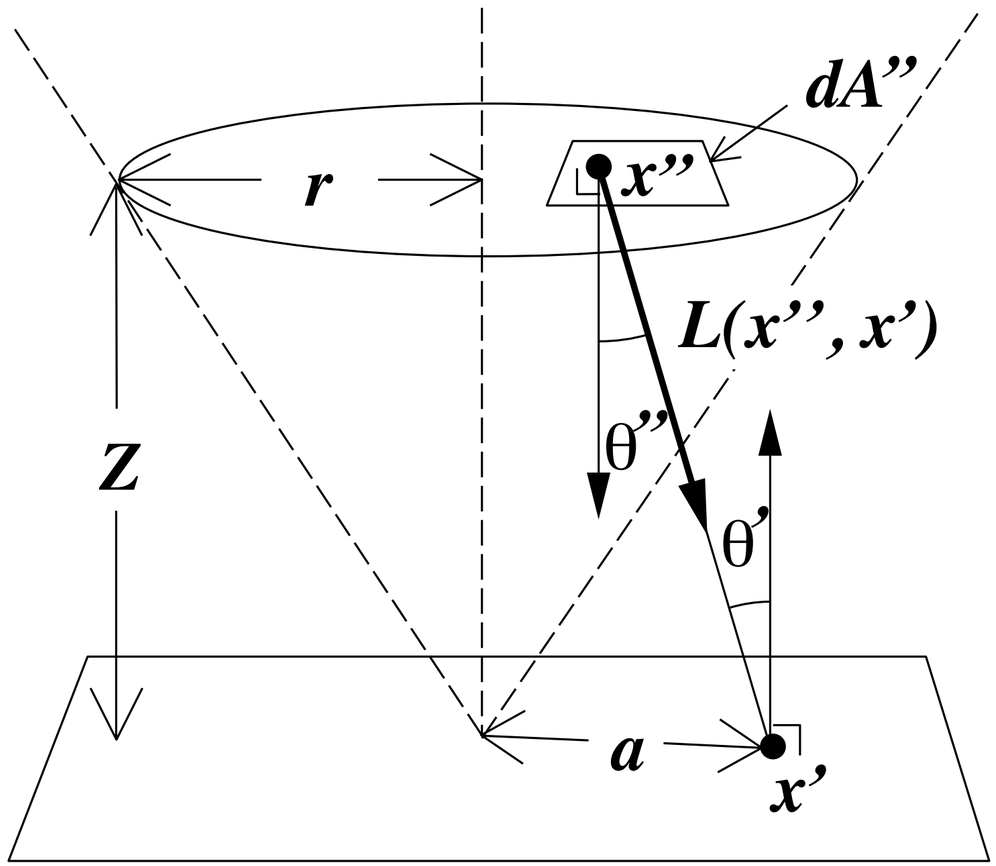
- At every sensor pixel, compute integral of rays incident on pixel area arriving from all paths through the lens

Algorithm (for a pixel)

- Choose N random positions in pixel
- For each position x′, choose a random position on the back element of the lens x″
- Trace a ray from x′ to x″, trace refractions through lens elements until it either misses the next element (terminate ray) or exits the lens (path trace through the scene)
- Weight each ray according to radiometric calculation on next slide to estimate power falling on the pixel



Radiometry for Tracing Lens Designs



$$E(x') = \int_{x'' \in D} L(x'' \to x') \frac{\cos\theta' \cos\theta''}{||x'' - x'||^2} dA''$$

$$= \frac{1}{Z^2} \int_{x'' \in D} L(x'' \to x') \cos^4\theta dA''$$

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Things to Remember

- Field of view depends on focal length and sensor size
- Perspective composition - focal length and camera position
- Exposure — aperture, shutter, gain (ISO)
- Deriving the thin lens equation from Gauss' ray diagram
- Defocus blur, circle of confusion, depth of field
- Using ideal thin lenses, or real compound lens optical designs, in ray tracing

Acknowledgments

Many thanks to Marc Levoy, who created many of these slides, and Pat Hanrahan.

- London, Stone, and Upton, Photography (9th ed.), Prentice Hall, 2008.
- Peterson, Understanding Exposure, AMPHOTO 1990.
- The Slow Mo Guys
- bobatkins.com
- Hari Subramanyan
- Canon EF Lens Work III

Extra

Auto Focus

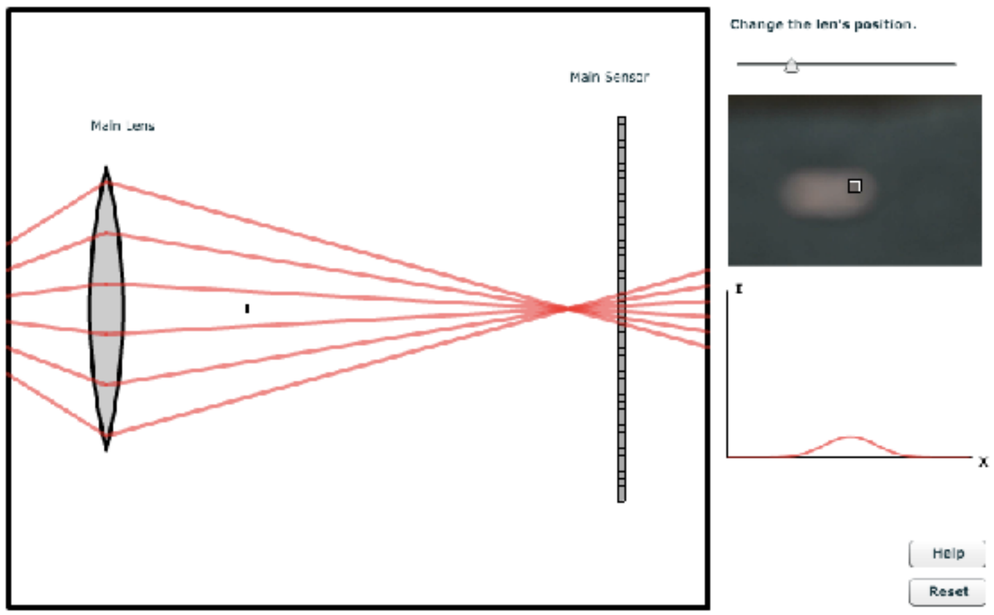
Contrast Detection Autofocus

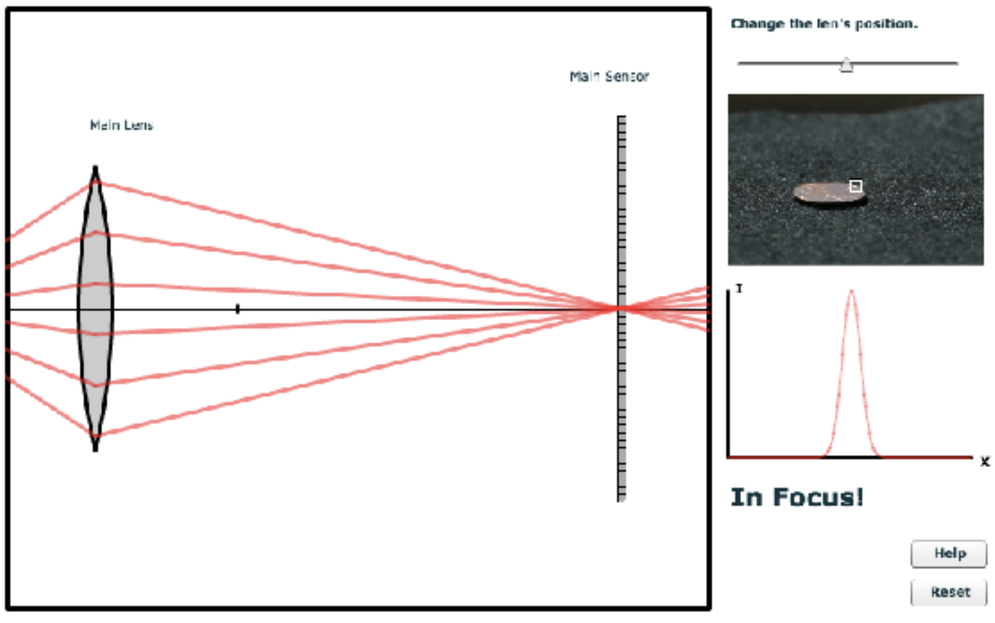
A target object is imaged through the lens to an image patch on the sensor.

The contrast of this image patch is high if the object is in focus, low otherwise.

The physical focus of the lens is adjusted until the contrast of this image patch is maximized.

Many ways to estimate how in-focus the image patch is: gradient, Sum Modified Laplacian (Nayar), variance…





Demo (Levoy, Willet, Adams)

https://graphics.stanford.edu/courses/cs178-10/applets/autofocusCD.html

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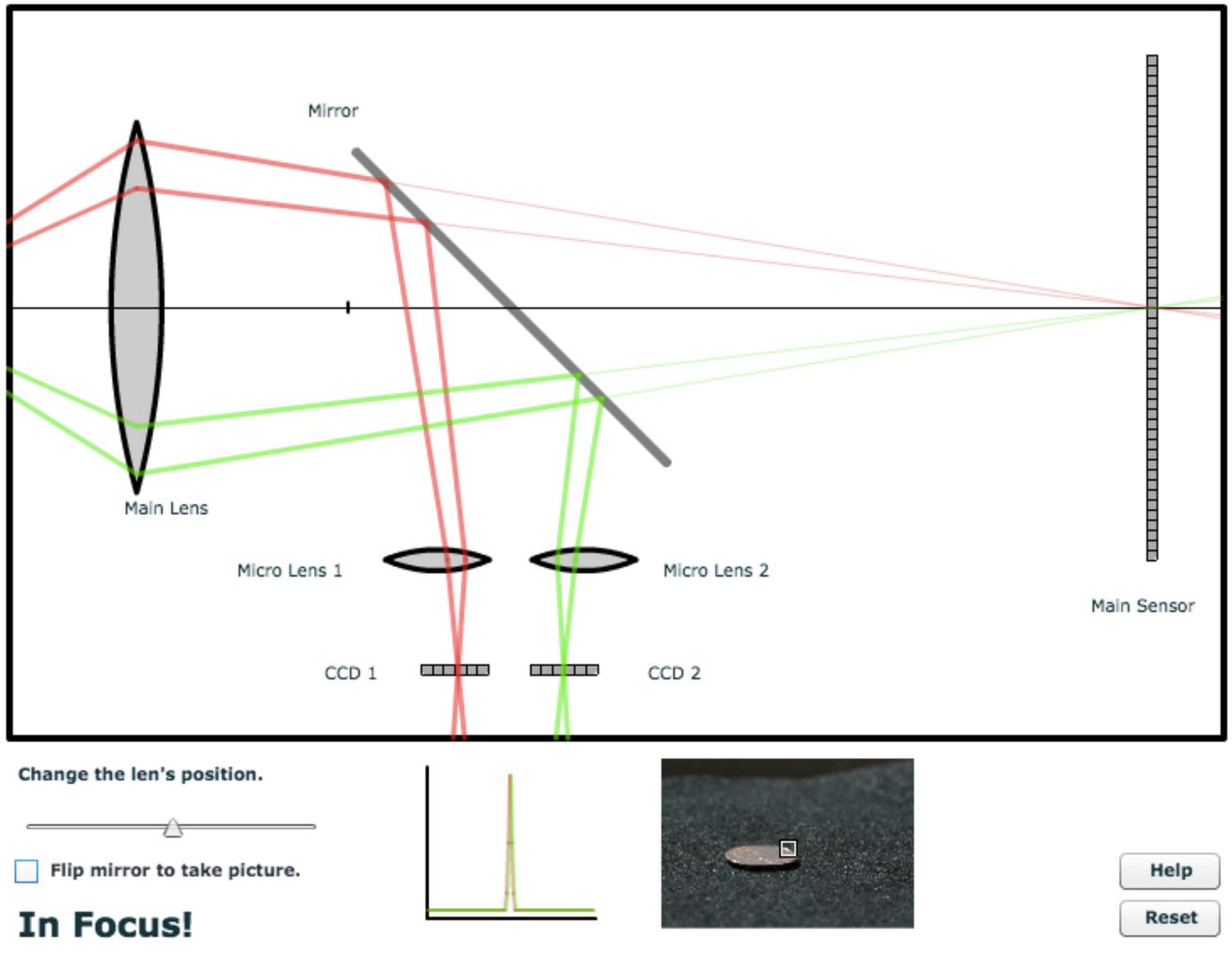
Phase Detection Autofocus

Ray bundles from a target object converge to points at different depths in the camera depending on the lens focus.

In a phase detection AF system ray bundles passing through different portions of the lens (red and green shown) are brought to focus on separate lenslets with separate AF sensors.

Depending on depth of focus point, the ray bundles converge to different positions on their respective AF sensors (see interactive demo).

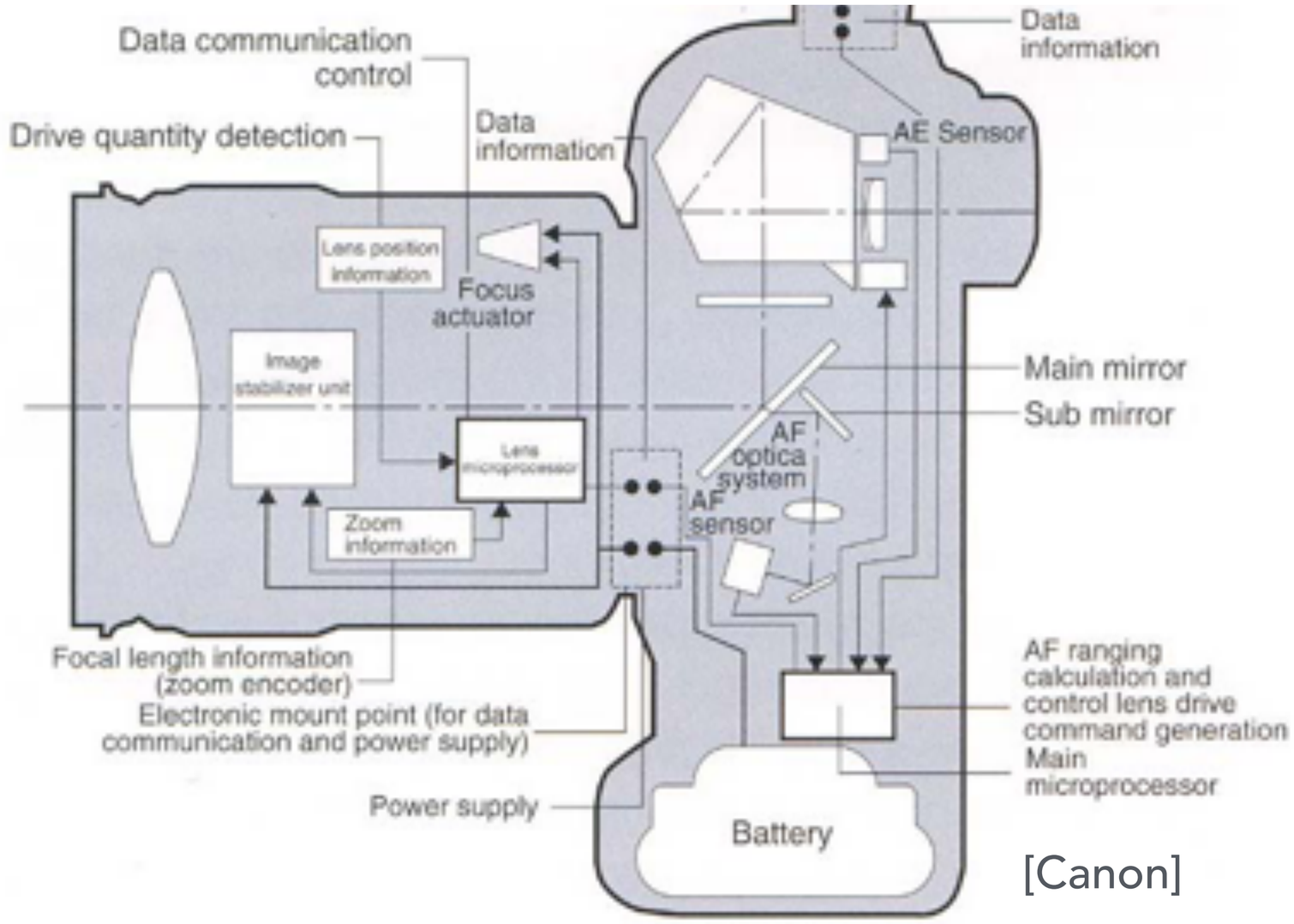
A certain spacing (disparity) between these images is “in focus”



Demo (Levoy, Willet, Adams)

https://graphics.stanford.edu/courses/cs178-10/applets/autofocusPD.html

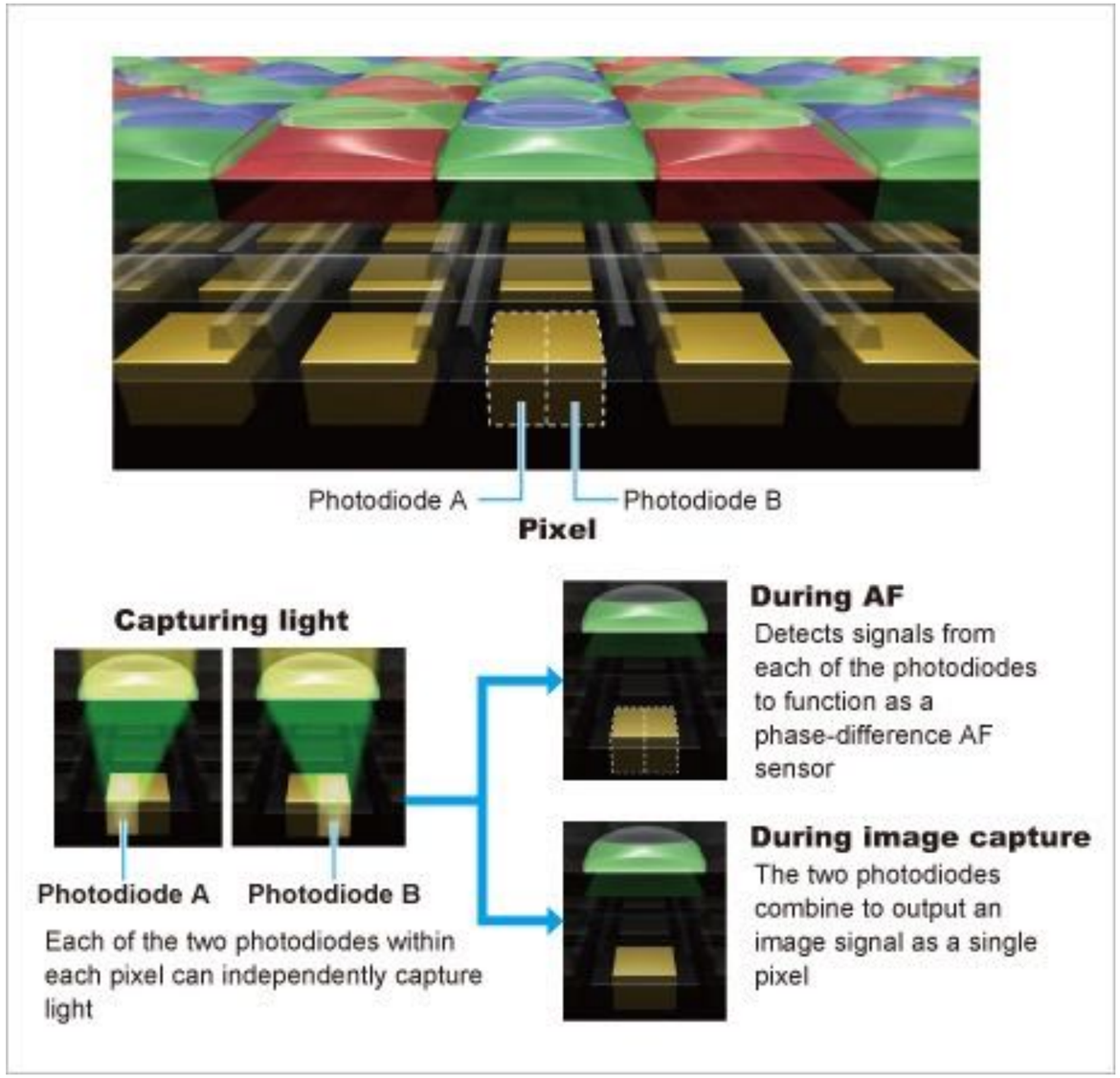
Phase Detection AF Used in DSLRs

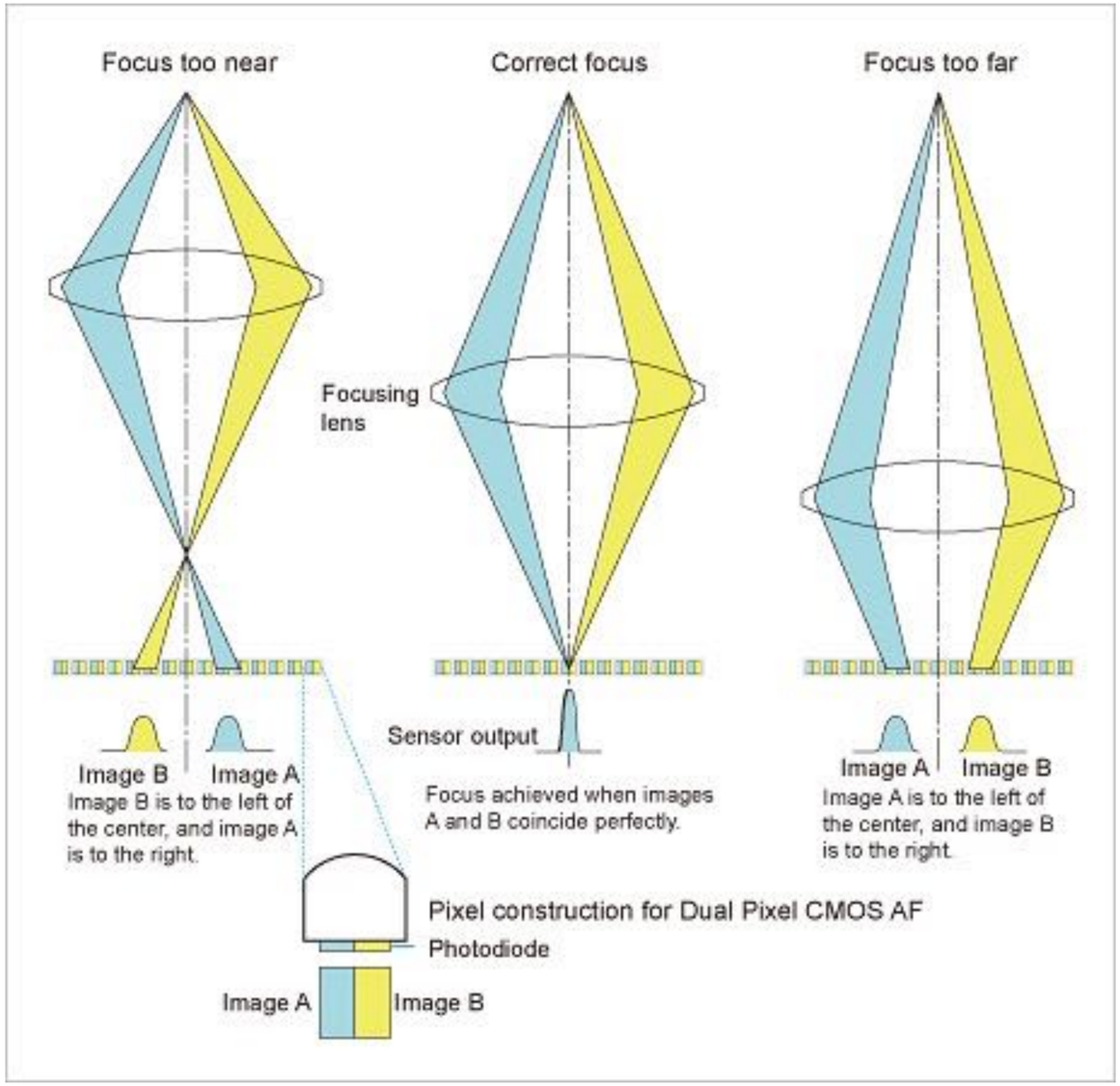


[Canon]

- Distance between phase-detect images correlates to distance in focus to target object (allows "jumping" to the right focus)
- Separate AF units cannot be used with "live view" or video recording

Phase Detection Pixels Embedded in Sensor





Canon

- Modern image sensors have small pixels, and may embed phase detection pixels directly into sensor image arrays